BANGABANDHU SHEIKH MUJIBUR RAHMAN
AVIATION AND AEROSPACE UNIVERSITY (BSMRAAU)



COURSE CURRICULUM
OF
BACHELOR OF SCIENCE IN AVIONICS ENGINEERING

DEPARTMENT OF AVIONICS ENGINEERING
BANGABANDHU SHEIKH MUJIBUR RAHMAN
AVIATION AND AEROSPACE UNIVERSITY (BSMRAAU)

PREFACE

Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University (BSMRAAU) began its journey as the country's first public higher educational institution specialised in aviation and aerospace studies. Through providing world-class education and capacity building, this university seeks to be acknowledged as a top institution for studies in aviation, space, and technology. In this regard BSMRAAU is offering various undergraduate and graduate programs for a wide range of students. Undergraduate students at BSMRAAU's Avionics Engineering Department are the target audience for this course curriculum. Although students were the main focus of this curriculum's development, student advisers and teachers will also find it to be a beneficial resource to administer their teaching-learning and assessment planning.

This curriculum provides general information about the B.Sc. in **Avionics Engineering programme**. The Outcome-Based Education (OBE) system is highlighted in the course outline, which provides semester-by-semester course material. Lists of core courses, courses offered by other departments and optional courses are presented along with a detailed course outline. It should be noted that the course outline provided is subject to revision and adjustment as the courses grow so that the courses we offer may keep up with the most recent advancements in the relevant subjects and are appropriate for undergraduate students. According to the policy of BSMRAAU, the curriculum will be revised in every 3 years.

We thank all the esteemed curriculum committee members and other BSMRAAU members for their earnest suggestions and hard work, both of which were crucial to the development of this curriculum. We look forward to further suggestions in enriching the curriculum.

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December 2022
BSMRAAU, Lalmonirhat
Bangladesh

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PART **A**

1. **Title of the Academic Program:**
   Bachelor of Science in Avionics Engineering

2. **Name of the University:**
   Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University (BSMRAAU)

3. **Vision of the University:**
   The vision of Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University is "To become a leading international university in the field of aviation, space and technological studies through capacity building".

4. **Mission of the University:**
   The mission of Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University is "To transform people into Aviation, Space and Technological professional and enthusiast by providing world class education".

5. **Name of the program offering entity:**
   Department of Avionics Engineering

6. **Vision of the program offering entity:**
   To create skilled and competent professionals in the field of Avionics Engineering with high morals to meet the national and global needs through creative research and innovations.

7. **Mission of the program offering entity:**
   Department of Avionics Engineering, BSMRAAU is working with the following missions in mind:

   a) To provide state-of-the-art education in Avionics Engineering, to produce qualified engineers, capable of solving real-world problems to meet the needs of industry and society.
   b) To contribute towards the creation of new knowledge through research and innovation in relevant fields of Avionics engineering and allied fields to address emerging national and global issues for the well-being of the society.
   c) To enable students in attaining required ethics with an attitude of entrepreneurial skills, ethical values and social consciences.
   d) To embed leadership qualities amongst the students to follow successful professional career paths and to pursue advanced studies in Aerospace and Avionics engineering.

8. **Objectives of the program offering entity:**

   Dept. of Avionics Engineering is to establish itself as the leading academic department for aviation technology on a national and international scale by providing students with the best undergraduate education possible in the disciplines of air traffic control, navigation, and communication.

   Societal Relevance and a Community Resource:

   a) Strengthening collaboration and partnership activities with the aviation community will increase undergraduate students' chances for academic achievement and get them ready for enrollment into the aviation community after graduation.
   b) Seek input from business and industry while creating academic programs to make sure that graduates are qualified to fill jobs in the aviation sector.

c) Expanding the graduate program will increase prospects for collaborative research with other universities with a focus on aviation, especially those certified by the Council on Aviation.

d) Expand the number of collaborations or collaborative relationships that BSMRAAU has with enterprises, organizations, governments, etc.

e) Campus Community and Learning Environment

f) Attract all qualifying undergraduate students, improve the number of accessible unique opportunities, boost student academic achievement, and expand and encourage staff and student involvement in volunteer and community work are all goals.

g) Increase the proportion of faculty members who offer professional advice or assistance to the public.

Core Academic Experiences

a) On the campus of BSMRAAU, resources should be developed and improved for teaching, research, creative work, and professional service.

b) Create chances for academics to conduct research and/or creative work with extramural support.

c) Increase the number of graduate students supported by extramural money and the proportion of teachers actively engaged in publications, practical projects, and professional service.

Professional Preparation

a) Continue educating and training students in a realistic airline environment so they can transition into flight positions with ease.

b) Continue to develop the emphasis in air transportation management by updating the curricula in response to advice from experts in the field, particularly the Avionics Program Industry Advisory Board.

c) Continue to develop the Air Traffic Management Program by updating the curricula to reflect advancements in air traffic controllers' and the aviation industry's utilization of technology, apparatus, and procedures.

Scholarship

a) To uphold and enhance the campus's intellectual and cultural climate and to find possibilities for faculty scholarship.

b) Boost the amount of outside funding generated for education, research, creative endeavors, and professional service.

c) Increase opportunities for collaborative research with industry and government agencies and identify areas of possible application for developing technologies to solve current industry problems, such as airport/aircraft noise, hazardous materials, pollution, airport/aircraft security, and air transportation education and training.

9. Name of the Degree:

Bachelor of Science in Avionics Engineering

10. Description of the Program:

Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University (BSMRAAU) offers a four-year, 160-credit B.Sc. in Avionics Engineering program to prepare students for careers in the aviation sector. The program focuses on instructing students in the correct methods for installing electrical, electronic, and computer equipment for an aircraft's control, navigation, and communication systems. The course covers a wide range of subjects, including electrical, electronics, computers, navigation, instruments, and communication, as well as themes like human factor, quality control, and the most recent developments in aviation technology. Since the program includes all necessary study materials, graduates can easily take on complex engineering problems and innovation activities when they graduate

11. Graduate Attributes:

According to Washington Accord (WA) a graduate must have the following attributes:

Attribute 1:	**Engineering Knowledge:** Apply mathematics, natural science, engineering fundamentals and engineering specialization to the solution of complex engineering problems
Attribute 2:	**Problem Analysis:** Identify, formulate, research literature & analyse complex engineering problems using first principles of mathematics, natural sciences and engineering sciences
Attribute 3:	**Design/Development of Solutions:** Design solutions for complex engineering problems and design systems, components or processes with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
Attribute 4:	**Investigation:** Conduct investigations of complex problems using research-based knowledge and research methods.
Attribute 5:	**Modern Tool Usage:** Create, select and apply modern engineering and IT tools including prediction and modelling to complex engineering problems.
Attribute 6:	**The Engineer and Society:** Apply reasoning to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems
Attribute 7:	**Environment and Sustainability:** Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems
Attribute 8:	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
Attribute 9:	**Individual and Team Work:** Function effectively as an individual, member or leader in diverse teams and in multi-disciplinary settings
Attribute 10:	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society able to comprehend, write, present, give and receive instructions.
Attribute 11:	**Project Management and Finance**: Demonstrate knowledge and understanding of engineering management principles and economic decision-making, apply to own work, as a member and leader in a team, manage projects and in multidisciplinary environments
Attribute 12:	**Lifelong learning:** Recognize the need, prepare and engage in independent and life-long learning

12. Program Educational Objective (PEO):

a) Our graduates will be able to solve critical technical problems related to Avionics Engineering.

b) Our graduates will be able to build up successful professional careers in the field of aviation (civil and military), government organizations, academia and military in the associated field.

c) Our graduates will be able to pursue continuous learning through professional development, practical training and specialized certifications.

d) Our graduates will be able to undertake postgraduate and doctorate and excel in academic and research careers.

e) Our graduates will be able to positively contribute to national and global socio-economic development.

13. Program Learning Outcome (PO):

Based on the suggestion of Board of Accreditation for Engineering and Technical Education (BAETE), Bangladesh, the Bachelor in Avionics Engineering (AVE) programme will have following program outcomes:

a) Engineering knowledge: Apply the knowledge of mathematics, natural science, engineering fundamentals and an engineering specialisation to the solution of complex engineering problems.

b) Problem analysis: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

c) Design/development of solutions: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental considerations.

d) Investigation: Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

e) Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations.

f) The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.

g) Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.

h) Ethics: Apply ethical principles and commit to professional ethics and responsibilities and the norms of the engineering practice.

i) Individual work and teamwork: Function effectively as an individual, as a member or leader in diverse teams and in multidisciplinary settings.

j) Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

k) Project management and finance: Demonstrate knowledge and understanding of the engineering management principles and economic decision-making and apply these to one's own work as a member and leader in a team, to manage projects and in multidisciplinary environments.

l) Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

14. Mapping Mission of the University with PEOs:

Mission 01: To provide world class education to attain capable, motivated and well-prepared graduate with aviation and aerospace engineering knowledge of highest quality.

Mission 02: To provide dissemination of knowledge in an environment of free movement of ideas by creating inquisitiveness among the students to reach their true potential.

Mission 03: To encourage students to consider professions in the aerospace, airline, and all other aviation-related and linked businesses, both locally and internationally.

PEOs	Mission 1	Mission 2	Mission 3
PEO 1	✓		
PEO 2	✓		✓
PEO 3		✓	
PEO 4		✓	
PEO 5			✓

15. Mapping POs with the PEOs:

POs	PEO 1	PEO 2	PEO 3	PEO 4	PEO 5
PO 1	✓				
PO 2	✓	✓			
PO 3			✓	✓	
PO 4			✓		
PO 5		✓	✓	✓	

PO 6					✓
PO 7	✓				✓
PO 8		✓			
PO 9		✓			✓
PO 10		✓			✓
PO 11		✓			✓
PO 12			✓		

16. Mapping courses with the POs:

Courses	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PL10	PO11	PO12
AVE 4101	✓	✓	✓									
AVE 4102	✓	✓							✓			
AVE 4201	✓	✓	✓									
AVE 4202	✓		✓						✓			
AVE 4301	✓	✓										
AVE 4302	✓	✓	✓						✓			
AVE 4401	✓	✓	✓									
AVE 4402	✓	✓	✓									
AVE 4403	✓		✓									
AVE 4404	✓	✓							✓			
AVE 4405	✓	✓	✓							✓		
AVE 4406	✓	✓									✓	
AVE 4501	✓	✓	✓									
AVE 4502	✓				✓							
AVE 4503	✓	✓	✓									
AVE 4504	✓		✓						✓			
AVE 4505	✓	✓	✓									
AVE 4510					✓				✓	✓		
AVE 4600	✓						✓			✓		✓
AVE 4601	✓	✓	✓									
AVE 4602	✓	✓			✓							✓
AVE 4603	✓	✓	✓									✓
AVE 4604	✓	✓	✓								✓	
AVE 4605	✓	✓	✓									✓
AVE 4607	✓	✓	✓									
AVE 4608	✓		✓		✓							
AVE 4609	✓		✓		✓							
AVE 4610			✓						✓			
AVE 4701	✓	✓	✓									✓

AVE 4702	✓	✓							✓			
AVE 4703	✓		✓			✓						
AVE 4704	✓								✓			
AVE 4705	✓										✓	
AVE 4706	✓										✓	
AVE 4707						✓	✓	✓				
AVE 4801	✓	✓	✓									
AVE 4802	✓	✓							✓			
AVE 4803	✓	✓		✓								
AVE 4804	✓	✓	✓	✓						✓		
AVE 4700			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ASE 4101	✓	✓		✓								
ASE 4102	✓		✓	✓					✓			
CSE 4201	✓			✓								
CSE 4202	✓		✓						✓			
ASE 4202	✓		✓						✓			
ASE 4341	✓		✓	✓								
ASE 4342	✓		✓	✓					✓			
ASE 4353	✓	✓		✓								
CSE 4305	✓		✓	✓								
CSE 4306	✓	✓	✓	✓								
ASE 4513	✓	✓		✓								
ASE 4514	✓	✓	✓		✓							
ASE 4807										✓	✓	✓
MAT 4101	✓	✓										
MAT 4203	✓	✓										
MAT 4305	✓	✓										
MAT 4407	✓	✓										
MAT 4509	✓	✓	✓	✓								
PHY 4101	✓	✓	✓									
PHY 4102	✓	✓	✓						✓			
CHM 4201	✓	✓	✓									
CHM 4202	✓	✓							✓			
HUM 4101										✓		✓
HUM 4102										✓		✓
HUM 4203		✓				✓				✓		
HUM 4413		✓									✓	

PART **B**

17. Structure of the Curriculum:

a. Duration of the Program: Year: 4 (Semester: 08)

b. Admission Requirements:

Admission in Bachelor Program:

1. Four Year BSc Program: The admission to the various B.Sc. programs of the University will be done for each session as decided by the Academic Council.

2. Eligibility:

   a. A candidate must pass SSC/Equivalent and HSC/Equivalent examination from science group. Candidates passed in the current year and previous one year eligible to apply.

   b. Candidates from science group must obtain a minimum GPA of 4.50 each in SSC/Equivalent and HSC/Equivalent with minimum A Grade in Mathematics, Physics, Chemistry, and English/English language subjects.

   c. Candidates from General Certificate of Education (GCE) in 'O' level examination background must pass in minimum 5 subjects. Candidates having 'C' or lower grades in more than 2 subjects out of 5 subjects including Mathematics, Physics, Chemistry, and English will not be eligible for admission test.

   d. Candidates from General Certificate of Education (GCE) in 'A' level examination background must pass in minimum 3 subjects. Candidates having 'C' or lower grades in more than 1 subject out of 3 subjects including Mathematics, Physics and Chemistry will not be eligible for admission test.

   e. Candidates for GCE (O & A level) background must have minimum 27 points in total based on the following scale:

Grade	Point
A*/A	5.00
B	4.00
C	3.50
D	3.00

   f) Candidates from International Baccalaureate (IB) must pass in minimum 6 (six) subjects having minimum 30 points based on the rating scale used in their curriculum (7, 6, 5, 4). Rating of 1, 2 and 3 will not be considered in point calculation.

   g) Students must have Mathematics in HSC as their compulsory/optional (4th subject) subject.

   h) Non-Resident Bangladeshi applicants from aboard shall apply through Embassies/High Commission. Educational qualifications are same as applicable for Bangladeshi students.

3. Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University (BSMRAAU) will invite application from the eligible candidates for admission in Bachelor Program once in a year aiming to start the session from January or as decided by the Academic Council.

4. BSMRAAU will offer four years BSc Program with 30 seats or as decided by Admission Committee in each session. The selection of the students for Bachelor Programs will be on a competitive basis of past academic records, and performances in the admission test with Viva-voce (if required).

5. Students must show the original Transcripts/Mark Sheets at the time of admission.

6. Students who all are finally selected for admission have to undergo a medical examination to be declared fit for the respective courses.

7. The university authority reserves the right to cancel/modify/change any information given in this advertisement. For any query, applicants may communicate at contact@bsmraau.edu.bd.

Qualifying Criteria for International Students:

1. Eligibility:

    a. Passed Higher Secondary School or Equivalent for admission into Under Graduate Programs and passed Under Graduate Program for admission into Post Graduate Program.

    b. GPA requirement for both the programs minimum 4.50 in 5 scale grading system and minimum 3.50 in 4 scale grading system.
    c. For Engineering Program applicant must have passed Higher Secondary School/Equivalent with Mathematics, Physics and Chemistry.

    d. Provide statement of purpose (SOP) for admission.

    e. Applicant will need to submit a list of two qualified referees with one in relation to academic attainment.

    f. Provide supporting paper (certificate and transcript) in support of academic qualifications.

2. English Language Requirement: Provide proof of English Proficiency, if English is not the first language to the country of citizen, as under:

    a. Minimum Score 6.5 in IELTS with No Score below 6.0 in individual part

    b. Minimum Score 1650 in SAT out of 2400 and Minimum Score of 1110 in SAT out of 1600.
    c. Minimum Score 86 in TOEFL with No Score below 21 in each skill band

    d. Validity of English Language Proficiency will be 2 years from test date.

    e. No admission test will be required for the students fulfilling the above-mentioned criteria. However, the selected candidate will be called for Viva-voce before granting the Confirmation of Enrolment (CoE).

Note: Eligibility and other requirements may be reviewed as and when required and are to be approved by the Academic Council.

c. Credit Requirement to complete the program Total Minimum: 154 Credits

d. Total Class Week in a Semester: 14 Weeks

e. Minimum CGPA Requirements for Graduation: 2.25

f. Maximum Academic Year of Completion: 6 Years

g. Category of Courses:

(I) List of Core Engineering Courses:

A total of 40 courses on avionics engineering topics are included in the program which covers 90 credit hours.

Course Code	Course Title	Type of Course	Contact Hours	Credit Hours	Semester
AVE 4101	Electrical Circuits Analysis I	Theory	3.00	3.00	1
AVE 4102	Electrical Circuits Analysis I Sessional	Sessional	3.00	1.50	1
AVE 4201	Electrical Circuits Analysis II	Theory	3.00	3.00	2
AVE 4202	Electrical Circuits Analysis II Sessional	Sessional	1.50	0.75	2
AVE 4301	Electronic Circuits I	Theory	3.00	3.00	3
AVE 4302	Electronic Circuits I Sessional	Sessional	3.00	1.50	3
AVE 4401	Electronic Circuits II	Theory	3.00	3.00	4
AVE 4402	Electronic Circuits II Sessional	Sessional	3.00	1.50	4
AVE 4403	Electric Machine and Drives	Theory	3.00	3.00	4
AVE 4404	Electric Machines and Drives Sessional	Sessional	3.00	1.50	4
AVE 4405	Digital Logic and Computer Design	Theory	3.00	3.00	4
AVE 4406	Digital Logic and Computer Design Sessional	Sessional	3.00	1.50	4
AVE 4501	Signals and Systems	Theory	3.00	3.00	5
AVE 4502	Signals and Systems Sessional	Sessional	1.50	0.75	5
AVE 4503	Microprocessor and Microcontroller Systems	Theory	3.00	3.00	5
AVE 4504	Microprocessor and Microcontroller Systems Sessional	Sessional	1.50	0.75	5
AVE 4505	Electromagnetic Field Theory	Theory	3.00	3.00	5
AVE 4510	Modelling & Simulation Sessional	Sessional	3.00	1.50	5
AVE 4600	Industrial Training	Sessional	(4 weeks)	1.00	6
AVE 4601	Communication Systems	Theory	3.00	3.00	6
AVE 4602	Communication Systems Sessional	Sessional	3.00	1.50	6
AVE 4603	Real Time Embedded Systems	Theory	3.00	3.00	6
AVE 4604	Real Time Embedded Systems Sessional	Sessional	1.50	0.75	6
AVE 4605	Microwave and Antennas	Theory	3.00	3.00	6
AVE 4607	Numerical Methods	Theory	3.00	3.00	6
AVE 4608	Numerical Methods Sessional	Sessional	1.50	0.75	6
AVE 4609	Control Systems	Theory	3.00	3.00	6
AVE 4610	Control Systems Sessional	Sessional	1.50	0.75	6
AVE 4701	Digital Signal Processing	Theory	3.00	3.00	7
AVE 4702	Digital Signal Processing Sessional	Sessional	1.50	0.75	7
AVE 4703	Aircraft Electrical, Instrument and Autopilot Systems	Theory	3.00	3.00	7
AVE 4704	Aircraft Electrical, Instrument and Autopilot Systems Sessional	Sessional	3.00	1.50	7
AVE 4705	Avionics System Design	Theory	3.00	3.00	7
AVE 4706	Avionics System Design Sessional	Sessional	1.50	0.75	7
AVE 4707	Engineering Ethics & Professionalism	Theory	2.00	2.00	7
AVE 4801	Radar System Engineering	Theory	3.00	3.00	8
AVE 4802	Radar System Engineering Sessional	Sessional	3.00	1.50	8
AVE 4803	Aircraft Communication, Navigation & Surveillance System	Theory	3.00	3.00	8
AVE 4804	Aircraft Communication, Navigation &	Sessional	3.00	1.50	8

Course Code	Course Title	Type of Course	Contact Hours	Credit Hours	Semester
	Surveillance System Sessional				
AVE 4700	Final Year Research and Design Project (FYRDP)	Sessional	9.00	4.50	7+8

(II) List of Other Engineering Courses:

A total of 13 courses (7 theory and 6 sessional) are included in the programme covering 30.25 credit hours.

Course Code	Course Title	Type of Course	Contact Hours	Credit Hours	Semester
ASE 4101	Introduction to Aeronautical Engineering	Theory	3.00	3.00	1
ASE 4102	Aeronautical Engineering Drawing	Sessional	3.00	1.50	1
CSE 4201	Computer Programming and Application	Theory	3.00	3.00	2
CSE 4202	Computer Programming and Application Sessional	Sessional	3.00	1.50	2
ASE 4202	Workshop Technology Sessional	Sessional	3.00	1.50	2
ASE 4341	Thermodynamics	Theory	3.00	3.00	3
ASE 4342	Thermodynamics Sessional	Sessional	1.50	0.75	3
ASE 4353	Engineering Mechanics (Statics & Dynamics)	Theory	4.00	4.00	3
CSE 4305	Data Structures and Algorithm	Theory	3.00	3.00	3
CSE 4306	Data Structures and Algorithm Sessional	Sessional	3.00	1.50	3
ASE 4513	Aerodynamics	Theory	3.00	3.00	5
ASE 4514	Aerodynamics Sessional	Sessional	3.00	1.50	5
ASE 4807	Industrial & Business Management	Theory	3.00	3.00	8

(III) List of Basic Sciences and Mathematics Courses:

A total of 5 mathematics courses and 2 basic sciences courses including 2 sessional are included in the program covering 23.25 credit hours.

Course Code	Course Title	Type of Course	Contact hours	Credit Hours	Semester
MAT 4101	Differential and Integral Calculus	Theory	3.00	3.00	1
MAT 4203	Ordinary and Partial Differential Equations	Theory	3.00	3.00	2
MAT 4305	Linear Algebra, Coordinate Geometry and Complex Variables	Theory	3.00	3.00	3
MAT 4407	Fourier and Laplace Transform	Theory	3.00	3.00	4
MAT 4509	Probability & Statistics	Theory	3.00	3.00	5
PHY 4101	Physics	Theory	3.00	3.00	1
PHY 4102	Physics Sessional	Sessional	3.00	1.50	1
CHM 4201	Chemistry	Theory	3.00	3.00	2
CHM 4202	Chemistry Sessional	Sessional	1.50	0.75	2

(IV) List of Humanities and Social Sciences (HASS) Courses:

A total of 4 courses from HASS are included in the programme covering 9.75 credit hours.

Course Code	Course Title	Type of Course	Contact hours	Credit Hours	Semester
HUM 4101	Communicative English	Theory	3.00	3.00	1
HUM 4102	Communicative English Sessional (Technical Report Writing)	Sessional	1.50	0.75	1
HUM 4203	Bangladesh Studies and Society	Theory	3.00	3.00	2
HUM 4413	Engineering Economics	Theory	3.00	3.00	4

(V) Elective Courses

Students of B.Sc. in Avionics Engineering can choose 2 elective courses along with 1 sessional covering 6.75 credit hours. Students can take optional courses from any of the following courses offered. Availability of elective courses is subject to availability of faculty and related laboratory facilities.

a) List of Departmental Electives Courses

Course No	Course Title	Type of course	Contact hours	Credit Hours
AVE 4721	Avionics Software Design	Theory	3.00	3.00
AVE 4723	Autonomous Systems, Guidance & Control	Theory	3.00	3.00
AVE 4725	Robotics and Mechatronics	Theory	3.00	3.00
AVE 4729	Remote Sensing and Digital Image Processing	Theory	3.00	3.00
AVE 4731	Avionics Human Factors Engineering	Theory	3.00	3.00
AVE 4733	RF Integrated Circuit	Theory	3.00	3.00
AVE 4821	Introduction to Satellite Communication Engineering	Theory	3.00	3.00
AVE 4822	Introduction to Satellite Communication Engineering Sessional	Sessional	1.50	0.75
AVE 4823	Space Avionics	Theory	3.00	3.00
AVE 4824	Space Avionics Sessional	Sessional	1.50	0.75
AVE 4825	VLSI Design	Theory	3.00	3.00
AVE 4826	VLSI Design Sessional	Sessional	1.50	0.75
AVE 4827	Optical Fibre Communication System	Theory	3.00	3.00
AVE 4828	Optical Fibre communication system Sessional	Sessional	1.50	0.75
AVE 4829	Avionics Cyber Security	Theory	3.00	3.00
AVE 4830	Avionics Cyber Security Sessional	Sessional	1.50	0.75
AVE 4831	Wireless Communication	Theory	3.00	3.00
AVE 4832	Wireless Communication Sessional	Sessional	1.50	0.75

b) List of Non-Departmental Elective Courses

Undergraduate students from the Department of Avionics Engineering can choose courses from the following list of Other/Aerospace engineering courses.

Course Code	Course Title	Semester	Contact Hours	Credit Hours
ASE 4551	Aircraft Systems	Theory	3.00	3.00

ASE 4552	Aircraft Systems Sessional	Sessional	1.50	0.75
ASE 4557	Aerospace Vehicle Dynamics & Control	Theory	3.00	3.00
ASE 4797	Orbital Mechanics	Theory	3.00	3.00
CSE 4781	Machine learning	Theory	3.00	3.00
CSE 4783	Artificial Intelligence	Theory	3.00	3.00
CSE 4785	Introduction to Quantum Computing	Theory	3.00	3.00
EEE 4881	Electrical Power Systems	Theory	3.00	3.00
EEE 4882	Electrical Power Systems Sessional	Sessional	1.50	0.75

(VI) Final Year Research and Design Project (FYRDP)

Final Year Research and Design Project (FYRDP) will have to be undertaken by students under a supervisor in partial fulfilment of the requirement of his degree. Credit hours allotted to the Final Year Research and Design Project (FYRDP) will be 9 credit hours having 18 contact hours.

18. Semester Wise Distribution of Courses:

The detailed course plan for Bachelor of Science in Avionics Engineering is presented below. The programme includes **39 theory courses** and **31 sessional courses** covering **117** and **43 credit** hours for theory and sessional respectively. **Total credit hours** for the programme are **160** in 8 semesters.

Undergraduate students of the Department of Avionics Engineering (AVE) must undertake the following course schedule, the semester-wise distribution of which is given below:

First Year, First Semester

Course Code	Course Title	Type	Contact Hours	Credit Hours
PHY 4101	Physics	Theory	3.00	3.00
AVE 4101	Electrical Circuits Analysis I	Theory	3.00	3.00
MAT 4101	Differential and Integral Calculus	Theory	3.00	3.00
ASE 4101	Introduction to Aeronautical Engineering	Theory	3.00	3.00
HUM 4101	Communicative English	Theory	3.00	3.00
	Subtotal (Theory)		**15.00**	**15.00**
PHY 4102	Physics Sessional	Sessional	3.00	1.50
AVE 4102	Electrical Circuits Analysis I Sessional	Sessional	3.00	1.50
ASE 4102	Aeronautical Engineering Drawing	Sessional	3.00	1.50
HUM 4102	Communicative English Sessional (Technical Report Writing)	Sessional	1.50	0.75
	Subtotal (Sessional)		**10.50**	**5.25**
	Total		**25.50**	**20.25**

First Year, Second Semester

Course Code	Course Title	Type	Contact Hours	Credit Hours
AVE 4201	Electrical Circuits Analysis II	Theory	3.00	3.00
CHM 4201	Chemistry	Theory	3.00	3.00
MAT 4203	Ordinary and Partial Differential Equations	Theory	3.00	3.00
CSE 4201	Computer Programming and Application	Theory	3.00	3.00
HUM 4203	Bangladesh Studies and Society	Theory	3.00	3.00
	Subtotal (Theory)		**15.00**	**15.00**

Course Code	Course Title	Type	Contact Hours	Credit Hours
AVE 4202	Electrical Circuits Analysis II Sessional	Sessional	1.50	0.75
CHM 4202	Chemistry Sessional	Sessional	1.50	0.75
ASE 4202	Workshop Technology Sessional	Sessional	3.00	1.50
CSE 4202	Computer Programming and Application Sessional	Sessional	3.00	1.50
	Subtotal (Sessional)		**9.00**	**4.50**
	Total		**24.00**	**19.50**

Second Year, First Semester

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 4301	Electronic Circuit I	Theory	3.00	3.00
ASE 4341	Thermodynamics	Theory	3.00	3.00
ASE 4353	Engineering Mechanics (Statics & Dynamics)	Theory	4.00	4.00
MAT 4305	Linear Algebra, Coordinate Geometry and Complex Variables	Theory	3.00	3.00
CSE 4305	Data Structures and Algorithm	Theory	3.00	3.00
Subtotal (Theory)			**16.00**	**16.00**
AVE 4302	Electronic Circuit I Sessional	Sessional	3.00	1.50
ASE 4342	Thermodynamics Sessional	Sessional	1.50	0.75
CSE 4306	Data Structures and Algorithm Sessional	Sessional	3.00	1.50
Subtotal (Sessional)			**7.50**	**3.75**
Total			**23.50**	**19.75**

Second Year, Second Semester

Course Code	Course Title	Type	Contact Hours	Credit Hours
AVE 4401	Electronic Circuits II	Theory	3.00	3.00
AVE 4403	Electric Machines and Drives	Theory	3.00	3.00
MAT 4407	Fourier and Laplace Transform	Theory	3.00	3.00
AVE 4405	Digital Logic and Computer Design	Theory	3.00	3.00
HUM 4413	Engineering Economics	Theory	3.00	3.00
Subtotal (Theory)			**15.00**	**15.00**
AVE 4402	Electronic Circuits II Sessional	Sessional	3.00	1.50
AVE 4404	Electric Machines and Drives Sessional	Sessional	3.00	1.50
AVE 4406	Digital Logic and Computer Design Sessional	Sessional	3.00	1.50
Subtotal (Sessional)			**9.00**	**4.50**
Total			**24.00**	**19.50**

Third Year, First Semester

Course Code	Course Title	Type	Contact Hours	Credit Hours
AVE 4501	Signals and Systems	Theory	3.00	3.00
AVE 4503	Microprocessor and Microcontroller Systems	Theory	3.00	3.00
AVE 4505	Electromagnetic Field Theory	Theory	3.00	3.00
ASE 4513	Aerodynamics	Theory	3.00	3.00
MAT 4509	Probability & Statistics	Theory	3.00	3.00
Subtotal (Theory)			**15.00**	**15.00**
AVE 4502	Signals and Systems Sessional	Sessional	1.50	0.75
AVE 4504	Microprocessor and Microcontroller Systems	Sessional	1.50	0.75

Course Code	Course Title	Type	Contact Hours	Credit Hours
	Sessional			
AVE 4510	Modelling & Simulation Sessional	Sessional	3.00	1.50
ASE 4514	Aerodynamics Sessional	Sessional	3.00	1.50
Subtotal (Sessional)			**9.00**	**4.50**
Total			**24.00**	**19.50**

Third Year, Second Semester

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 4601	Communication Systems	Theory	3.00	3.00
AVE 4603	Real Time Embedded Systems	Theory	3.00	3.00
AVE 4605	Microwave and Antennas	Theory	3.00	3.00
AVE 4607	Numerical Methods	Theory	3.00	3.00
AVE 4609	Control Systems	Theory	3.00	3.00
Subtotal (Theory)			**15.00**	**15.00**
AVE 4600	Industrial Training	Sessional	(4 week)	1.00
AVE 4602	Communication Systems Sessional	Sessional	3.00	1.50
AVE 4604	Real Time Embedded Systems Sessional	Sessional	1.50	0.75
AVE 4608	Numerical Methods Sessional	Sessional	1.50	0.75
AVE 4610	Control Systems Sessional	Sessional	1.50	0.75
Subtotal (Sessional)			**9.50**	**4.75**
Total			**24.50**	**19.75**

Fourth Year, First Semester

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 4701	Digital Signal Processing	Theory	3.00	3.00
AVE 4703	Aircraft Electrical, Instrument and Autopilot Systems	Theory	3.00	3.00
AVE 4705	Avionics System Design	Theory	3.00	3.00
AVE 4707	Engineering Ethics & Professionalism	Theory	2.00	2.00
AVE 47XX	Elective - I	Theory	3.00	3.00
Subtotal (Theory)			**14.00**	**14.00**
AVE 4702	Digital Signal Processing Sessional	Sessional	1.50	0.75
AVE 4704	Aircraft Electrical, Instrument and Autopilot Systems Sessional	Sessional	3.00	1.50
AVE 4706	Avionics System Design Sessional	Sessional	1.50	0.75
AVE 4700	Final Year Research and Design Project (FYRDP)	Sessional	9.00	4.50
Subtotal (Sessional)			**15.00**	**7.50**
Total			**29.00**	**21.50**

Fourth Year, Second Semester

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 4801	Radar System Engineering	Theory	3.00	3.00
AVE 4803	Aircraft Communication, Navigation & Surveillance System	Theory	3.00	3.00
ASE 4807	Industrial & Business management	Theory	3.00	3.00

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 48XX	Elective - II	Theory	3.00	3.00
Subtotal (Theory)			**12.00**	**12.00**
AVE 4802	Radar System Engineering Sessional	Sessional	3.00	1.50
AVE 4804	Aircraft Communication, Navigation & Surveillance System Sessional	Sessional	3.00	1.50
AVE 48XX	Elective - II Sessional	Sessional	1.50	0.75
AVE 4700	Final Year Research and Design Project (FYRDP)	Sessional	9.00	4.50
Subtotal (Sessional)			**16.50**	**8.25**
Total			**28.50**	**20.25**

19. Summary of Departmental, Non-departmental, Basic Sciences, Mathematics and Humanities Courses

Semester	Theory		Sessional		Total	
	Credit Hours	Contact Hours	Credit Hours	Contact Hours	Credit Hours	Contact Hours
Semester 1	15.00	15.00	5.25	10.50	20.25	25.50
Semester 2	15.00	15.00	4.50	9.00	19.50	24.00
Semester 3	16.00	16.00	3.75	7.50	19.75	23.50
Semester 4	15.00	15.00	4.50	9.00	19.50	24.00
Semester 5	15.00	15.00	4.50	9.00	19.50	24.00
Semester 6	15.00	15.00	4.75	9.50	19.75	24.50
Semester 7	14.00	14.00	7.50	15.00	21.50	29.00
Semester 8	12.00	12.00	8.25	16.50	20.25	28.50
Total	**117.00**	**117.00**	**43.00**	**86.00**	**160.00**	**203.00**

20. Distribution of Credit Hours for Different Categories of Courses

Semester	Humanities	Math	Basic	Engineering		Elective	FYRDP	Total
				Core	Other			
Semester 1	3.75	3.00	4.50	4.50	4.50	-	-	20.25
Semester 2	3.00	3.00	3.75	3.75	6.00	-	-	19.50
Semester 3	-	3.00	-	4.50	12.25	-	-	19.75
Semester 4	3.00	3.00	-	13.50	-	-	-	19.50
Semester 5	-	3.00	-	12.00	4.50	-	-	19.50
Semester 6	-	-	-	19.75	-	-	-	19.75
Semester 7	-	-	-	14.00	-	3.00	4.50	21.50
Semester 8	-	-	-	9.00	3.00	3.75	4.50	20.25
Total Credit Hours	**9.75**	**15.00**	**8.25**	**81.0**	**30.25**	**6.75**	**9.00**	**160.00**
	6.09%	9.37%	5.16%	50.63%	18.9%	4.22%	5.63%	100%

21. List of Elective Courses

a. Elective -I [One Theory]

Course No	Course Title	Type	Contact hours	Credit Hours
AVE 4721	Avionics Software Design	Theory	3.00	3.00
AVE 4723	Autonomous Systems, Guidance & Control	Theory	3.00	3.00
AVE 4725	Robotics and Mechatronics	Theory	3.00	3.00
AVE 4729	Remote Sensing and Digital Image Processing	Theory	3.00	3.00
AVE 4731	Avionics Human Factors Engineering	Theory	3.00	3.00
AVE 4733	RF Integrated Circuit	Theory	3.00	3.00
CSE 4781	Machine Learning	Theory	3.00	3.00
CSE 4783	Artificial Intelligence	Theory	3.00	3.00
CSE 4785	Introduction to Quantum Computing	Theory	3.00	3.00
ASE 4557	Aerospace Vehicle Dynamics & Control	Theory	3.00	3.00
ASE 4797	Orbital Mechanics	Theory	3.00	3.00

b. Elective-II [One Theory + One Sessional]

Course No	Course Title	Type	Contact hours	Credit Hours
AVE 4821	Introduction to Satellite Communication Engineering	Theory	3.00	3.00
AVE 4823	Space Avionics	Theory	3.00	3.00
AVE 4825	VLSI Design	Theory	3.00	3.00
AVE 4827	Optical Fibre Communication System	Theory	3.00	3.00
AVE 4829	Avionics Cyber Security	Theory	3.00	3.00
AVE 4831	Wireless Communication	Theory	3.00	3.00
ASE 4551	Aircraft Systems	Theory	3.00	3.00
AVE 4822	Introduction to Satellite Communication Engineering Sessional	Sessional	1.50	0.75
AVE 4824	Space Avionics Sessional	Sessional	1.50	0.75
AVE 4826	VLSI Design Sessional	Sessional	1.50	0.75
AVE 4828	Optical Fibre communication system Sessional	Sessional	1.50	0.75
AVE 4830	Avionics Cyber Security Sessional	Sessional	1.50	0.75
AVE 4832	Wireless Communication Sessional	Sessional	1.50	0.75
ASE 4552	Aircraft Systems Sessional	Sessional	1.50	0.75

PART **C**

22. Description of Courses of B.Sc. in Avionics Engineering Program

a. Core Engineering Courses

Course Code: AVE 4101 Course Title: Electrical Circuits Analysis I Credits: 3.00

Pre-requisite: None.

Rationale of the Course of the Course:

To learn and familiarise the basics of electrical circuit components as well as the analysis of DC circuits.

Course Objectives:

1. Enable the students to learn the basic electrical circuit elements and applications.
2. Facilitate students to grasp electrical network theorems and apply those theorems in solving complex circuit networks.
3. Familiarise the students with the principles of DC circuits.
4. Enable students to understand the ac circuit and their practical applications in day-to-day life uses.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain different circuit elements and electrical circuitries characteristics.
2. Interpret simplification of complex circuits to speed up the solving process by applying different circuit theorems.
3. Analyse resistive circuits by using different circuit analysis techniques.
4. Apply the concepts of energy storage elements in a circuit.

Course Contents:

Circuit variables and elements: Voltage, current, power, energy, independent and dependent sources, resistance. Basic laws: Ohm's law, Kirchoff's current and voltage laws.

Simple resistive circuits: Series and parallel circuits, voltage and current division, wye-delta transformation.
Techniques of circuit analysis: Nodal and mesh analysis including supernode and supermesh. Network theorems: Source transformation, Thevenin's, Norton's and superposition theorems with applications in circuits having independent and dependent sources, maximum power transfer condition and reciprocity theorem.
Energy storage elements: Inductors and capacitors, series parallel combination of inductors and capacitors.
First order circuits: Natural and step responses. Second order circuits

Text Books:

1. Fundamentals of Electric Circuit- Alexander & Sadiku.

Reference Books:

1. Introductory Circuit Analysis - R.L. Boylestad; Prentice Hall of India Private Ltd.
2. Introductory Circuits for Electrical & Computer Engineering - James. W. Nilson; Prentice Hall of India Private Ltd.
3. Alternating Current Circuits – Russell & George F. Corcoran; John Wiley and Sons

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3		✓										
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Flipped Classroom	Class Test/Assignment, Final Exam
CO4	Project Based Learning	Project Report and Presentation

Course Code: AVE 4102 Course Title: Electrical Circuits Analysis I Sessional Credits: 1.50

Pre-requisite: None

Rationale of the Course of the Course:

To learn and familiarise the basics of electrical circuit components as well as the analysis of DC circuits practically.

Course Objectives:

1. Familiarise the students with electrical circuit components, and the proper use of lab equipments
2. Introduce with different circuit techniques and theorems practically.
3. Provide the necessary knowledge about the design and implementation of any desired circuit.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Explain basic electrical components like resistors, capacitors, digital and analogue measuring equipment etc.
2. Describe the properties of basic electrical networks.
3. Apply the knowledge of basic electrical components and networks practically.
4. Analyse the differences between theoretical knowledge with practical observations.

Course Contents:

Experiment no. 01: Familiarisation with Electrical Circuit Components, Instruments and Verification of

Ohm's Law
Experiment no. 02: Verification of KVL and Voltage Divider Rule.
Experiment no. 03: Familiarisation with parallel DC circuits and Kirchhoff's current Law with Current Divider Rule.
Experiment no. 04: Perform to Delta-Wye Conversion
Experiment no.05: Verification of Superposition Theorem
Experiment no. 06: Verification of Thevenin's Theorem.
Experiment no. 07: Verification of Norton's Theorem
Experiment no. 08: Perform to Maximum Power Transfer Theorem
Experiment no. 09: Perform Capacitor testing and Capacitors Series and Parallel Combination
Experiment no. 10: Study of Transient Behavior of RC and RL Circuits.
Experiment no. 11: Charging and Discharging of RC circuits

Text Books:

1. Fundamentals of Electric Circuit- Charles K. Alexander & Matthew N. O. Sadiku.

Reference Books:

1. Introductory Circuit Analysis - R.L. Boylestad; Prentice Hall of India Private Ltd.
2. Introductory Circuits for Electrical & Computer Engineering - James. W. Nilson; Prentice Hall of India Private Ltd.

Alternating Current Circuits – Russell & George F. Corcoran; John Wiley and Sons

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3									✓			
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4201 Course Title: Electrical Circuits Analysis II Credits: 3.00

Pre-requisite: Electrical Circuits Analysis I (AVE 4101)

Rationale of the Course of the Course:
To learn and familiarise the basics of electrical AC circuits as well as the analysis of different AC networks.

Course Objectives:

1. Familiarise with the terminology associated with AC electrical circuits like sinusoidal signal, frequency, power factor, sinusoidal voltage, current, power and other related terminologies.

2. Teach students to identify the ways to solve AC circuit related problems
3. Provides knowledges about magnetically coupled circuit, passive filters, and three phase circuit

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain different AC circuit elements (for both single-phase and three-phase), storage elements, and magnetically coupled circuits power calculation and to apply various theorems/ concepts to solve circuit problems.
2. Analyse single phase AC circuits by using analysis techniques and network theorems.
3. Analyse different types of passive filters using RLC circuits in terms of their design and application.

Course Contents:

Generation of AC voltage and sinusoidal waveforms: Sinusoidal functions: Instantaneous current, voltage, power, cycle, time period, frequency and angular frequency, effective current and voltage, form factor, peak factor.
Complex quantities: Impedance, admittance, reactance and susceptance, Phasors algebra.
AC power analysis: Instantaneous and average power, maximum average power transfer, complex power, real power, reactive power, apparent power, power triangle, power calculation of R-L, R-C and R-L-C circuits, power factor and power factor correction.
Analysis of single-phase AC circuits: Series and parallel RL, RC and RLC circuits, nodal and mesh analysis, application of network theorems in AC circuits, circuits with non-sinusoidal excitations, transients in ac circuits.
Introduction to passive filter: Low pass filter, high pass filter, band pass filter and band stop filter.
Resonance in AC circuits: Series and parallel resonance, magnetically coupled circuits.
Analysis of three-phase circuits: Generation of poly-phase voltage, three-phase supply, balanced and unbalanced circuits, three-phase power calculation, power factor and power factor improvement in three-phase system.

Text Books:

1. Fundamentals of Electric circuits- Charles K. Alexander & Matthew N. O. Sadiku

Reference Books:

1. Alternating Current Circuits – Russell & George F. Corcoran; John Wiley and Sons.
2. Introductory Circuits for Electrical & Computer Engineering - James. W. Nilson; Prentice Hall of India Private Ltd.
3. A Text Book of Electrical Technology- B L Theraja and A K Theraja; S.Chand & Company Ltd.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										

CO3			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Computer Simulations	Class Test/Assignment, Final Exam

Course Code: AVE 4202 Course Title: Electrical Circuits Analysis II Sessional Credits: 0.75

Pre-requisite: Electrical Circuits Analysis I (AVE 4101) and sessional (AVE 4102)

Rationale of the Course:

Students will be able to compare the performance of ac circuits and knowledge will help to build a strong foundation for further development of their project work in the final year.

Course Objectives:

1. Familiarise the students with electrical AC circuits
2. Provide the knowledge of basic electrical components and networks practically.
3. Explain the power measurement process of single phase and three phase ac circuits experimentally.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Verify electrical laws and theorems experimentally
2. Design different types of passive filters and analyse their response.
3. Evaluate the performance of a resonant circuit with the variation in frequency.

Course Contents:

Experiment no. 01: Familiarization with alternating current (ac) waves and verification of Ohm's law in AC circuit.
Experiment no. 02: Verification of KCL and KVL in AC circuits
Experiment no. 03: Verification of Superposition Principle.
Experiment no. 04: Verification of Thevenin's theorem and maximum power transfer theorem.
Experiment no. 05: Measurement of three phase power by using two wattmeter method.
Experiment no. 06: Introduction to the use of PSPICE in AC circuit.
Experiment no. 07: Nodal analysis of a circuit by using Pspice simulation tool.
Experiment no. 08: Determining Thevenin's equivalent circuit by using Pspice.
Experiment no. 09: PSPICE simulation of transient and parametric analysis of series RLC circuit using step and pulse input.
Experiment no. 10: PSPICE simulation of transient and parametric analysis of series RLC circuit using sine input.
Experiment no. 11: Different types of filters and their characteristics with different input frequencies.

Text Books:

1. Fundamentals of Electric circuits- Charles K. Alexander & Matthew N. O. Sadiku

Reference Books:

1. Alternating Current Circuits – Russell & George F. Corcoran; John Wiley and Sons.
2. Introductory Circuits for Electrical & Computer Engineering - James. W. Nilson; Prentice Hall of India Private Ltd.
3. A Text Book of Electrical Technology- B L Theraja and A K Theraja; S.Chand & Company Ltd.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2			✓									
CO3									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4301 Course Title: Electronic Circuits I Credits: 3.00

Pre-requisite: Electrical Circuit Analysis-I (AVE 4101), Electrical Circuit Analysis-II (AVE 4201)

Rationale of the Course:
One of the core requirements for students studying avionics engineering is to develop an in-depth understanding of basic electronic circuits that include electronic devices such as diodes, BJTs, FETs, MOSFETs and Op-amps. The course aims to develop students' skills for analysis of such electronic circuits.

Course Objectives:

1. Ensure that students understand the fundamental behaviour of semiconductor devices, such as diodes, BJTs, and MOSFETs.
2. Give students the opportunity to use DC and ac analysis to investigate circuits using semiconductor devices and passive components.
3. Allow students to examine the biassing mechanisms of MOSFET, FET, and BJT amplifiers.
4. Encourage students to conduct thorough analyses of Op-Amp functions and their uses.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain the operation and terminal characteristics of diodes, BJTs, JFETs and MOSFETs

2. Analyse the diode, BJT, and MOSFET circuits with DC only or DC and AC sources.
3. Analyse the BJT and MOSFET amplifier circuits to evaluate amplifiers' performance parameters.
4. Describe the properties of op amps and the analysis and design of simple circuits using them.

Course Contents:

Semiconductor diode: equivalent circuits; P-N junction as a circuit element: Intrinsic and extrinsic semiconductors, operational principle of p-n junction diode, contact potential, current-voltage characteristics of a diode; Diode circuits: Half wave and full wave rectifiers, rectifiers with filter capacitor, characteristics of a Zener diode, Zener shunt regulator, clamping and clipping circuits.

Bipolar junction transistor (BJT): BJT characteristics and regions of operation, BJT as an amplifier. Single stage mid-band frequency BJT amplifier circuits: Voltage and current gain, input and output impedance of a common base, common emitter, and common collector amplifier circuits.

Junction field-effect-transistor (JFET) and Metal-oxide-semiconductor field-effect-transistor (MOSFET)

JFET: Structure and physical operation of JFET, transistor characteristics, and pinch-off voltage. Differential and multistage amplifiers: Description of differential amplifiers, small signal operation, differential and common mode gains, RC coupled mid-band frequency amplifier.

MOSFET: Structure and physical operation of an enhancement MOSFET, threshold voltage, Body effect, current- voltage characteristics of an enhancement MOSFET, biasing discrete and integrated MOS amplifier circuits, single-stage MOS amplifiers, MOSFET as a switch, CMOS inverter

Operational amplifiers (Op-Amp): Properties of ideal and practical Op-Amps, non-inverting and inverting amplifiers, inverting integrators, differentiator, weighted summer, and other applications of Op-Amp circuits, effects of finite open loop gain and bandwidth on circuit performance, logic signal operation of Op-Amp.

Textbooks:

1. Electronic Devices and Circuits Theory – Robert L Boylestad
2. Microelectronic Circuits, Adel S. Sedra & Keneth C. Smith; Oxford University Press.
3. Operational Amplifier and Integrated Circuit – Robert F. Coughlin, Frederick F. Driscoll

Reference Books:

1. Electronic Devices - Thomas L Floyd
2. Electronic Devices and Circuits - David A. Bell
3. Principles of Electronics – V K Mehta

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2		✓										
CO3		✓										

CO4	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Flipped Classroom	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4302 Course Title: Electronic Circuits I Sessional Credits: 1.50

Prerequisite: AVE 4102 & AVE 4202

Rationale of the Course

This subject is classified under the core avionics engineering group and intended to teach the students practical application of basic Electronics i.e., design, build and test electronic circuits using the knowledge, skills and electronic test and measurement instruments in experimental research. These hands-on experiences will help the students in Data Communications, Embedded Systems, Space Avionics, Robotics etc.

Course Objectives:

1. Help students use the appropriate basic laboratory equipment for conducting electronic circuit analysis according to common engineering practice.
2. Provide practical knowledge on demonstrating various types of electronic circuit with correct practice for valid outcome.
3. Familiarise the students with the characteristics of several passive and active components using standard circuit analysis.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Explain the basic electronic components like diode, Bipolar Junction Transistor (BJT), Op-Amp, digital and analogue measuring equipment etc.
2. Apply the knowledge of basic electronic components and networks practically.
3. Analyse the characteristics of different electronic devices such as diodes, Transistors etc., and simple circuits like rectifiers, amplifiers, etc.
4. Evaluate the percentage and causes of differences between theoretical knowledge with the practical observations.

Course Contents:

Experiment no 01: Study of I-V characteristics of a PN junction diode
Experiment no 02: Study of different rectifier circuits by using PN junction diodes.
Experiment no 03: Study of I-V characteristics of a Zener junction diode

Experiment no 04: Study of voltage regulator circuit by using Zener diode.
Experiment no 05: Study the input and output characteristics of n-p-n Bipolar Junction Transistor (BJT) in Common Base Configuration (CBC).
Experiment no 06: Study the input and output characteristics of n-p-n Bipolar Junction Transistor (BJT) in Common Emitter Configuration (CEC).
Experiment no 07: Study and design a single stage small signal amplifier circuit by using Bipolar Junction Transistor (BJT).
Experiment no 08: Study the drain and transfer characteristics of Junction Field Effect Transistor (JFET).
Experiment no 09: Study the drain and transfer characteristics of Metal Oxide Semiconductor Field Effect Transistor (MOSFET).
Experiment no 10: Study and design a non-inverting amplifier circuit by using an op-amp.
Experiment no 11: Study and design an inverting amplifier circuit by using an op-amp.
Experiment no 12: Study and design differential and integral amplifier using op-amp.

Text and Ref Books:

1. Microelectronic Circuits, Adel S. Sedra & Keneth C. Smith; Oxford University Press.
2. Electronic Devices and Circuit Theory, R.L Boylestad; Prentice Hall of India Private Ltd. Private Ltd.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2			✓									
CO3									✓			
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4401 Course Title: Electronic Circuits II Credits: 3.00

Pre-requisite: Electronic Circuit I (AVE 4301)

Rationale of the Course:
This subject focuses on how to create electronic systems with 'building block' circuits using bipolar transistors and FETs and looks at the use and operation of amplifiers. It also looks at how to design feedback systems, oscillator and power amplifier circuits.

Course Objectives:

1. Familiarise students about the frequency response of amplifier circuit using both BJT and MOSFET
2. Helps students to understand the concept and use of feedback circuits, oscillator circuit, Filter Circuits, Power Electronics and Amplifier Circuit
3. Train students to realise the fundamental operation and application of various Optoelectronic Devices

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Demonstrate amplifier circuits by using BJT, FET and MOSFET and analyse the frequency response of these amplifier circuits.
2. Explain the properties and function of feedback circuits and application of such circuits.
3. Analysis power amplifier circuit using Bipolar Junction Transistor (BJT) and Op-amp.
4. Design various filter circuits and their uses in electrical circuits.

Course Contents:

Frequency response of amplifiers: Poles, zeros and Bode plots, amplifier transfer function, techniques of determining 3 dB frequencies of amplifier circuits, frequency response of single-stage and cascade amplifiers, frequency response of differential amplifiers.

Positive and Negative Feedback: Typologies of Negative Feedback, Amplifier Frequency Response, Effect of Coupling, Internal Capacitance in case of BJT amplifier, Miller's Theorem, Decibel, 0dB References, Bode Plot, The Critical Frequency.
Oscillators or Signal Generators: Introduction to Oscillator, Positive feedback, Condition of Oscillator, Phase Shift Oscillator, The Wei-Bridge Oscillator, Resonant circuit Oscillators. Crystal Oscillator, VCO, Introduction to 555 Timer and its operation, Waveform generator.

Power Electronics and Amplifiers: SCR, DIAC, TRIAC, UGT, Classification of power amplifiers, Collector efficiency, Transformer coupled class A amplifier; Class-B push-pull amplifier, Class-C amplifier, Tuned amplifier, class D, E & S amplifier.

Active Filters: Explanation of Low, High, Band Pass and Band Stop Filter Response, Response Characteristics, Damping Factor, Critical Frequency and Roll-Off Rate, Single Pole Filter, Sallen-Key Low Pass and High Pass filter, Cascaded Filter, Multiple Feedback Band-Pass and Band Stop Filter, State Variable Band-Pass and Band Stop Filter

Optoelectronic Devices: PN photodiode, Phototransistor, Solar cell, Photoconductive cell, Photovoltaic, Sensors, LED, LCD, Alphanumeric display, Photo couplers, Photodiode, LDR

Text Books:

1. Microelectronic Circuits – Adel S. Sedra & Keneth C. Smith; Oxford University Press.
2. Electronic Devices and Circuit Theory - R.L Boylestad; Prentice Hall of India Private Limited

Ref Books:

1. Electronic Devices - Thomas L Floyd
2. Electronic Devices and Circuits – David A Bell
3. Principles of Electronics – V K Mehta

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2	✓											
CO3		✓										
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Flipped Classroom	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4402 Course Title: Electronic Circuits II Sessional Credits: 1.50

Pre-requisite: Electrical Circuit Analysis II Sessional (AVE 4202)

Rationale of the Course:

This subject is classified under the core avionics engineering group and intended to teach the students practical application of advanced Electronics i.e. design, build and test electronic circuits using the knowledge, skills and electronic test and measurement instruments in experimental research. These hands-on experiences will help the students in Data Communications, Embedded Systems, Space Avionics, Robotics etc.

Objective:

1. Help students to know about the design and implementation of any desired electronic circuit.
2. Support students to generate desired output of any electronic circuit
3. Teach comparative reasoning of the theoretical and practical values of circuits.

Course Outcomes (CO):

Upon completion of all experiments, the students will be able to:

1. Explain the advanced electronic components like Power transistor, oscillator IC, different optoelectronic devices.
2. Apply the knowledge of advanced electronic components into designing electronic circuits practically.
3. Analyse the characteristics of different amplifier circuits (both small signal and large signal) and frequency response of these circuits.
4. Design various filter circuits and oscillator circuits by using different electronic components.

Course Contents:

Experiment No 01: Study of two stage RC coupled amplifier circuit using BJT.
Experiment No 02: Study the frequency response of BJT based amplifier circuit and find the gain vs frequency curve.
Experiment No 03: Study the frequency response of JFET/MOSFET based amplifier circuit and find the gain vs frequency curve.
Experiment No 04: Study and Analysis the behaviour of voltage series feedback amplifier circuit.
Experiment No 05: Study and Analysis the behaviour of current shunt feedback amplifier circuit.
Experiment No 06: Design and analysis Colpitts Oscillator Circuit using BJT
Experiment No 07: Design and analysis Hartley Oscillator Circuit using BJT
Experiment No 08: Design and analysis RC phase shift Oscillator Circuit using op-amp
Experiment No 09: Design and analysis Wein Bridge Oscillator Circuit using op-amp
Experiment No 10: Study of class A power amplifier circuit.
Experiment No 11: Study of class B push pull amplifier circuit.
Experiment No.: 12 Design and study of a -40 dB/decade low pass Butterworth filter circuit.

Text and Ref Books:

1. Microelectronic Circuits, Adel S. Sedra & Keneth C. Smith; Oxford University Press.
2. Electronic Devices and Circuit Theory, R.L Boylsted; Prentice Hall of India Private Ltd. Private Ltd.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2			✓									
CO3		✓										
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4403 Course Title: Electric Machines and Drives Credits: 3.00

Pre-requisite: Electrical Circuits Analysis II (AVE 4201)

Rationale of the Course:

This course is designed to deliver clear concepts about transformer, DC and AC machines: types, constructions, working principles, advantages, disadvantages, practical applications, different types of mathematical manipulations etc.

Course Objectives:

1. Familiarise the students with different aspects of the transformer and induction motor.
2. Teach them about the construction of D.C machines, different windings, their merits and demerits.
3. Train about different types of DC and AC machines, their characteristics, and industrial applications.
4. Explain the performance and controlling process of special motors, and their applications.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Describe the fundamentals and basic concepts of transformers, DC machines, AC machines and special machines.
2. Explain the construction and working principle of transformer, DC machines, AC machines and special machines.
3. Identify and solve transformer DC machines, AC machines and special machines related problems.
4. Recognize suitable electrical machines for a given application.

Course Contents:

Transformer: Principle of operation, construction, E.M.F equation, behaviour during loading, ideal transformer, leakage reactance and equivalent circuit of a transformer, losses and efficiency, determination of parameters by short circuit and open circuit test, Auto transformer, three phase transformer connection.

DC Generator: Principle of electromagnetic induction (Faraday's law, Lenz's law), Construction, Types, EMF equation, build-up of a self-excited shunt generator, load-voltage characteristic, efficiency, different types of loss, theory and analysis of armature reaction, eddy current, Parallel operation of DC generator.

DC motor: Operation principle, construction, torque, counter emf, torque-speed characteristics, starting and speed control.

Three Phase Induction Motor: Construction and operating principle, rotating magnetic field, synchronous speed, torque in induction motor, slip and its effect on rotor frequency and voltage, equivalent circuit of an induction motor, developed torque, classification, speed characteristic, determination of induction motor parameters by tests, starting, braking and speed control, Induction generator.

Single Phase Induction Motor: Theory of operation, starting and torque speed characteristic.

Alternator: Construction, armature(stator) and rotating field (exciter), windings, equivalent circuit, Parallel operation: necessary conditions, synchronising.

Synchronous motor: Construction, operation, starting, effect of variation of excitations, V curves, inverted V curves, power factor adjustment, synchronous capacitor and power factor correction.

Special Machines: Construction, operation, and controlling of special machines.

Text and Ref Books:

1. Electric Machinery Fundamental - Stephan J. Chapman; McGraw-Hill.
2. A Textbook of Electrical Technology (volume-II)- B.L Theraja, A.K Theraja.
3. Direct and Alternating Current Machinery– Rosenblatt, Friedman.
4. Electric Machine and Transformers – Irving L. Kosow; Prentice Hall of India.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2	✓											
CO3			✓									
CO4	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4404 Course Title: Electric Machines and Drives Sessional Credits: 1.50

Pre-requisite: Electrical Circuits Analysis II Sessional (AVE 4202)

Rationale of the Course:
This course is designed to teach about practical experiments on different concepts and theorems on DC, AC, and special machines, and transformers.

Course Objectives:

1. Enable, train and evaluate the ability of the students to perform the analysis of DC, AC, special machines, and transformers.
2. Empower students to determine the parameters of DC, AC, special machines, and transformers by performing experiments.
3. Enrich the capability of analysing the performance of different types of rotating machines.

Course Outcomes (CO):

1. Familiarize with different types of rotating electrical machines and their basic characteristics through experimental work.
2. Perform various tests in order to find different parameters of transformer and induction motor
3. Analyse the performance criteria of electrical machines and transformers to select for a suitable application.

Course Contents:

Experiment No 1 A general procedure for conducting experiments with different types of machines and transformers.

Experiment No 2 Study of open circuit test and short circuit test of a single-phase transformer.
Experiment No 3 Study the effect of the stator rheostat starter as a method of starting a three-phase squirrel cage induction motor.
Experiment No 4 Study of no-load test and block rotor test of an induction motor.
Experiment No 5 Study of speed control methods of three- phase induction motor.
Experiment No 6 Study of no-load magnetising curve of a separately excited dc generator.
Experiment No 7 Study of speed vs armature current characteristics curve of a dc shunt motor.
Experiment No 8 Study of speed vs armature current characteristics curve of a dc series motor.
Experiment No 9 To synchronise an alternator to an infinite bus by various methods.
Experiment No 10 V curve and inverted V curve of a synchronous motor.
Experiment No 11 Study of controlling stepper motor for particular application.

Text and Ref Books:

1. Electric Machinery Fundamentals - Stephan J. Chapman; McGraw-Hill.
2. A Textbook of Electrical Technology (volume-II)- B.L Theraja, A.K Theraja.
3. Direct and Alternating Current Machinery– Rosenblatt, Friedman.
4. Electric Machine and Transformers – Irving L. Kosow; Prentice Hall of India.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2									✓			
CO3		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4405 Course Title: Digital Logic and Computer Design Credits: 3.00

Pre-requisite: None

Rationale of the Course:

To learn and familiarise the basics of digital logics as well as the analysis and design of computers.

Course Objectives:

1. Impart basic ideas regarding logic gates used for digital avionics systems.
2. Teach the combinational and sequential logic circuit and its application in avionics computer design.
3. Provide knowledge and insights on computer-aided design tools for logic design, analysis and synthesis for avionics application.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain the structure of various number systems, combinational and sequential circuits and its applications in digital avionics system design.
2. Analyse logic networks using modern engineering techniques and tools (such as K-maps, state tables) and hardware description language (Verilog, VHDL).
3. Design the elements of a computer, addressing modes, instruction set, assembly language programming of microprocessors and its application in avionics.
4. Execute and report on a project working in a group both as a member and as a leader.

Course Contents:
Introduction to number systems and codes.

Analysis and synthesis of digital logic circuits: Basic logic functions, Boolean algebra, combinational logic circuits, minimization of combinational logic. Modular combinational circuit, Multiplexer, demultiplexer, decoder, encoder, comparators, Introduction to programmable logic devices.

Sequential circuits: Different types of latches, flip-flops. Shift registers, counters and their applications. Introduction to memory devices and their structure.

Computer-aided design tools for logic design: Hardware Description Languages: Verilog/VHDL, Schematic entry, State diagram entry, Hardware description language entry, Compilation to logic networks, Simulation, Mapping to programmable logic devices

Elements of computers: Arithmetic circuits, Arithmetic and logic units, Memory Array, Logic Array-PLAs, FPGAs, register and bus structures, Controllers/Sequencers, Microprogramming

Text Books:

1. Digital Design and Computer Architecture: ARM Edition- Sarah Harris and David Harris, Morgan Kaufmann Publishers Inc.
2. Digital Logic and Computer Design- M Morris Mano; Prentice Hall of India Private Ltd.

Reference Books:

1. Digital Fundamentals - Floyd; Prentice Hall International, Inc.
2. Aircraft digital electronics and computer systems -Mike Tooley, 3rd Edition, Routledge
3. Microprocessor and Microprocessor Based System Design - Dr M. Rafiquzzaman; Universal Book Stall New Delhi.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2		✓										
CO3			✓									

CO4										✓		

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Project Based Learning	Project Report and Presentation

Course Code: AVE 4406 Course Title: Digital Logic and Computer Design Sessional Credits: 1.50

Pre-requisite: None

Rationale of the Course:

This subject is classified and intended to teach the students basics, concepts, principles and working of digital circuits putting forth the use of a transistor as a switch, number systems, Boolean Algebra, logic gates, counters, timers and so on.

Course Objectives:

1. Provide basic knowledge on basic logic gates and sequential digital systems in a hardware description language.
2. Teach on how to include design for test structures in a sequential digital system.
3. Support students to generate tests for a combinational digital circuit and validate a digital system using a simulator.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Demonstrate that theoretical device/circuit operation can be implemented in properly constructed digital circuits
2. Analyse and correlate standard electronic test equipment such as logic gates, digital multimeters, and power supplies and other digital equipment to test, and implement digital circuits.
3. Conduct comparative study between theoretical result and actual result in an experiment.
4. Design project-based digital circuit.

Course Contents:

Experiment No 1 Familiarisation and use of truth table of basic logic Gates.
Experiment No 2 De Morgan's Laws using the Logic Gates
Experiment No 3 Truth tables and simplification using Boolean algebra
Experiment No 4 Design of adder & subtractor circuits using basic gates
Experiment No 5 Design and implementation of encoder and decoder circuits

Experiment No6 Design and implementation of BCD to seven-segment decoder circuit using logic gates.
Experiment No 7 Design and implementation of Multiplexer & Demultiplexer circuit using logic gates
Experiment No 8 Design and implementation of various types of clocked flip-flop circuits using logic gates.
Experiment No 9 Design and implementation of up and down counters
Experiment No 10 Verilog HDL Programming and Simulation
Experiment No 11 Verilog HDL Programming and Simulation
Experiment No 12 Design an arithmetic and logic unit (ALU) in System Verilog.

Text Books:

1. Digital Design and Computer Architecture: ARM Edition- Sarah Harris and David Harris, Morgan Kaufmann Publishers Inc.
2. Digital Logic and Computer Design- M Morris Mano; Prentice Hall of India Private Ltd.

Reference Books:

1. Digital Fundamentals - Floyd; Prentice Hall International, Inc.
2. Aircraft digital electronics and computer systems -Mike Tooley, 3rd Edition, Routledge

Microprocessor and Microprocessor Based System Design - Dr M. Rafiquzzaman; Universal Book Stall New Delhi.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2		✓										
CO3		✓										
CO4											✓	

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Project Based Learning	Project Report and Presentation

Course Code: AVE 4501 Course Title: Signals and Systems Credits: 3.00

Pre-requisite: Differential and Integral Calculus (MAT 4101), Ordinary and Partial Differential Equations (MAT 4203)

Rationale of the Course:
To understand and apply the basics of electrical signals as well as the analysis and design of systems.

Course Objectives:

1. Support students to describe signals mathematically and understand how to perform mathematical operations on signals. The operations should include operations on the dependent as well as independent variables.
2. Familiarize students with commonly used signals such as the unit step, ramp, impulse function, sinusoidal signals and complex exponentials.
3. Teach classifications of different signals such as continuous-time vs. discrete-time, periodic vs. non-periodic, energy signal vs. power signal, odd vs. even, conjugate symmetric vs. anti-symmetric and system properties - linearity, time invariance, presence or absence of memory, causality, bounded-input bounded-output stability and inevitability.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Describe continuous-time signals and discrete-time signals.
2. Demonstrate about the application and use of mathematical transforms and state-variables in order to solve electrical engineering problems.
3. Analyse electrical engineering signals and circuit problems.
4. Design various electrical systems using different transforms and monitor the performance.
5. Analyse systems for a discrete-time (DT) processing of continuous-time (CT) signals using software.

Course Contents:

Classification of signals and systems: signals - classification, basic operation on signals, elementary signals, representation of signals using impulse function; systems – classification.
Properties of Linear Time Invariant (LTI) systems: Linearity, causality, time invariance, memory, stability, inevitability.

Time domain analysis of LTI systems: Differential equations - system representation, order of the system, solution techniques, zero state and zero input response, system properties; impulse response - convolution integral, determination of system properties; state variable - basic concept, state equation and time domain solution.

Frequency domain analysis of LTI systems: Fourier series- properties, harmonic representation, system response, frequency response of LTI systems.

Fourier transformation- properties, system transfer function, system response and distortion-less systems. Applications of time and frequency domain analyses: solution of analogue electrical and mechanical systems, amplitude modulation and demodulation, time-division and frequency-division multiplexing.

Laplace transformation: properties, inverse transform, solution of system equations, system transfer function, system stability and frequency response and application.

Solution of analog electrical and mechanical systems using MATLAB.

Text and Ref Books:

1. Continuous and Discrete Signals & Systems, S.S. Soliman & M. D. Srinath; Prentice Hall of India Private Ltd.

2. Signal and System (Continuous & Discrete), R.E. Ziemer; Pearson Education Asia.
3. Feedback Control System, Phillips & Horbour; Prentice Hall.
4. Signals and Systems, Alan V. Oppenheim and Alan S. Willsky; Prentice Hall

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2	✓											
CO3		✓										
CO4			✓									
CO5		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Inquiry-Based Learning	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4502 Course Title: Signals and Systems Sessional Credits: 0.75

Pre-requisite: MAT 4101: Differential and Integral Calculus, MAT 4203: Ordinary and Partial Differential Equations

Rationale of the Course:

To understand and apply the basics of electrical signals as well as the analysis and design of systems.

Course Objectives:

1. Teach the basics of MATLAB from signals and systems perspective.
2. Help students simulate the generation of signals and operations on them.
3. Familiarize the students Fourier, Laplace and Z transforms.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Generate matrices of various dimensions, various signals and sequences and perform operations on signals.
2. Obtain the frequency domain representation of signals and sequences using Fourier transform, Laplace and z transform.
3. Apply the concept of convolution and correlation.
4. Produce various types of noise and measure various characteristics of noise.

Course Contents:

Experiment no. 01: Familiarization with basic operations on matrices.
Experiment no. 02: Generation of various signals and sequences.
Experiment no. 03: Operation on signals and sequences.
Experiment no. 04: Application of Fourier transforms and inverse Fourier transforms.
Experiment no.05: Study of Laplace transforms.
Experiment no. 06: Study of Z-transformation.
Experiment no. 07: Convolution operation between signals and sequences.

Text and Ref Books:

1. Continuous and Discrete Signals & Systems, S.S. Soliman & M. D. Srinath; Prentice Hall of India Private Ltd.
2. Signal and System (Continuous & Discrete), R.E. Ziemer; Pearson Education Asia.
3. Signals and Systems Using MATLAB, Luis F. Chaparro, Aydin Akan, Academic Press.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1					✓							
CO2					✓							
CO3	✓											
CO4					✓							

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4503 Course Title: Microprocessor and Microcontroller Systems Credits: 3.00

Prerequisite: Digital Logic and Computer Design (AVE 4405)

Rationale of the Course:
To learn and familiarize the basics of internal architecture of microprocessors and microcontrollers, its language and interfacing between different parts of the memory of the microprocessor and microcomputer.

Course Objectives:

1. Explain the internal architecture of microprocessors and microcontrollers.
2. Teach the programming of microprocessors and microcontrollers.

3. Illustrate interfacing of embedded peripherals.

Course Outcomes (CO):

1. Explain internal modules of microprocessors and microcontrollers.
2. Write assembly code for microprocessor and Embedded C code for microcontroller.
3. Interface I/O devices to microcontrollers.
4. Design an embedded system using microcontroller.

Course Contents

UNIT I THE 8086 MICROPROCESSOR
Introduction to 8086 — Microprocessor architecture — Addressing modes — Instruction set and assembler directives — Assembly language programming — Modular Programming — Linking and Relocation — Stacks — Procedures — Macros — Interrupts and interrupt service routines — Byte and String Manipulation.

UNIT II 8086 SYSTEM BUS STRUCTURE
8086 signals — Basic configurations — System bus timing –System design using 8086 — I/O programming — Introduction to Multiprogramming — System Bus Structure — Multiprocessor configurations — Coprocessor, Closely coupled and loosely Coupled configurations — Introduction to advanced processors.

UNIT III I/O INTERFACING
Memory Interfacing and I/O interfacing — Parallel communication interface — Serial communication interface — D/A and A/D Interface — Timer — Keyboard /display controller — Interrupt controller — DMA controller — Programming and applications Case studies: Traffic Light control, LED display , LCD display, Keyboard display interface and Alarm Controller.

UNIT IV MICROCONTROLLER
Architecture of 8051 — Special Function Registers (SFRs) — I/O Pins Ports and Circuits — Instruction set — Addressing modes — Assembly language programming.

UNIT V INTERFACING MICROCONTROLLER
Programming 8051 Timers — Serial Port Programming — Interrupts Programming — LCD & Keyboard Interfacing — ADC, DAC & Sensor Interfacing — External Memory Interface- Stepper Motor and Waveform generation — Comparison of Microprocessor, Microcontroller, PIC and ARM processors

Text Books:

1. Barry B. Brey (2009), "The Intel Microprocessors: 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, Pentium Pro Processor, Pentium II, Pentium III, Pentium 4, and Core2 with 64-bit Extensions : Architecture, Programming, and Interfacing", Pearson Education India, ISBN – 9788131726228
2. Mohammad Ali Mazidi (2007) "The 8051 Microcontroller And Embedded Systems Using Assembly And C", Pearson education, ISBN – 9788131710265

Ref. Books:
1. Alan Clements, "Principles of Computer Hardware", Oxford University Press, 3rd edition, 2003, ISBN – 9781420009545
2. John. B. Peatman, "Design with PIC Microcontrollers", Pearson education Inc., 2003, ISBN – 9780070591165
3. Dogan Ibrahum (2000) "Microcontroller Projects in C for the 8051", Newnes, ISBN – 9780750646406

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2	✓											
CO3		✓										
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy:

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Flipped Classroom	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Inquiry-Based Learning	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4504 Course Title: Microprocessor and Microcontroller Systems Sessional Credits: 0.75

Prerequisite: Digital Logic and Computer Design Sessional (AVE 4406)

Rationale of the Course:
To learn and familiarize the basics of internal architecture of microprocessors and microcontrollers, its language and interfacing between different parts of the memory of the microprocessor and microcomputer.

Course Objectives:

1. Execute variety of assembly language programs of Intel 8086 including arithmetic and logical, sorting, searching, and string manipulation operations.
2. Develop and execute the assembly language programs for interfacing Intel 8086 with peripheral devices.
3. Illustrate interfacing of embedded peripherals practically.

Course Outcomes (COs):

1. Learn the internal organization of popular 8086/8051 microprocessors/microcontrollers.
2. Interface I/O devices to microcontrollers.
3. Apply the concepts in the design of microprocessor/microcontroller-based systems in real time applications

Course Contents
Experiment no. 1: Programs for 16-bit arithmetic operations for 8086 (Using various addressing modes)
Experiment no. 2: Write a program for sorting an array for 8086.
Experiment no. 3: Write a program for string manipulations

Experiment no. 4: Write a program for digital clock design using 8086.
Experiment no. 5: Interfacing ADC and DAC to 8086
Experiment no. 6: Parallel communication between two microprocessors using 8255
Experiment no. 7: Serial communication between two microprocessors using 8251
Experiment no. 8: Write a program using arithmetic, logical and bit manipulation instruction of 8051
Experiment no. 9: Write a program and verify timer/counter in 8051
Experiment no. 10: Write a program and verify interrupt handling in 8051
Experiment no. 11: Communication between 8051 kit and PC
Experiment no. 12: Interfacing LCD to 8051
Experiment no. 13: Data Transfer from peripheral to memory through DMA controller 8237/8257

Text Books:

1. Barry B. Brey (2009), "The Intel Microprocessors: 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, Pentium Pro Processor, Pentium II, Pentium III, Pentium 4, and Core2 with 64-bit Extensions: Architecture, Programming, and Interfacing", Pearson Education India, ISBN – 9788131726228
2. Mohammad Ali Mazidi (2007) "The 8051 Microcontroller and Embedded Systems Using Assembly And C", Pearson education, ISBN – 9788131710265

Ref. Books:

1. Alan Clements, "Principles of Computer Hardware", Oxford University Press, 3rd edition, 2003, ISBN – 9781420009545
2. John. B. Peatman, "Design with PIC Microcontrollers", Pearson education Inc., 2003, ISBN – 9780070591165
3. Dogan Ibrahum (2000) "Microcontroller Projects in C for the 8051", Newnes, ISBN – 9780750646406

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1	✓											
CO2			✓									
CO3									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4505 Course Title: Electromagnetic Field Theory Credits: 3.00

Prerequisite: None

Rationale of the Course:

To learn and familiarise with the basics of electro- magnetic field theories and implement that knowledge in the field of communication.

Course Objectives:

1. Familiarize the students with the behaviour of electrostatic and electromagnetic fields and their application in electrical and electronics engineering fields.
2. Teach the Maxwell's equation in integral and differential form, their interpretation, and applications.
3. Provide knowledge on the propagation of EM waves in free space, conductors & dielectrics.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Use different coordinate systems, Coulomb's Law, and Gauss Law for the evaluation of electric fields produced by different charge configurations.
2. Calculate the energy and potential due to a system of charges & explain the behaviour of electric fields across boundary conditions.
3. Explain the Poisson's, Laplace equations and behaviour of steady magnetic fields.
4. Describe the behaviour of magnetic fields and magnetic materials.
5. Assess time varying fields and propagation of waves in different media.

Course Contents:

Vector Analysis:
Scalars and Vectors, Vector algebra, Cartesian coordinate system, Vector Components, and unit vectors. Scalar field and Vector field. Dot product and Cross product, Gradient of a scalar field. Divergence and Curl of a vector field. Co – ordinate systems: cylindrical and spherical, relation between different coordinate systems. Expression for gradient, divergence, and curl in rectangular, cylindrical, and spherical coordinate systems. Numerical.

Electrostatics:
Coulomb's law, Electric field intensity and its evaluation for (i) point charge (ii) line charge (iii) surface charge (iv) volume charge distributions. Electric flux density, Gauss law and its applications. Maxwell's first equation (Electrostatics). Divergence theorem. Numerical.

Energy and Potential:
Energy expended in moving a point charge in an electric field. The line integral. Definition of potential difference and potential. The potential field of a point charge and of a system of charges. Potential gradient. The dipole. Energy density in the electrostatic field. Numerical.

Conductor and Dielectrics:
Current and current density. Continuity of current. Metallic conductors, conductor's properties and boundary conditions. Perfect dielectric materials, capacitance calculations. Parallel plate capacitor with two dielectrics with dielectric interface parallel to the conducting plates. Numerical.

Poisson's and Laplace Equations:
Derivations and problems, Uniqueness theorem.

Steady magnetic fields:
Biot - Savart's law, Ampere's circuital law. The Curl. Stokes theorem. Magnetic flux and flux density. Scalar and vector magnetic potentials. Numerical.

Magnetic forces:
Force on a moving charge and differential current element. Force between differential current elements.

Force and torque on a closed circuit. Numerical.

Magnetic Materials and Magnetism:

Nature of magnetic materials, magnetization, and permeability. Magnetic boundary conditions. Magnetic circuit, inductance, and mutual inductance. Numerical.

Time Varying Fields and Maxwell's Equations:

Faraday's law, Displacement current. Maxwell's equations in point form and integral form. Numerical.

Uniform plane wave:

Wave propagation in free space and in dielectrics. Pointing vector and power considerations. Propagation in good conductors, skin effect. Numerical.

Textbooks:

1. Engineering Electromagnetics – W. H. Hayt Jr& John A. Buck; Tata McGraw-Hill Publishing Company Ltd
2. Principles of Electromagnetics Matthew N. O. Sadiku Oxford 6th Edition, 2015

Reference Books:

1. Fields and Waves in Communication Electronics - Simon Ramo; John Wiley & Sons.
2. Fundamentals of Engineering Electromagnetic - D.K. Cheng; Prentice Hall of India Private Ltd.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1		✓										
CO2			✓									
CO3	✓											
CO4	✓											
CO5		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4510 Course Title: Modelling & Simulation Sessional Credits: 1.50

Prerequisite: Computer Programming and Application (CSE 4201) and Sessional (CSE 4202)

Rationale of the Course:

To learn the basics of MATLAB/SIMULINK, COMSOL, CST.

Course Objectives:

1. Train of students in the field of modelling and simulation-supported development methodology
2. Teach the usage of different loops in MATLAB and the usage of Simulink
3. Validation of the simulations through real tests

Course Outcomes (CO):

1. Demonstrate associated skills regarding using MATLAB and simulation software.
2. Create suitable simulation models in MATLAB/Simulink
3. Compute engineering problems using MATLAB and COMSOL software

Course Contents:

Experiment 01 Introduction to MATLAB
Experiment 02 Programming in MATLAB: Introduction, Branching statements, loops, functions, additional data types, plots, arrays, inputs/outputs etc.
Experiment 03 Matrix operations and Matrix Applications
Experiment 04 Introduction to MATLAB Simulink
Experiment 05 Solving Engineering Problems Using MATLAB and Simulink
Experiment 06 Introduction to COMSOL Multiphysics software (Installation, run, and usage)
Experiment 07 Design and simulate of conventional optical fibre using COMSOL Multiphysics software
Experiment 08 Design and simulate of photonic crystal fibre by using COMSOL Multiphysics software.
Experiment 09 Introduction to CST Studio Suite (Installation, run, and usage)
Experiment 10 Design, analyse and optimize of electromagnetic components by using CST Studio Suite.
Experiment 11 Design and Simulation of antennas by using CST software.

Text and Ref book:

1. Palm, William John. *Introduction to MATLAB for Engineers*. New York: McGraw-Hill, 2011.
2. Higham, Desmond J., and Nicholas J. Higham. *MATLAB guide*. Society for Industrial and Applied Mathematics, 2016.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12
CO1									✓			
CO2					✓							
CO3										✓		

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam

CO3	Project Based Learning	Assignment and Presentation, Final Exam

Course Code: AVE 4600 **Course Title: Industrial Training** **Credits: 1.00**

Pre-requisite: Students should complete all courses up to 3rd Year.

Rationale of the Course:

To provide the experience for the students of the industrial environment and organisation as well as the functionality of the engineers in industries.

Course Objectives:

1. Provide the opportunity to practice engineering professionalism and engineering communication skills which include daily interaction with the working environment and technical writing.
2. Provide the experience of a true working environment for an Avionics Engineer.
3. Facilitate the student to work individually and in a team.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain the structure and management of an industry/organization to apply this knowledge in the individual's professional life.
2. Devise the industrial works like maintenance, planning, engineering service and aircraft inspection with practical experience.
3. Apply the industrial training knowledge further in a project work.
4. Describe various engineering aspects of different industries/organizations through group presentation.
5. Develop the responsibility of safety, legal, environmental and cultural issues in Avionics engineering practices**.**

Criterion:

1. Time duration 4 week
2. Assessment and presentation, observation and evaluation by officers of that industry.
3. The industry has to be relevant to Avionics Engineering.

Mapping of Course Outcomes to Program Outcomes:

COs	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO12**
CO1	✓											
CO2												✓
CO3												✓
CO4										✓		
CO5							✓					

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Industry Professional)	Project Report and Presentation
CO2	Standard Lecture (Industry Professional)	Project Report and Presentation
CO3	Industry Project Based Learning	Project Report and Presentation
CO4	Industry Experience	Project Report and Presentation
CO5	Standard Lecture (Industry Professional)	Project Report and Presentation

Course Code: AVE 4601 Course Title: Communication Systems Credits: 3.00

Pre-requisite: None

Rationale of the Course:

The course objective is to cover the principles of analogue and digital communication systems involving different modulation and coding schemes in the background of noise and interference

Objective:

1. Supporting students to apply deterministic and statistical Fourier concepts to transmission propagation, power spectral density and modulation/coding characteristics.
2. Teach development of the functional blocks of coding/modulation and demodulation/decoding for analogue and digital communication systems.
3. Provide knowledge on the analogue-to-digital conversion process with emphasis on Nyquist Sampling Criteria, line coding, pulse shaping and optimum detection functions.
4. Train the students to do comparative study on the performance of binary, M-array and spread spectrum digital communications in the presence of noise and interference and design of linear systematic block coding and its error detection/correction capability is also discussed.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain different modulation and demodulation techniques used in analogue communication systems
2. Evaluate and solve basic communication problems
3. Analyse transmitter and receiver circuits
4. Examine design issues, advantages, disadvantages and limitations of analogue communication systems

Course Contents:

Overview of communication systems: Basic principles & fundamental elements; Noise: Sources & characteristics.

Information theory: Measure of information, channel capacity of a continuous system and channel capacity;
Communication systems: Analogue and Digital.

Continuous wave modulation: Transmission types, Amplitude modulation: Introduction, double side band, single side band, vestigial side band, quadrature, spectral analysis of each type, envelope and synchronous detection.

Angle modulation: Instantaneous frequency, frequency modulation (FM) and phase modulation (PM), spectral analysis, demodulation of FM and PM. Pulse modulation: Sampling theorem, Nyquist criterion, aliasing, instantaneous and natural sampling. Pulse amplitude modulation: Principle, bandwidth requirements.

Pulse code modulation (PCM): Quantization principle, quantization noise, non-uniform quantization, signal to quantization error ratio, differential PCM, demodulation of PCM. Delta modulation (DM): Principle, adaptive DM, line coding – formats and bandwidths. Digital modulation: Amplitude-shift keying - Phase-shift keying (PSK): Frequency-shift Keying (FSK)

Multiplexing: Time division multiplexing (TDM) - principle, receiver synchronization, frame synchronization, TDM of multiple bit rate systems, frequency division multiplexing - principle, de-multiplexing, wavelength-division multiplexing.

Aircraft Communication System: Intercommunication System, VHF/UHF Communication, HF Communication, Satellite Communication, Emergency Locator Transmitter.

Text Book:

1. Principles of Electronic Communication Systems by Louis Frenzel

Ref Books:

1. Digital and Analog Communication System, Leon W. Couch; Pearson Education.
2. Communication System, Somon Haykin; John Wiley & Sons, Inc.
3. Modern Digital & Analog Communication System, B. P. Lathi; Oxford University Press.
4. Telecommunication Switching Systems and Networks, Thiagarajan Viswanathan; Prentice Hall of India Private Ltd.
5. Electronic Communication Systems, Kennedy & Davis; Tata McGraw Hill.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3		✓										
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4602 Course Title: Communication Systems Sessional Credits: 1.50

Pre-requisite:

Rationale of the Course:

To learn and familiarise the basics of communication as well as the analysis and implementation of various communication methods.

Course Objective:

1. Share basic understanding of different Analog Modulation.
2. Enlighten the students regarding the principles of communication in various practical fields.
3. Guide to apply different Digital Modulation techniques.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Explain the basic theory of different types modulation techniques of communication
2. Apply the basic theory of modulation using different equipment.
3. Compare the numerical results with software results to design a communication system as per requirement
4. Analyse the effect of noise in different communication techniques

Course Contents:

Experiment No: 1- Familiarise the students about oscilloscope (Analog and Digital) and function generator/oscillator
Experiment No: 2- Study the simple periodic analog signal and measure different parameters of the signal
Experiment No: 3- Study the digital (binary) signal and measure different parameters of the signal
Experiment No: 4: Study and design the Analog to Digital (A/D) and Digital to Analog (D/A) converter
Experiment No: 5- Study the amplitude Modulation (AM) technique and find the modulation index.
Experiment No: 6- Study the amplitude Demodulation technique by Diode detector method
Experiment No: 7- Study the Frequency Modulation (FM) technique and find the frequency deviation.
Experiment No: 8- Study the Frequency Demodulation technique.
Experiment No: 9- Study the DSB-SC and SSB amplitude Modulation (AM) techniques
Experiment No: 10- Study the DSB-SC and SSB amplitude Demodulation Techniques
Experiment No: 11- Study the Delta Modulation and Demodulation
Experiment No: 12 - Study the Pulse Amplitude Modulation and Demodulation Techniques
Experiment No: 13- Study different Digital Modulation Techniques (ASK, FSK, PSK, etc.)

Text and Ref Books:

1. Digital and Analog Communication System - Leon W. Couch; Pearson Education.
2. Communication System – Somon Haykin; John Wiley & Sons, Inc.
3. Modern Digital & Analog Communication System - B. P. Lathi; Oxford University Press.
4. Telecommunication Switching Systems and Networks – Thiagarajan Viswanathan; Prentice Hall of India Private Ltd.

Electronic Communication Systems

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2					✓							
CO3												✓
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4603 Course Title: Real Time Embedded Systems Credits: 3.00

Pre-Requisite: Microprocessor and Microcontroller Systems (AVE 4503)

Rationale of the Course:

Real-time systems are prevalent today and the ability to design and use real-time systems makes an engineer more marketable.

Course Objectives:

1. Provides the opportunities to be skilful in the analysis of scheduling algorithms for real-time computing, including the techniques to evaluate worst-case delays and utilization bounds.
2. Guide to apply model-driven development approaches to construct target execution environments for simulation analysis and rapid prototyping, and to verify real-time requirements.

Course Outcome (CO):

On successful completion of this course students will be able to:

1. Apply the terminology, and list applications, of real time systems.
2. Demonstrate the ability to work within the constraints imposed by the real-time aspects of systems.
3. Solve practical design problems into real time task models for the purpose of analysis, evaluation or implementation.
4. Apply simple real time functions using a real time operating system and a programming language suitable for embedded real-time systems.
5. Use real-time methodology to multiprocessor, and distributed systems.

Course Content:

PIC Microcontroller

Architecture – Features – Resets –Memory Organizations: Program Memory, Data Memory – Instruction Set – simple programs. Interrupts –I/O Ports –Timers- CCP Modules- Master Synchronous serial Port (MSSP)- USART –ADC- I2C

Embedded Processors

ARM processor- processor and memory organization, Data operations, Flow of Control, CPU Bus configuration, ARM Bus, Memory devices, Input/output devices, Component interfacing, designing with microprocessor development and debugging, Design Example: Alarm Clock.

Embedded Programming

Programming in Assembly Language (ALP) Vs. High level language – C program elements, Macros and Functions – Use of pointers – NULL pointers – use of function calls – multiple function calls in a cyclic order in the main function pointers – Function queues and interrupt service Routines queues pointers – Concepts of Embedded programming in C++ – Object oriented programming – Embedded programming in C++, C program compilers – Cross compiler – optimization of memory codes.

Embedded System design

Embedded System project management – Embedded system design and Co-Design Issues in System Development process – Design cycle in the development phase for an embedded system – Uses of Target system or its emulator and In-Circuit Emulator – Use of software Tools for Development of an embedded system – Use of scopes and logic analyzers for system hardware tests – Issues in Embedded System Design.

Real Time Operating Systems

Operating system services –I/O subsystems – Network operating systems –Interrupt Routines in RTOS Environment – RTOS Task scheduling models, Interrupt – Performance Metric in Scheduling Models – IEEE standard POSIX functions for standardization of RTOS and inter-task communication functions – List of Basic functions in a Pre-emptive scheduler – Fifteen-point strategy for synchronization between processors, ISRs, OS Functions and Tasks – OS security issues- Mobile OS.

Text Books:

1. Raj Kamal, Embedded Systems Architecture, Programming and Design, Tata McGraw-Hill, New Delhi, 2003.*ISBN* 0-07-049470-3
2. Wayne Wolf, Computers as Components: Principles of Embedded Computing System Design, Morgan Kaufman Publishers, 2001.ISBN=0123884365

Ref. Books:

1. Frank Vahid and Tony Givargi Embedded System Design: A Unified Hardware/Software Introduction's, John Wiley & Sons, 2000.
2. John B Peatman, Design with PIC Microcontrollers, Prentice Hall of India, 2007 ISBN=0130462136
3. Ajay V Deshmukh, Microcontroller Theory and Applications, Tata McGraw Hill,2007 ISBN=0070585954

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1		✓										
CO2												✓
CO3			✓									
CO4		✓										
CO5	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Flipped Classroom	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4604 Course Title: Real Time Embedded Systems Sessional Credits: 0.75

Pre-requisite: Microprocessor and Microcontroller Systems Sessional (AVE 4504)

Rationale of the Course:
To learn and familiarize the basics of how a computer works, its language and interfacing between different parts of the memory of the microprocessor.

Course Objectives:

1. Teach basic architecture of microprocessors and interfacing by programming languages to control the operation of microprocessors.
2. Facilitate to Design embedded systems and interfacing peripheral I/O devices and memory devices with microprocessors.

Course Outcomes (CO):

1. Identify the functionality of development boards to implement embedded applications
2. Design an electronic circuit for diverse I/O devices used in real time embedded applications.
3. Develop a product with all sub systems of functional requirements in optimal hardware and software components.
4. Develop communication between I/O and memory devices with microprocessor and thereby successfully accomplish error estimation, execution, signal conditioning/ conversion, automatic processing and finally transmission of data with various communication protocols.

Experiment no. 1: Introduction to embedded systems lab content
Experiment no. 2: Programming using Arithmetic and logical instructions of 8051
Experiment no. 3: Timer programming of 8051.
Experiment no. 4: Counter programming of 8051.
Experiment no. 5: Serial port programming of 8051.
Experiment no. 6: Interfacing of Stepper motor.
Experiment no. 7: Interfacing of Temperature sensor and Relay control
Experiment no. 8: Using of more complex memory and branch type instructions such as LDMFD/STMFD, B and BL.
Experiment no. 9: Programmable peripheral interrupt controlling by 8259 PIC
Experiment no. 10: Implement the lighting and winking LEDs of the ARM I/O port via programming.

Experiment no. 11: Embedded System Design using UML State Machines
Experiment no. 12: Build Real Time embedded applications with Free RTOS

Text and Ref Books:

1. Real-Time Embedded Systems: Design Principles and Engineering Practices, Xiaocong Fan, Publisher: Newnes
2. Assembly Language Programming and Organization of IBMPC – Ytha Yu and Charles Marut, International Edition, McGraw Hill, Inc
3. The Intel Microprocessors – Architecture Programming and Interfacing – Barry B Brey, Pearson Education, Inc., Upper Saddle River, New Jersey 07458.
4. Digital Computer Electronics – Albert P. Malvino and JeraldA. Brown, 3rd Edition

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3											✓	
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4605 Course Title: Microwave and Antennas Credits: 3.00

Pre-requisites: Electromagnetic Field Theory (AVE 4505)

Rationale of the Course:
This course provides basic knowledge of designing of transmission lines and wave guides. The various modes of propagations through transmission lines and wave guides are included. Students will become familiar with the usage of active and passive components of microwave systems. Measurements of various parameters of microwave systems are also part of the subject

Course Objectives:

1. Provide knowledge regarding the microwave properties, its transmission media and the basics of antenna theory.
2. Share insights of different microwave devices for several applications
3. Upskill the student to select antennas for specific applications

Course Outcome (CO):

After the course the participants should be able to;

1. Describe the fundamental principles of microwave transmission
2. Analyse various parameters related to microwave transmission lines and waveguides
3. Utilize microwave devices for several applications
4. Compute various antenna parameters necessary for building a RF system
5. Set various antenna configurations according to the applications.

Course Content:

Microwave Tubes:

Introduction, Reflex Klystron Oscillator, Mechanism of Oscillations, Modes of Oscillations, Mode Curve (Qualitative Analysis only).

Microwave Transmission Lines: Microwave Frequencies, Microwave devices, Microwave Systems, Transmission Line equations and solutions, Reflection Coefficient and Transmission Coefficient, Standing Wave and Standing Wave Ratio, Smith Chart, Single Stub matching.

Microwave Network theory: Introduction, Symmetrical Z and Y—Parameters for reciprocal Networks, S matrix representation of Multi-Port Networks.

Microwave Passive Devices:

Coaxial Connectors and Adapters, Attenuators, Phase Shifters, Waveguide Tees, Magic tees.

Strip Lines:

Introduction, Micro Strip lines, Parallel Strip lines, Coplanar Strip lines, Shielded Strip Lines.

Antenna Basics:

Introduction, Basic Antenna Parameters, Patterns, Beam Area, Radiation Intensity, Beam Efficiency, Directivity and Gain, Antenna Apertures, Effective Height, Radio Communication Link, Antenna Field Zones.

Point Sources and Arrays: Introduction, Point Sources, Power Patterns, Power Theorem, Radiation Intensity, Arrays of two isotropic point sources, Linear Arrays of n Isotropic Point Sources of equal Amplitude and Spacing.

Electric Dipoles:

Introduction, Short Electric Dipole, Fields of a Short Dipole, Radiation Resistance of a Short Electric Dipole, Thin Linear Antenna (Field Analyses)

Loop and Horn Antenna:

Introduction, Small loop, The Loop Antenna General Case, The Loop Antenna as a special case, Radiation resistance of loops, Directivity of Circular Loop Antennas with uniform current, Horn antennas.

Antenna Types:

The Helix geometry, Helix modes, Practical Design considerations for the mono-filar axial mode Helical Antenna, Yagi-Uda array, Parabolic reflector.

Textbooks:

1. Microwave Engineering — Annapurna Das, Sisir K Das, TMH,Publication, 2“, 2010.

2. Microwave Devices and circuits- Samuel Y Liao, Pearson Education

3. Antennas and Wave Propagation- John D. Krauss, Ronald J Marhefl<a, Ahmad S Khan, 4th Edition, McGraW Hill Education, 2013

Reference Books:

1. Microwave Engineering - David M Pozar, John Wiley India Pvt. Ltd., 3rd Edn, 2008.

2. Microwave Engineering — Sushrut Das, Oxford Higher Education, 2nd Edn, 2015

3. Antennas and Wave Propagation — Harish and Sachidananda: Oxford University Press, 2007

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3												✓
CO4			✓									
CO5	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Montessori	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4607 **Course Title: Numerical Methods** **Credits: 3.00**

Pre-requisite: Differential and Integral Calculus (MAT 4101)

Rationale of the Course:

The course will develop numerical methods aided by technology to solve algebraic, transcendental, and differential equations, and to calculate derivatives and integrals and also understanding of the elements of error analysis for numerical methods.

Course Objectives:

1. Help students to analyse appropriate numerical methods to solve algebraic and transcendental equations as well as generating approximate function
2. Develop the capability to use appropriate numerical methods to solve a differential equation and to evaluate a derivative at a value
3. Acquaint the students about numerical methods to solve a linear system of equations so that students are able to perform an error analysis for various numerical methods.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Demonstrate understanding of common numerical methods, number representation and errors and how they are used to obtain approximate solutions to otherwise intractable mathematical problems.
2. Apply numerical methods to obtain approximate solutions to mathematical problems.
3. Associate numerical methods for various mathematical operations and tasks, such as interpolation, differentiation, integration, the solution of linear and nonlinear equations, and the solution of differential equations.
4. Perform an error analysis for various numerical methods and prove results for various numerical root finding methods

Course Contents:

Roots of polynomials and transcendental equations; Determinants and matrices; Eigenvalues and Eigenvectors; Solution of simultaneous linear equations; Solution of linear and non-linear algebraic equations; Solution of ordinary differential equations; Solution of partial differential equation; Introduction to the use of scalar, vector and matrix variables; The manipulation of matrix variables in arithmetic functions.

Interpolation methods; Numerical differentiation and integration; Solving equations by finite differences; Graph plotting and curve fitting; Applications in structural mechanics.

Iterative solutions for non-linear problems; Use fundamental programming concepts to solve mathematical problems; Develop computer programs to solve simple engineering and mathematical problems.

Engineering analysis by using graphical tools in MATLAB and MS Excel; Use of spreadsheet; Data structures; Graphing; Recursion; Packages for data manipulation.

Text and Ref Books:

1. Numerical Methods, S. Balachandra Rao and C.K. Shantha; Stosius Inc.
2. Numerical Methods for Engineers, Steven C. Chopra, Raymond P. Carale; Tata McGraw-Hill Publishing Company Ltd.
3. Applied Numerical Analysis, Curtis F. Gerald, Patrick O. Wheatley; Addison-Wesley Publishing Company User's Guide for Student Edition of MATLAB, Duane Hanselman & Bruce Littlefield, Prentice Hall, NJ, 1997.
4. Introductory Methods of Numerical Analysis, S.S. Sastry

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3		✓										
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Computer Simulations	Class Test/Assignment, Final Exam

Course Code: AVE 4608 Course Title: Numerical Methods Sessional Credits: 0.75

Pre-requisite: Computer Programming and Application Sessional (CSE 4202), Modelling & Simulation Sessional (AVE 4510)

Rationale of the Course:

To learn and familiarize the basics of common numerical methods as well as the analysis and implementation and application of numerical methods.

Course Objectives:

1. Support students to Guide to Familiarize the student with software MATLAB for numerical computations and familiarizing students with the MATLAB Desktop, basic commands through the Command window and output through the Graph window
2. Provide a foundation in use of this software for real time applications
3. Prepare the students to use MATLAB/LABVIEW in their project works.

Course Outcomes (CO):

1. Demonstrate understanding of common numerical methods and how they are used to obtain approximate solutions to otherwise intractable mathematical problems.
2. Apply numerical methods to obtain approximate solutions to mathematical problems.
3. Analyse practical Numerical problems and approaching to solve accordingly applying MATLAB

Course Contents:

Experiment 01 To find the roots of nonlinear equations using the Bisection method.
Experiment 02 To find the roots of nonlinear equations using False Position method
Experiment 03 To find the roots of nonlinear equations using Newton-Raphson method.
Experiment 04 To find the roots of nonlinear equations using the Secant method.
Experiment 05 To find the roots of nonlinear equations using Newton's method
Experiment 06 To find the roots of nonlinear equations using the Modified Bisection method.
Experiment 07 To solve the system of linear equations using Basic Gauss Elimination method.
Experiment 08 To solve the system of linear equations Using Gauss - Jordan Method
Experiment 09 To fit a straight line and a polynomial using curve fitting regression method.
Experiment 10 To find the value of f(x) for x using Lagrange interpolation method.

Text and Ref Books:

1. Numerical Methods, S. Balachandra Rao and C.K. Shantha; Stosius Inc.
2. Numerical Methods for Engineers, Steven C. Chopra, Raymond P. Carale; Tata McGraw-Hill Publishing Company Ltd.
3. Applied Numerical Analysis, Curtis F. Gerald, Patrick O. Wheatley; Addison-Wesley Publishing Company User's Guide for Student Edition of MATLAB, Duane Hanselman& Bruce Littlefield, Prentice Hall, NJ, 1997.
4. Introductory Methods of Numerical Analysis, S.S. Sastry

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3					✓							

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4609 Course Title: Control Systems Credits: 3.00

Prerequisite: Electronic Circuits II (AVE 4401), Signal and Systems (AVE 4501)

Rationale of the Course:
To learn and familiarize the basics of control systems as well as the application of this area of electrical engineering.

Course Objectives:

1. Enable students to analysis and synthesis of linear, non-linear, and discrete control systems employing the state space approach.
2. Help students to determine different design components for a specific control system

Course Outcome (CO):

Students who successfully complete the course will be able to:

1. Describe continuous-time systems in the discrete-time domain
2. Design digital controller using a state space approach
3. Use CAD tools to support the control design
4. Design observers to support the control in a state space approach

Course Content:

The course content covers the following topics:

Introduction to control systems. Linear system models: Transfer function, block diagram and signal flow graph (SFG). State variables: SFG to state variables, transfer function to state variable and state variable to transfer function. Feedback control system: Closed loop systems, parameter sensitivity, transient characteristics of control systems, effect of additional pole and zero on the system response and system types and steady state error. Routh stability criterion. Analysis of feedback control system: Root locus method and frequency response method. Design of feedback control system: Controllability and observability, root locus, frequency response and state variable methods. Digital control systems: introduction, sampled data systems, stability analysis in Z-domain.

Text Books:

1. Control Systems Engineering 7th Ed – Normal N Nise

Reference Books:

1. Computer-Controlled Systems: Theory and Design, Third Edition, Dover Books on Electrical Engineering, Karl A Astrom, Bjorn Wittenmark, ISBN10: 0486486133, ISBN-13: 978-0486486130.
2. Digital Control of Dynamic Systems, by Franklin, Powell, Workman, Addison Wesley.
3. Digital Control: A State Space Approach, by Vaccaro, McGraw Hill.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3					✓							
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy

CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Cooperative Learning	Class Test/Assignment, Final Exam

Course Code: AVE 4610 Course Title: Control Systems Sessional Credits: 0.75

Prerequisite: Electronic Circuits II (AVE 4401), Signal and Systems (AVE 4501)

Rationale of the Course:

To learn and familiarize the basics of high voltage engineering as well as the application of this area of electrical engineering practically.

Course Objective:

1. Educate students about the basic ideas of control system using small prototyped modules
2. Help students design practical control system.

Course Outcome (CO):

Students who successfully complete the course will be able to:

1. Evaluate the concepts learned in the theoretical classes in practical small systems.
2. Design an actual large-scale system fulfilling all control system constraints.
3. Measure different parameters of a practical control system.

Course Content:

The course content covers the following topics:

Experiment no. 1: Introduction to MATLAB for control systems
Experiment no. 2: Mathematical modelling of physical systems.
Experiment no. 3: Modelling of physical systems using MATLAB.
Experiment no. 4: Linear time invariant systems and representation
Experiment no. 5: Block diagram reduction
Experiment no. 6: Performance of 1st order and second order systems
Experiment no. 7: Design of PID controllers.

Text Books:

1. Control Systems Engineering 7th Ed - Nise

Reference Books:

1. Computer-Controlled Systems: Theory and Design, Third Edition, Dover Books on Electrical Engineering, Karl A Astrom, Bjorn Wittenmark, ISBN10: 0486486133, ISBN-13: 978-0486486130.
2. Digital Control of Dynamic Systems, by Franklin, Powell, Workman, Addison Wesley.
3. Digital Control: A State Space Approach, by Vaccaro, McGraw Hill.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1									✓			
CO2			✓									
CO3									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4701 Course Title: Digital Signal Processing Credits: 3.00

Pre-requisite: Signal and Systems (AVE 4501)

Rationale of the Course:

To learn and familiarize the discrete signals and systems and designing various filters

Course Objective:

1. Educate the students about discrete time systems and the design techniques for FIR and IIR digital filters.
2. Teach the conversion for analogue signal to digital signal in signal processing.
3. Acquaint the student with the properties of random signal, digital signal processing.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Describe the basic concepts of signals, signal processing and digital signals;
2. Analyse signals and systems in discrete time, including use of the z-transform;
3. Explain the Fourier transform and convolution to filter signals and explain the properties of the discrete-time Fourier transform;
4. Design finite impulse response (FIR) filters to satisfy a desired frequency response
5. Design infinite impulse response (IIR) filters using impulse invariance method and bilinear transformation method

Course Contents:

Introduction to digital signal processing (DSP): Discrete-time signals and systems, analog to digital conversion, impulse response, finite impulse response (FIR) and infinite impulse response (IIR) of discrete-time systems, difference equation, convolution, transient and steady state response.

Discrete transformations: Discrete Fourier series, discrete-time Fourier series, discrete Fourier transform (DFT) and properties, fast Fourier transform (FFT), inverse fast Fourier transform. Z transformation - properties, transfer function, poles and zeros and inverse Z transform. Correlation: Circular convolution,

auto-correlation and cross correlation.

Digital Filters: FIR filters - linear phase filters, specifications, design using window, optimal and frequency sampling methods. IIR filters – specifications, design using impulse invariant, bi-linear Z transformation, least-square methods and finite precision effects.

Introduction to MATLAB Simulink application in DSP. Implementation of DSP in RADAR Engineering.

Text and Ref Books:

1. Digital Signal Processing, John G.Proakis& Dimitris Manolakis; Prentice Hall.
2. Digital Signal Processing Using MATLAB, Vinay K. Langle& John G. Proakis; CL-Engineering.
3. Digital Signal Processing, Thomas J. Cavicchi; John Wiley & Sons.
4. Digital Signal Processing A practical approach, Emmanuel C. Ifeachor& Barrie W. Jervis; Prentice Hall.
5. Signal and System (Continuous & Discrete), Rodger E. Ziemer, W. H. Tranter & D. R. Fannin; Pearson Education.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3												✓
CO4			✓									
CO5			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Seminar	Presentation
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4702 Course Title: Digital Signal Processing Sessional Credits: 0.75

Pre-requisite: Signal and Systems Sessional (AVE 4502)

Rationale of the Course:

To learn and familiarize the basics of digital signals as well as the analysis and design of different analog and digital filters.

Course Objective:

1. Teach students to practically design FIR and IIR filters by MATLAB to meet specific magnitude and phase requirements.
2. Help students to determine if a DT system is linear, time-invariant, causal, and memoryless, determine asymptotic, marginal and BIBO stability of systems given in frequency domain.
3. Help students to use computers and MATLAB to create, analyse and process signals, and to simulate and analyse systems sound and image synthesis and analysis, to plot and interpret magnitude and phase of LTI system frequency responses.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Explain the basic sampling process, quantization to convert the analogue signal into digital signal using software and analytically
2. Apply the theory of modulation, using MATLAB and analyse the voice signal
3. Analyse the solution technique of different control system for varying input signal using MATLAB Simulink
4. Explain the basic theory of Fourier series, Fourier Transform, Z transform and interpret the analytical results using MATLAB.

Course Contents:

Experiment No. 1: Review of Basic MATLAB Functions and recollecting the lessons of Numerical Analysis Sessional
Experiment No. 2: Study of Sampling and Quantization and creating 3d surfaces
Experiment No. 3: Number Conversion Technique and Amplitude Modulation
Experiment No. 4: Review of Basic MATLAB Simulink and processing the signal
Experiment No. 5: Study of z transform and FM, DSBSC modulation and voice signal analysis
Experiment No. 6: Analysis and interpretation of Fourier series and Fourier Transform using MATLAB

Text and Ref Books:

1. Digital Signal Processing, John G.Proakis & Dimitris Manolakis; Prentice Hall.
2. Digital Signal Processing Using MATLAB, Vinay K. Langle & John G. Proakis; CL-Engineering.
3. Digital Signal Processing, Thomas J. Cavicchi; John Wiley & Sons.
4. Digital Signal Processing-A practical approach, Emmanuel C. Ifeachor & Barrie W. Jervis; Prentice Hall.
5. Signal and System (Continuous & Discrete), Rodger E. Ziemer, W. H. Tranter & D. R. Fannin; Pearson Education.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3									✓			
CO4	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4703 Course Title: Aircraft Electrical Instrument and Autopilot Systems
Credits: 3.00

Pre-requisite:

Rationale of the Course:

To learn and familiarize with the basics of electro- magnetic field theories and implement that knowledge in the field of communication.

Course Objectives:

1. Share in depth knowledge on the generation of AC and DC power and supply in aircraft, bus bar, generator, alternator, fuses and circuit breakers in aircraft power supply and the concept of emergency power supply, aircraft batteries.
2. Educate students regarding various safety requirements, aircraft electrical wiring and lighting.
3. Provide insights regarding the performance of Full Authority Digital Engine Control (FADEC) System and the basic aspects of Hydraulic systems, Pneumatic systems, brake systems.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:
1. Explain different electrical instrument components of an Aircraft
2. Ensure various safety requirements, aircraft electrical wiring and lighting
3. Design an Autopilot System of an Aircraft

Course Contents:

Aircraft Electrical Systems: AC and DC power generations and supply in aircraft, bus bar, generator, alternator, fuses and circuit breakers in aircraft power supply, Concept of emergency power supply, aircraft batteries, types, capacity etc. external power supplies, Auxiliary Power Unit (APU), Components of power distribution, safety requirements, aircraft electrical wiring and lighting.

Aircraft Electronic Systems: Integrated Cockpit Display System: Introduction, Cockpit Display System, Glass Cockpit, Display Unit, HUD, HDD, HMD, IEEE smart sensors.

Engine Control and Monitoring System: Principles of Operation, Engine Indications and Monitoring, Full Authority Digital Engine Control (FADEC) System.

Emergency Systems: Warning Systems, Fire Detection and Suppression, Emergency Oxygen, Passenger Evacuation, Cockpit Voice Recorder & Flight Data Recording System, Ice & Rain Protection System, Emergency power sources, Emergency landing.

Airplane control systems: Push pull rod system, operating principles, Cable and pulley system, Power

assisted and fully powered flight controls, digital fly by wire systems. Introduction to Hydraulic systems, Pneumatic systems, brake system, anti-skidding, landing gear systems, Engine Fuel systems, Air conditioning and pressurisation system, de-icing and anti- icing system.

Text Books:

1. Aircraft Systems (3rd edition) -- Ian Moir, Allan Seabridge; WILEY Publications.

Reference Books:

1. Handbooks of Airframe and Powerplant Mechanics; US dept. of Transportation, Federal, Aviation Administration, The English Book Store, New Delhi, 1995
2. Aircraft Electrical Systems- EHJ Pallette, 3rd edition, Pearson Education.
3. Aircraft Electronics- F Terry White, White Publications.
4. Fundamentals of Aircraft Electronics- Scott Kenney, Avotek Information Resources.
5. Aircraft Communications and Navigation Systems - David Wyatt, Mike Tooley: Routledge

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2						✓						
CO3			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4704 Course Title: Aircraft Electrical Instrument and Autopilot Systems Sessional Credits: 1.50

Pre-requisite:

Rationale of the Course:
To familiarize the basics of electro-mechanical components and their operations.

Course Objectives:
1. Teach the students about Aircraft Electrical Systems and Electronic Systems
2. Help students practically interface Flight Management System (FMS) and Electronic Flight Instrument System (EFIS) system.

Course Outcomes:
1. Practically identify of Aircraft Electrical Systems and Electronic Systems
2. Demonstrate FMS and EFIS system

3. Simulate Avion Autopilot

Course Contents:

Experiment no. 1: Familiarization with Pressure Transducer (Strain Gauge)
Experiment no. 2: Flow Rate Control of Water by Feedback Transducer.
Experiment no. 3: Study of negative temperature coefficient transducer (NTC) and its linearity by feedback method
Experiment no. 4: Errors in measurement and basic statistical sampling
Experiment no. 5: Study of an 8 bit Analog to Digital converter
Experiment no. 6: Study of Aircraft Communication System (Analog Modulation)
Experiment no. 7: Study of Aircraft Communication System (Digital Modulation)
Experiment no 8: Study of Flight Management System (FMS)
Experiment no 9: Study of the Electronic Flight Instrument System (EFIS)
Experiment no. 10: Simulation of Avion Autopilot

Text Books:

1. Aircraft Systems (3rd edition) -- Ian Moir, Allan Seabridge; WILEY Publications.

Reference Books:

1. Handbooks of Airframe and Powerplant Mechanics; US dept. of Transportation, Federal, Aviation Administration, The English Book Store, New Delhi, 1995
2. Aircraft Electrical Systems- EHJ Pallette, 3rd edition, Pearson Education.
3. Aircraft Electronics- F Terry White, White Publications.
4. Fundamentals of Aircraft Electronics- Scott Kenney, Avotek Information Resources.
5. Aircraft Communications and Navigation Systems - David Wyatt, Mike Tooley: Routledge

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4705 Course Title: Avionics Systems Design Credits: 3.00

Pre-requisite: Digital Logic and Computer Design (AVE 4405) and Real Time Embedded Systems (AVE 4603)

Rationale of the Course:

To learn and familiarize with the digital avionics system used in modern aircraft and spacecraft system.

Course Objectives:

1. Teach avionics architectures and their evolution to today's highly capable integrated modular avionics architectures.
2. Help to explore the features and evolution of significant data bus networks used in today's modern aircraft and spacecraft.
3. Familiarize with avionics system development – including hardware and software integration, regulations, development processes and analyses that are employed in the development of modern aircraft and spacecraft systems.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Describe modern avionics architectures with its merits and limitations
2. Explain the features and evolution of avionics systems in modern aircraft and spacecraft
3. Design an avionics system integrating different hardware and software components

Course Contents:

Avionics System Functionalities-Civil and Fighter Aircrafts, Avionics Subsystem- Navigation Subsystems, Landing Aids, Surveillance Subsystems, Communication Subsystems, Flight Control System.

Overview of avionics architecture evolution- distributed analogue architectures, distributed digital architectures, federated digital architectures, Integrated modular avionics architectures. Avionics architecture examples- The Airbus A320 Avionics Architecture, The Boeing 777 Avionics Architecture.

Digital data bus-basic principles of bus technologies including encoding techniques, attributes and topologies. Example- ARINC 429, MIL-STD-1553B, ARINC 629, ARINC 664, CANbus. AFDX etc.

Emergency Systems: Integration of different emergency system i.e Warning Systems, Fire Detection and Suppression, Emergency Oxygen, Passenger Evacuation, Cockpit Voice Recorder & Flight Data Recording System, Ice & Rain Protection System, Emergency power sources, Emergency landing.

Avionics system development –Hardware and software integration, regulations, development processes.

Text Books:

1. Civil Avionics Systems (3rd edition) -- Ian Moir, Allan Seabridge; WILEY Publications.
2. Avionics systems: design, development and integration, P.N.A.P. Rao

Reference Books:

1. Digital avionics handbook, Cary R. Spitzer, 3rd edition CRC Press, 2015
2. Aircraft Electrical Systems- EHJ Pallette, 3rd edition, Pearson Education.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3											✓	

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Project Based Learning	Project Report and Presentation

Course Code: AVE 4706 Course Title: Avionics Systems Design Sessional Credits: 1.50

As per the theory contents the sessional will be conducted.

Course Code: AVE 4707 Course Title: Engineering Ethics & Professionalism Credits: 2.00

Pre-Requisite: None

Rationale of the Course: To learn and understand the basics of ethics and professionalism

Course Objectives:

1. Instill the moral values that ought to guide their profession.
2. Resolve the moral issues in the profession.
3. Infer moral judgment concerning the profession.
4. Correlate the concepts in addressing the ethical dilemmas.
5. Judge a global issue by presenting an optimum solution.

Course Outcomes:

1. Distinguish between ethical and non-ethical situations.
2. Practice moral judgment in conditions of dilemma.
3. Demonstrate social and professional responsibility in an avionics engineering situation
4. Develop concepts based on moral issues and enquiry regarding environmental perspectives.
5. Apply risk and safety measures in avionics engineering fields.

Course Content:

Scope of sociology, Industrial revolution, Society and population, Social pathology, Nature of social change, Sociology of development, Urban ecology.

Definition and scopes of Ethics. Different branches of ethics. Social change and the emergence of new technologies. History and development of engineering ethics. Science and technology- necessity and application. Study of ethics in engineering. Applied ethics in engineering.

Human qualities of an engineer. Obligation of an engineer to the clients. Attitude of an engineer to other

engineers. Measures to be taken in order to improve the quality of engineering profession.

Ethical expectation: Employers and employees, inter-professional relationship, Professional Organization - maintaining a commitment of ethical standards. Desired characteristics of a professional code. Institutionalization of ethical conduct. Socialization; Poverty social exclusion and welfare; Women and Development; Crime, deviance and social control; Environment and risk; Sustainable development; Rural sociology; Family urbanization and industrialization; Urban ecology; Collective behaviour and social movements;

Reading Material:

1. Ethics and Professionalism in Engineering by Kristin L. Gilroy and Richard H McCuen
2. Engineering Ethics: Real World Case Studies by Steven K. Starrett (Author), Amy L. Lara (Author), Carlos Bertha (Author)
3. Sociology (4th edition), Anthony Gidders, Excel Media, India.
4. Sociology: Primary Principles, C. N. Shankar Raw, S. Chand Co Ltd.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1								✓				
CO2								✓				
CO3						✓						
CO4							✓					
CO5						✓						

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment/Case Study, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment/Case Study, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4801 Course Title: Radar System Engineering Credits: 3.00

Rationale of the Course:
To learn and familiarize the concept of radio engineering

Pre-requisite: Electronic Circuits II (AVE 4401), and Digital Signal Processing (AVE 4701)

Course Objectives:
1. Share insights on the principle involved in radar system and the various types of radar and areas of applications
2. Help students compute radar parameters

3. Guide to solve problems relating to radar

Course Outcomes (CO)
Upon completion of the course, the students will be able to:

1. Describe theoretical principles of different equipment of RADAR system.
2. Apply basic radar knowledge in handling different radar system in aircraft communication and navigation both in peacetime and wartime scenarios.
3. Design basic radar systems to contribute to the development of society.
4. Analyse the performances of different antenna and different radar system

Course Contents:

Introduction to Radar: Radar Principle, Functional block diagrams, Radar range equation, Radar frequencies, Pulse repetition frequency and Range ambiguity, Minimum detectable signal.

Radar cross-section of targets: Detection and tracking, jamming techniques.

Doppler Effect: Continuous wave and frequency modulation radars, moving target indicator and phase-Doppler radars.

Radar transmitter: Magnetron oscillator, klystron amplifier and travelling wave tube amplifier.

Radar antenna: Antenna parameters, radiation pattern and aperture distribution.

Radar receivers: Displays and duplexers.

Electronic Warfare: Electronic counter measures, electronic counter measures. Introduction to Airborne Radar

Text and Ref Books:

1. Introduction to RADAR systems, M. Sholnik, McGraw-Hill International.
2. Principle of Radar, Tomay, Prentice Hall of India.
3. Radar design, principles, signal processing and the environment, Fred E Nathanson, Prentice Hall of India Private Ltd.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3			✓									
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4802 Course Title: Radar System Engineering Sessional Credits: 1.50

Pre-requisite:

Rationale of the Course: To get hands on experience on various concepts of radio engineering

Objective:

1. Familiarize the students with the fundamentals of Radar and the different types of Radar and their operation.
2. Inspire the students regarding the need for signal detection in Radar and various Radar signal detection techniques.
3. Practically teach different Radio Navigation techniques.

Course Outcomes (CO)

Upon completion of the course, the students will be able to:

1. Analyse the performance of simple tracking radar systems
2. Apply problem solving approaches to work challenges and make decisions using sound engineering methodologies.
3. Analysis of the principles of Synthetic Aperture Radar, its use in geophysical remote sensing and surveillance applications, and the digital processing used to form SAR images
4. Explain the principles of radio navigation systems (including secondary radar and GPS)

Course Contents:

Experiment No. 1: Detection of stationary targets using parabolic antenna and study the influence of Sensitivity Time Control (STC) on display.
Experiment No. 2: Detection of moving targets using parabolic antenna and estimation of beam- width.\
Experiment No. 3: Detection of moving targets using patch antenna.
Experiment No. 4: Study of the effect of short pulses on range.
Experiment No. 5: Study of range resolution and multi- path effect.

Text and Ref Books:

1. Introduction to RADAR systems, M. Sholnik, McGraw-Hill International.
2. Principle of Radar, Tomay, Prentice Hall of India.
3. Radar design, principles, signal processing and the environment, Fred E Nathanson, Prentice Hall of India Private Ltd.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1		✓										
CO2									✓			
CO3		✓										
CO4	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4803 Course Title: Aircraft Communication, Navigation & Surveillance System Credits: 3.00

Pre-requisite: Communication Systems (AVE 4601)

Rationale of the Course:

This course is provided to gather knowledge about communication, navigation and guidance systems of an aircraft for their proper implementation in future workplace or studies.

Course Objectives:

1. Provide a fundamental understanding and knowledge of conventional and modern design and working principles of radar, guidance and navigation for air vehicles.
2. Provide the basic mathematical concepts of radar, navigation by NDB, VOR, GPS and Inertial Navigation approaches, and guidance laws.
3. Provide an expansive view into the technological trends of future aircraft navigation and guidance systems designs.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Demonstrate an understanding of the fundamental methods of navigation, radio direction finding, automatic direction finder and radio compass.
2. Analyse various navigation components like VOR, DME, ILS, MLS, Secondary Radar, TCAS etc.
3. Interpret the working principle of Doppler navigation, its beam configuration and frequency spectrum analysis.
4. Distinguish between different radio navigation components such as non-directional beacon and radio direction finding, loop antenna, sense aerial, radio altimeter etc.
5. Outline the basic operating principle and function of the Flight management system.

Course Contents:

Introduction: Block diagram of navigation system, Types of navigation, Coordinate Frames, Coordinate transformation- Direction cosine matrix, Euler angles.

Methods of navigation: Dead Reckoning (DR) Computation: Position, Radio Fix, LOS distances measurement, ranging, course computation (range & bearing), Inertial Navigation System (INS), Sensors-Accelerometers, Gyroscopes (mechanical, ring laser, fiber optic), Inertial measurement unit (IMU), INS Errors.

Radio Navigation: Non Directional Beacon and Automatic Direction Finder, Loop Antenna, Sense Aerial, Ambiguity Resolve, Limitations; Radio Altimeter, Ground Proximity Warning System; VHF Omnidirectional Range- LightHouse Principle, Frequency Spectrum, Wave Equation, Errors and limitations; Distance measuring equipment- DME operation, Mathematical Relations, Signal Equation (Gaussian Pulse), Mode of operation; Instrument Landing System-Mathematical Relations, Antenna Array Arrangement, Beam pattern Geometry of LOC and GS, Development and concepts of Microwave Landing System; Doppler Navigation- Principle of operation, Beam configuration, Fly by wire systems, Hyperbolic Navigation-LORAN-C; Air Data Navigation; Satellite Navigation.

Secondary Radar: Concept of Secondary Radar, MODE-A,C,S Signal Format and types of transmission, ATC-RBS Interrogation-Reply Pulse, Technical features.

Traffic Alert Collision Avoidance System: Introduction, Basic operating principle, Block diagram and system description, Controls and display.

Flight Management System: Introduction, basic operating principle, block diagram and system description, controls and display.

Text and Reference Books:

1. Avionics Fundamentals, Jeppesen, Highflyn.
2. Principles of Avionics, Albert Helfrick, Avionics Communication.
3. Digital Avionics Systems Principles and Practice, R. Spitzer, The Blackburn Press.
4. Antennas and Wave propagation, 4th Edition, John D Kraus, Ronald J Marhefka, McGraw-Hill
5. Avionics Navigation Systems, Myron Kayton, Wiley-Interscience.
6. Elements of Electronic Navigation, N S Nagaraja, McGraw-Hill.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2				✓								
CO3		✓										
CO4	✓											
CO5	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final

		Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4804 Course Title: Aircraft Communication, Navigation & Surveillance System Sessional Credits: 1.50

Pre-requisite:

Rationale of the Course: None

This subject aims at providing a basic understanding of navigation guidance and communication, with special attention to the signal processing aspect and overall system integration for further workplace.

Course Objectives:

1. Teach the fundamentals of the various guidance techniques and their properties.
2. Provide insights on position and attitude estimation.
3. Share examples of current and planned implementations and applications of navigation instruments and their working mechanism.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Discuss the construction and working principle of the equipment by practically observing and operating them.
2. Apply the formulas and equations learned in the theory course in calculating various parameters from the obtained data.
3. Analyse the data obtained from this equipment by plotting graphs or other means.
4. Evaluate the performance of different navigation equipment based on the theoretical knowledge.
5. Design any model related to this subject and demonstrate that model for further lab use.

Course Contents:

Experiment No 01: Familiarization with DME operation and its terminologies using a DME trainer set.

Experiment No 02: Familiarization with ILS operation and terminologies and ILS components Testing using an ILS Trainer set.

Experiment No 03: Familiarization with Radio Altimeter and simulating a return signal through a test set.

Experiment No 04: Familiarization with autopilot operation and its terminologies and autopilot Testing using a Trainer set.

Experiment No 05: Familiarization with GPS operation and terminologies and GPS Receiver Testing using a GPS Simulator test set.

Experiment No 06: Familiarization with Analog Modulation for Aircraft Communication System

Experiment No 07: Familiarization with Digital Modulation for Aircraft Communication System

Experiment No 08: Familiarization with Aircraft Navigation System

Experiment No 09: Familiarization with Equipment Code

Text and Reference Books:

1. Avionics Fundamentals, Jeppesen; Highflyn.
2. Principles of Avionics, Albert Helfrick; Avionics Communication.
3. Digital Avionics Systems Principles and Practice, R. Spitzer; The Blackburn Press.
4. Antennas and Wave propagation, 4th Edition, John D Kraus, Ronald J Marhefka; McGraw-Hill
5. Avionics Navigation Systems, Myron Kayton; Wiley-Interscience.
6. Elements of Electronic Navigation, N S Nagaraja; McGraw-Hill.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1										✓		
CO2	✓											
CO3		✓										
CO4				✓								
CO5			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Applied Research	Quiz, Viva, Lab Report, Mid Term Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Applied Research	Quiz, Viva, Lab Report, Final Exam
CO5	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: AVE 4700 Course Title: Final Year Research and Design Project (FYRDP) Credits: 9.00

Pre-requisite: Student should complete all courses up to 6th semester

Rationale of the Course:

The main course is to expose and teach students the aspects and techniques related to performing a research project in the aerospace engineering fields.

Course Objectives:

1. Assess the culmination of all program outcomes
2. Provide an industry-oriented project experience
3. Assess the capability of the students to solve complex engineering problems and complex engineering activities

Course Outcomes (CO):

Upon completion of the course, the students will be able:

1. To develop the ability to undertake complex problem identification, formulation and solution utilising a system approach.
2. To design and conduct experiments of complex engineering problems with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.
3. To conduct investigations, devise appropriate measurement, analyse and interpret data and form reliable conclusions.
4. To apply modern engineering and IT tools to find the solution of well-defined Avionics engineering problems.
5. To develop effective all-round communication skills.
6. To solve the problems effectively in a multi-disciplinary and multi-cultural environment both as a team member and a leader with effective managerial skills.
7. To be able to apply ethical principles in all professional and research work.
8. To develop engineers with universal socio-cultural adaptability who can undertake responsibilities in the global arena without compromising on environmental safety.
9. To develop the capacity to undertake life-long learning.
10. Apply appropriate research methodology to find novel conclusions and insights for producing a scientific publication

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1			✓									
CO2						✓						
CO3				✓								
CO4					✓							
CO5										✓		
CO6											✓	
CO7								✓				
CO8							✓					
CO9												✓
CO10									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Applied Research	Project Report and Presentation
CO2	Applied Research	Project Report and Presentation

CO3	Applied Research	Project Report and Presentation
CO4	Applied Research	Project Report and Presentation
CO5	Applied Research	Project Report and Presentation
CO6	Applied Research	Project Report and Presentation
CO7	Applied Research	Project Report and Presentation
CO8	Applied Research	Project Report and Presentation
CO9	Applied Research	Project Report and Presentation
CO10	Applied Research	Project Report and Presentation

b. Other Engineering Courses

Course Code: ASE 4101 Course Title: Introduction to Aeronautical Engineering Credits: 3.00

Pre-requisite: None.

Rationale of the Course:

The purpose of this course is to serve as an introduction into the basics of aircraft aerodynamic-characteristics, components, structures and avionics systems.

Course Objectives:

1. Provide the knowledge about basic Aeronautical Engineering and the aerodynamic characteristics of aircraft.
2. Assist them to identify the forces acting on aircraft and learn how to analyse them.
3. Interpret them about the aircraft basic structure, different aircraft component configurations.
4. Explain the students about the mechanics of flight and fight performance.
5. Present aircraft avionics systems regarding instrumentation, communication and navigation System.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain basics of Aeronautical Engineering.
2. Identify the forces acting on aircraft.
3. Interpret the aircraft structure and basic configurations.
4. Explain about mechanics of flight.
5. Illustrate aircraft instrumentation, communication and navigation system.

Course Contents:

Introduction to Aeronautical Engineering: Classification of aircraft, Different parts of aircraft (airframe, engine, avionics systems, communication systems, instrumentation and navigation systems) and their function.

Introduction to Aerodynamics: Standard atmosphere, Dimensional analysis, Bernoulli's theorem for incompressible flows and its applications in aeronautical engineering. Local and free stream characteristics.

Aerofoil Classification and Characteristics: Pressure distribution over aerofoil and its variation with angle of attack. Centre of pressure and its movement, Forces and moments acting on aerofoil, centre of gravity, centre of pressure and aerodynamic centre concepts. Characteristics of Lift drag and pitching moment curves. Stall and its effects.

Flight Mechanics: Aircraft manoeuvres- Take off, climb, cruise, glide, descend and landing. Aircraft performance parameters such as endurance, aircraft ceiling and range. Aircraft control surfaces and High lift devices.

Text and Ref Books:

1. Airframe and Power Plant – C A Zweng; Galotia Publications.
2. Spacecraft Systems Engineering –Peter Fortescue and John Stark; John Wiley and Sons.
3. Introduction to Flight -John D Anderson Jr; Tata McGraw-Hill.
4. Introduction to Aerospace structural Analysis –David H Allen, Publisher; Wiley and Sons.
5. Avionics Navigation Systems, 2nd Ed – Myron Kayton
6. Aerodynamics – Clancy
7. Flight without Formulae – Kermode
8. Fundamentals of Aerodynamics- John D. Anderson; McGraw-Hill
9. Principles of Avionics – 6th Ed.– Albert Helfrick.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3		✓										
CO4	✓											
CO5				✓								

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Flipped Classroom	Class Presentation

Course Code: ASE 4102 Course Title: Aeronautical Engineering Drawing Credits: 1.50

Pre-requisite: None

Rationale of the Course:

It will be useful for designing and drawing accurate schematics for simple blocks, orthographic and isometric representations, dimensioning, etc., which will be helpful during project, work in later semesters, as well as professionally.

Course Objectives:

1. Introduce the principles and perspectives of geometric drawing that includes the standardisation, drafting, dimensions etc.
2. Help the student demonstrate the technique of engineering graphics as a basis of engineering communication and expression of idea and thought
3. Enable them to use proper and standard technique in lettering, basic geometric constructions, sketching, dimensioning methods to describe size, shape and position accurately on an engineering drawing.
4. Assist to create orthographic projection auxiliary, sectional views, and apply 3Dpictorials to choose the best view to present the drawings.
5. Illustrate to produce final drawings during the design process including assembly, machine and working drawings.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Demonstrate use of appropriate standards and conventions in drawing sheet preparation and layout
2. Apply the correct methods of referencing relevant specifications in the interpretation of aeronautical engineering drawings.
3. Apply the correct conventions and techniques in drawing sectional, auxiliary views.
4. Develop isometric and oblique pictorial form from third angle orthogonal drawings

Course Contents:

Class 1: Introduction, Familiarization with drawing tools and types of projections.
Class 2: Drawing orthographic views of simple blocks
Class 3: Drawing orthographic views of objects with round features.
Class 4: Drawing orthographic views of objects with fillets, rounds.
Class 5: Class test on orthographic views.
Class 6: Drawing sectional views.
Class 7: Class test on sectional views.
Class 8: Drawing auxiliary views.
Class 9: Drawing oblique views.
Class 10: Class test on auxiliary and oblique views.
Class 11: Drawing isometric views of simple blocks
Class 12: Drawing orthographic views of objects with round features, fillets and rounds.
Class 13: Class test on isometric views
Class 14: Final Quiz.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3				✓								

CO4									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam Final Exam

Course Code: CSE 4201 Course Title: Computer Programming and Application Credits: 3.00

Pre-requisite: None.

Rationale of the Course:
This course is designed to introduce the fundamental principles, mechanism of programming skills and develop basic programming skills to design and develop computer programs.

Course Objectives:
1. Provide the basic idea of computer programming in C/C++.
2. Assist to learn how to solve problems with structured programming using C and OOP using C++.

Course Outcomes (CO):
Upon completion of the course, the students will be able to:
1. Explain the fundamental concepts and purpose of computer programming.
2. Identify classes, objects, members of a class and the relationships among them needed for a specific problem.
3. Develop programming skills with respect to program design and development.

Course Contents:

Introduction to digital computers, Introduction to programming languages, Algorithms and flowcharts, Structured Programming using C,Variable and constants, operators, expressions, Control statements, Function, Arrays, Pointers, Structure & Union, User defined data types,Input output and files, Advantages of OOP over structured programming, Object oriented Programming using C++, Introduction, Classes, and objects, Constructors, Destructos,Encapsulation, Inheritance, Polymorphism.

Text and Ref Books:

1. Programming in ANSI C - E. Balagurusamy
2. Schaum's Outline of Programming with C - Byron S. Gottfried
3. C, The Complete Reference – Schildt, H; McGraw-Hill.
4. Object Oriented Programming With C++ - E. Balagurusamy
5. C++ Programming: An Object-Oriented Approach - Behrouz A. Forouzan

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Flipped Classroom	Class Test/Assignment, Final Exam

Course Code: CSE 4202 Course Title: Computer Programming and Application Sessional
Credits: 1.50

Pre-requisite: None

Rationale of the Course:

To deliver the basic knowledge of C and C++ programming language as a preparation for students in using programming tools to solve engineering related mathematical problems.

Course Objectives:

1. The course is designed to provide practical knowledge of C language.
2. Assist them to develop logics which will help them to create programs in C.
3. Familiarise them with the basic programming constructs using other languages like C++.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Solve problems systematically using structured programming and object oriented programming.
2. Analyse the fundamental principles, typical characteristics and mechanisms of a structured programming language experimentally.
3. Develop complete programs for simple to moderate problems individually.

Course Contents:

Lesson 01: Application of *printf ()* and *scanf ()* functions in C programming.
Lesson 02: Application of conditional, relational, and logical operators C programming.
Lesson 03: Application of Loops in C programming.
Lesson 04: Application of Array in C programming
Lesson 05: Application of String in C programming.
Lesson 06: Application of Functions in C programming.
Lesson 07: Application of Structure in C programming
Lesson 08: Application of Pointer in C programming.
Lesson 09: Application of File in C programming.

Lesson 10: Application of Class and object in C++.
Lesson 11: Application of Constructor and destructor in C++.
Lesson 12: Application of encapsulation, polymorphism and inheritance in C++.

Text and Ref Books:

1. Programming in ANSI C - E. Balagurusamy
2. Schaum's Outline of Programming with C - Byron S. Gottfried
3. C, The Complete Reference – Schildt, H; McGraw-Hill.
4. Object Oriented Programming With C++ - E. Balagurusamy
5. C++ Programming: An Object-Oriented Approach - Behrouz A. Forouzan

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: ASE 4202 Course Title: Workshop Technology Sessional Credits: 1.50

Rationale of the Course:

To provide instructions and practical experience in basic mechanical workshop methods, manufacturing process, tools and machines.

Course Objectives:

1. Provide an introduction to the basic operations of bench fitting tools.
2. Familiarise the student with the various types of manual machines and introduce them to different operations of machining.
3. Teach them to learn about different types of moulding.
4. Enable them to know about various types of Welding and introduce them to different types of welding joints.
5. Provide knowledge on different tools of wood shop and make them familiar with different type of wood joints.

Course Outcomes (CO):

1. Identify how to work in workshop environments.
2. Organise practical work in the engineering workshop in various environments.

3. Demonstrate practical knowledge to produce engineering works like- foundry, carpentry, welding etc.

Course Content:

Foundry: Introduction to foundry, tools and equipment; Patterns: function, pattern making; Moulding: moulding materials sand preparation, types of mould, procedure; Cores: types, core making materials; Metal melting and casting; Inspection of casting and casting defects. Welding: Metal joints, riveting, grooving, soldering, welding; Welding practice: electric arc steel, aluminium; Types of electrode; Welding defects: visual, destructive and non-destructive tests of welding. Gas welding and equipment; Types of flame; Welding of different types of material; Gas welding defects; Test of gas welding.

Reading Material:

1. Building Materials- Gurcharan Singh, Standard Publishes-Distributors-Del (2008).
2. Engineering Materials- M. A. Aziz.
3. Machine Shop Practice- James Anderson, W. A. Chapman.
4. Shop Theory (6th ed)- Anderson and Tatro, McGraw-Hill Education (1976).

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

Course Code: ASE 4341 Course Title: Thermodynamics Credits: 3.00

Pre-requisite: Physics (PHY 4101)

Rationale of the Course:

To introduce the fundamental concepts of energy, work and heat, as well as to provide understanding on the thermodynamic concepts, first and second thermodynamic laws.

Course Objectives:

1. Explain the student about basic concepts including the First Law of Thermodynamic and to derive the corollaries of the First Law.
2. Teach them to solve problems for each thermodynamic process using steam or air.

3. Describe them about the Second Law of Thermodynamics and its corollaries, entropy and explain thermodynamic processes based on T-s diagram.
4. Assist to determine the performance of various steam and air thermodynamics cycles.

Course Outcomes (CO):

Upon completion of the course, the students will be able to

1. Develop the basic concepts of Thermodynamics.
2. Describe Thermodynamics Laws.
3. Apply the basics of thermodynamics to thermal equipment.

Course Contents:

Fundamental concepts and first law: Concept of continuum, macroscopic approach, thermodynamic systems; closed, open and isolated. Property, state, path and process, quasi-static process, work, modes of work, Zeroth law of thermodynamics- concept of temperature and heat, internal energy, specific heat capacities, enthalpy - concept of ideal and real gases. First law of thermodynamics - applications to closed and open systems - steady flow processes.

Second law and entropy: Second law of thermodynamics; kelvin Planck and Clausius statements of second law. Reversibility and irreversibility - Carnot theorem, Carnot cycle using steam, reversed Carnot cycle, efficiency, COP - thermodynamic temperature scale - Clausius inequality, concept of entropy, entropy of ideal gas, principle of increase of entropy.

Thermodynamic availability: Basics; energy in non- flow processes: expressions for the energy of a closed system – equivalence between mechanical energy forms and energy – flow of energy associated with heat flow – exergy, consumption and entropy generation - exergy in steady flow processes: expressions for exergy in steady flow processes – exergy dissipation and entropy generation.

Properties of pure substance: Properties of pure substances; thermodynamic properties of pure substances in solid, liquid and vapor phases, Use of property tables, phase rule, PVT surfaces, standard Rankine cycle.

Air standard and Refrigeration cycles: Equations of state for ideal gases, Properties of gases and vapours; Properties of atmospheric air; non-flow and flow processes; air standard cycles; Brayton, Otto and Diesel cycles. Refrigeration cycles; phase change of working substance. Thermodynamic

relations and equations of state; Mass and energy balance for a combustion reaction; Mixtures of gases and vapours; Fuels and combustion.

Textbook:

1. Thermodynamics, Yunus A. Cengel, Michael A. Boles
2. Fundamentals of Thermodynamics, R E Sonntag, C. Borgnakke, G J. Van Wylen; John Wiley & Sons, Inc, 5th edition, 2000.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1			✓									
CO2	✓											

CO3				✓								

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: ASE 4342 Course Title: Thermodynamics Sessional Credits: 0.75

Pre-requisite:

Rationale of the Course

To describe the concepts of heat, work, and energy and correctly use thermodynamic terminology.

Course Objectives:

1. Teach about the identity of an unknown metal.
2. Assist the student to calculate the approximate specific heat of unknown metal.
3. Describe them about the relationship between the heat that is transferred and change in temperature.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Describe and correctly use thermodynamic terminology.
2. Define the concepts of heat, work, and energy.
3. Explain fundamental thermodynamic properties.
4. Analyse basic thermodynamic cycles.

Course Contents:

The course will include the following experiments:

1. Determination of Flash Point of Liquid Fluid
2. Study of Sling Psychrometer
3. Viscosity Test of Liquid Substance.
4. Determination of Carbon Residue of a given fuel.
5. Proximate Analysis of Coal
6. Study of Different Speed Measuring Devices
7. Study of a Refrigeration and Air Conditioning Unit.
8. Study and Calibration of pressure gauge by Dead Weight Tester.
9. Determination of the Calorific value of Fuel.
10. Determination of Calorific value of Gaseous by Gas Calorimeter.
11. Concept of pressure and pressure sensor behaviour.

Text and Ref Books:

1. Thermodynamics, Yunus A. Cengel, Michael A. Boles

2. Fundamentals of Thermodynamics, R E Sonntag, C. Borgnakke, G J. Van Wylen; John Wiley & Sons, Inc, 5th edition, 2000.
3. Thermodynamics, Kenneth Wark, 6th Ed; McGraw-Hill, Singapore, 1999.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2				✓								
CO3			✓									
CO4									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: ASE 4353 Course Title: Engineering Mechanics (Statics and Dynamics)
Credits: 4.00

Prerequisite: None.

Rationale of the Course:

To provide the students with the basic knowledge in the mechanics of rigid body which will be helpful while studying the strength of materials, aircraft structures etc.

Course Objectives:

1. Enable the student to express and resolve the position and force into vector unit components.
2. Assist them to determine the forces in the members of trusses and frames using the method of joints and sections.
3. Illustrate them to draw and describe the free-body diagram and to solve the problems using the equations of equilibrium.
4. Teach to determine the location of centre of gravity and centric for a system and to determine the moment of inertia for an area.
5. Explain about the application of Newton's laws of motion and conservation principles to solve real life.
6. Make them understand the principles and methods used in analysing motion of a particle.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:
1. Explain basic kinematics concepts – displacement, velocity, and acceleration (and their angular

counterparts).
2. Demonstrate use of basic dynamics concepts- Work-Energy principle, Impulse-Momentum principle to solve dynamics problems
3. Apply scalar and vector analytical techniques for analysing forces in statically determinate structures.
4. Calculate the location of "centre of gravity" and "centroids" for bodies of arbitrary shape.
5. Evaluate equilibrium of particles and bodies in real world problems.

Course Contents:
Basic concepts of mechanics; Statics of particles and rigid bodies; Properties of forces, moments, couples, and resultants; Analysis of two- and three-dimensional problems; Centroids of lines, areas, and volumes; Forces in truss, frames, and cables; Friction.

Moments of inertia of areas and masses; Relative motion.

Planar mechanisms, linkages, mobility; instant centers of rotation, Kennedy's theorem; Velocity and acceleration polygons; Euler's first law; angular momentum and Euler's second law.

Kinetics of particles: Newton‟s second law of motion; Principles of work, energy, impulse, and momentum; System of particles; Kinematics of rigid bodies.

Kinetics of plane motion of rigid bodies: forces and acceleration; Principles of work and energy.

Text and Ref Books:
1. Vector Mechanics for Engineers: Statics and Dynamics - Ferdinand P. Beer, E Russell Jr. Johnstone; McGraw-Hill Companies, 5th edition 1988.
2. Engineering Mechanics - Timoshenko, D H Young, J V Rao
3. Engineering Mechanics - Andrew Pytel, JaonKiusaloas
4. Engineering Mechanics, Statics and Dynamics - Joseph F Shelley; McGraw-Hill, 1980.
5. Engineering's Mechanics - J.L. Merian& LG Kraige

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3		✓										
CO4				✓								
CO5				✓								

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: CSE 4305 Course Title: Data Structures and Algorithm Credits: 3.00

Pre-requisite: Computer Programming and Application (CSE 4201)

Rationale of the Course:
This course covers the fundamentals of computer programming including data structures such as arrays, lists, stacks, queues, and trees, and algorithms such as list manipulation, sorting, graph searches, and tree traversals. In the present context of the avionics field, it is imperative that avionics engineers are aware of these fundamentals and able to apply these to their relevant fields.

Course Objectives:

1. Provide ideas on designing algorithms for solving problems that use data structures such as arrays, linked lists, stacks, queues, graphs, and trees, and algorithms such as those that are used for list manipulation, graph manipulation (e.g., depth-first search), sorting, searching, and tree traversals
2. Assist them to implement algorithms in C or C++ using a good programming style.

Course Outcome (CO):

Based on satisfactory completion of the course, a student should be able to:

1. Develop the understanding of basic data structures (such as an array-based list, linked list, stack, queue, binary search tree) and algorithms.
2. Demonstrate the ability to analyse, design, apply and use data structures and algorithms to solve engineering problems and evaluate their solutions.
3. Formulate new solutions for problems or improve existing code using learned algorithms and data structures.

Course Content:
Introduces fundamental data structures, algorithms, and abstract data types. Main topics include data structures such as arrays, linked lists, stacks, queues, graphs, and trees, and algorithms such as those that are used for list manipulation, graph searches, sorting, searching, and tree traversals.

Reading Material:
1. Data Structures with C - Seymour Lipschutz
2. Data Structures Using C++ -D. S. Malik (2nd Edition)
3. Data Structures and Algorithm Analysis in C++ - Mark Allen Weiss (4th Edition)
4. Data structures and algorithms with C- Debdutta Pal, Suman Halder

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											

CO2			✓									
CO3				✓								

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: CSE 4306 Course Title: Data Structures and Algorithm Sessional Credits: 1.50

Pre-requisite: Computer Programming and Application (CSE 4201)

Rationale of the Course:
Data structure is a logical & mathematical model of storing & organising data in a particular way in a computer. In system application programming the methods & techniques of data structures are widely used. The study of data structure helps the students in developing logic & structured programs.

Course Objectives:

1. Illustrate general understanding of basic data structures.
2. Assist them to developing programming skills for advanced data structures such as trees, hash tables, priority queues etc.

Course Outcomes:

1. Demonstrate the advantages and disadvantages of specific algorithms and data structures by experimental work.
2. Select basic data structures and algorithms for autonomous realisation of simple programs or program parts
3. Initiate practical knowledge to determine and demonstrate bugs in programs.
4. Formulate new solutions for problems or improve existing code using learned algorithms and data structures.

Course Content:

Experiment no. 1: Array List (Implementing List using Array)
Experiment no. 2: Binary Search
Experiment no. 3: Linked List Implementation
Experiment no. 4: Double Linked List Implementation
Experiment no. 5: Implementing Stack using Array
Experiment no. 6: Implementation of Queue using Array, Circular Queue

Experiment no. 7: Sorting Algorithm: Quick Sort
Experiment no. 8: Sorting Algorithm: Merge Sort
Experiment no. 9: Operation on Binary Search Tree (Insertion, Deletion, Traversal, Merging)
Experiment no. 10: Operation on Heap Trees/Priority Queue (MAX and MIN Heap)
Experiment no. 11: Implementation of Graph Algorithms: BFS
Experiment no. 12: Implementation of Graph Algorithms: DFS

Text and Ref Books:

1. Data Structures with C - Seymour Lipschutz
2. Data Structures Using C++ -D. S. Malik (2nd Edition)
3. Data Structures and Algorithm Analysis in C++ - Mark Allen Weiss (4th Edition)
4. Data structures and algorithms with C- Debdutta Pal, Suman Halder

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3				✓								
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: CSE 4513 Course Title: Aerodynamics Credits: 3.00

Pre-requisite: Fundamentals of Aeronautical Engineering, Fundamentals of Fluid Mechanics, Mechanics of Solid

Rationale of the Course:

This course introduces the students with the fundamental principles of aerodynamics for understanding stability and control, aircraft performance etc.

Course Objectives:

1. Teach the fundamental principles of incompressible and compressible fluid mechanics and aerodynamics.
2. Train to apply these principles to real systems such as pipe flows, automobiles and aircraft.

3. Explain them about the sources of friction, induced, wave, and pressure drag.
4. Teach about different aspects of flight characteristics that relate to lift, drag, thrust and power.
5. Enable them to perform calculations involving lift, drag in relation to various aspects of flight and aircraft performance.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain the methods of fluid flow analysis i.e. theoretical, experimental and computational.
2. Analyse the concept potential theory and its application to incompressible and inviscid flows.
3. Apply of theoretical techniques to analyse the simple viscous flows.
4. Apply the numerical methods for solution of complex flow situations.
5. Describe implications errors and stability analysis of numerical methods.

Course Contents:

Inviscid flows
Models of fluid flow, continuity and momentum equations applied to inviscid flows, drag momentum theory, concept of streamlines, stream tubes, streak line, path lines. Angular velocity, strain and vorticity, potential theory applied to Inviscid flows, elementary flows, their combination and applications. Solution of flows past bodies using Panel methods.

Theory of 2D aerofoils: Kutta-Joukowski theorem, Kutta condition, Kelvin circulation theorem. Classical thin aerofoil theory. Types of flow separation and inviscid flow characteristics over a 2D aerofoil. Inviscid & incompressible flow over finite wings, Prandtl's lifting line theory, lift distribution over finite wings, effect of aspect ratio; Different types of drags.

Viscous Flows

Qualitative aspects of viscous flows, Navier-Stokes equations, modification N-S equation for different flows, Exact solutions of N-S equations, Aerodynamic heating, Prandtl Boundary Layer theory; Boundary Layer equations and their solutions. Skin friction and skin friction drag.

Laminar flow past flat plate. Concept free shear flows viz. jet, wake and mixing streams. Flow past cylinder and spheres and their applications. Boundary layer separation and its effects. Flow control techniques. Methods to reduce different types of drag. Introduction to turbulence, concept of turbulence modelling, Prandtl mixing length theory.

Text and Ref Books:

1. Mechanics of Fluids - Irving H. Shames
2. Mechanics of Fluids - B. S. Messy
3. Fundamentals of Aerodynamics - John D Anderson; McGraw-Hill.
4. Aerodynamics for Engineering Students –E. L Houghton, P.W. Carpenter, S.H. Collicot and D.T. Valentine; Elsevier.
5. Computational Fluid Mechanics and Heat Transfer – Anderson

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											

CO2	✓											
CO3		✓										
CO4		✓										
CO5				✓								

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: CSE 4514 Course Title: Aerodynamics Sessional Credits: 1.50

Pre-requisite: Fundamentals of Fluid Mechanics (ASE 4411)

Rationale of the Course:

This course introduces the students with the fundamental principles of aerodynamics for understanding stability and control, aircraft performance etc.

Course Objectives:

1. Provide the fundamental principles of incompressible and compressible fluid mechanics and aerodynamics.
2. Explain them about the sources of friction, induced, wave, and pressure drag.
3. Teach about different aspects of flight characteristics that relate to lift, drag, thrust and power.
4. Enable students to perform calculations involving lift, drag in relation to various aspects of flight and aircraft performance.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Conduct experiments, and then analyse and interpret results successfully.
2. Demonstrate that theoretical characteristics of low-speed aerodynamics can be implemented through the wind tunnel operations and flow visualization techniques.
3. Analyse the Pressure and velocity distribution along the radius of a forced vortex and interpret the results
4. Demonstrate coefficient of drag for a right circular cylinder
5. Evaluate the performance of lift and drag characteristics of NACA-0012 aerofoil.

Course Contents:

Experiments include topics of theory of flight, low speed aerodynamics, wind tunnel operations and flow

visualization techniques; Pressure and velocity distribution along the radius of a forced vortex, coefficient of drag for a right circular cylinder, and lift and drag characteristics of NACA-0012 aerofoil; Experiments in the open circuit subsonic wind tunnel, wind tunnel calibration, model testing and data reduction for obtaining important aerodynamic and stability parameters of an aircraft; Familiarization with the use of supersonic wind tunnel and its flow visualization system.

Text and Ref Books:

1. Mechanics of Fluids - Irving H. Shames
2. Mechanics of Fluids - B. S. Messy
3. Fundamentals of Aerodynamics - John D Anderson; McGraw-Hill.
4. Aerodynamics for Engineering Students –E. L Houghton, P.W. Carpenter, S.H. Collicot and D.T. Valentine; Elsevier.
5. Computational Fluid Mechanics and Heat Transfer – Anderson

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3		✓										
CO4			✓									
CO5					✓							

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO5	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: CSE 4807 Course Title: Industrial and Business Management Credits: 3.00

Pre-requisite: None

Rationale of the Course:
To learn the principles and techniques of industrial management which is essential for doing job in industry

Course Objectives:

1. Teach about central theories within the field of industrial management.
2. Illustrate the short description about the transactions of a company.

3. Explain about the cash flow and the financial tools for a company.
4. Provide a basic understanding of contemporary organisational forms

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Gain knowledge about financial concepts used in making business decisions.
2. Apply quantitative measurements to solve business problems related to depreciation, percentages, future value of money and to be able to make better business decisions.
3. Understand the roles of Manager and how to use motivation in workplace.
4. Gain knowledge about the methods and processes of plant layout & production planning and control.
5. Understand project scheduling selection and operations management.

Course Contents:

Organization and management: system approach to organization, organization theory and organizing practices, basics of organizing.

Personnel and human resource management in business human factors and motivation, leadership, group decision making and communication, Job gradation, process of performance appraisal and reward Systems.

Emergence of management thought and the patterns of management analysis scientific management and Taylorism, Modern operational-management theory, emergence of the behavioural sciences, recent contributors to management thought.

Cost management elements of cost of products, cost centres and allocation of overhead costs;
Management accounting: marginal costing, standard costing, cost planning and control, budget and budgetary control; Development and planning process; Annual development plan; National budget.
Financial management: objectives, strategy, financing, performance analysis of enterprises, investment appraisal, criteria of investment.

Marketing management: marketing concept, marketing organization; Industrial and consumer selling; Channel decisions; Advertising decisions; new product strategy.
Elements of production planning and control; plat location and layout; Types of production system; Functions of production, planning and control with an overview of different types of manufacturing systems; Factors affiliated with different fields of production including product characteristics and economic analysis.

Forecasting methods and their application; Aggregate planning; Master production scheduling; MRP; Coding and standardization; Capacity planning; Inventory management - ABC analysis; Production scheduling techniques - CPM and PERT; Line balancing; Capacity planning.

Types of information systems and it„s benefits at different management levels; Computers in production planning and control; MRPII and JIT.

Text and Ref Books:

1. Management - Jams A. F. Stoner, R. Edward Freeman, Daniel R. Gilbert,
2. Management - Stephen P. Rubbins, Mar Conlter, Robin Stuart kotze.
3. Operation Management - Chase & Jacob.
4. Managerial Accounting - Garrison, Noreen.
5. Product Design & Development - Ulrich &Eppinger
6. Product Design Methods & Practices - Henry W Stoll

7. Developing New Products with TQM - Charles Gevirtz

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1											✓	
CO2											✓	
CO3										✓		
CO4												✓
CO5											✓	

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO6	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

c. Basic Sciences and Mathematics Courses

Course Code: MAT 4101 Course Title: Differential and Integral Calculus Credits: 3.00

Pre-Requisite:

Rationale of the Course:

To provide an understanding of the basic concepts of differential and integral calculus.

Course Objective:

1. To develop fundamental ideas on limits, continuity and derivative.
2. To gain knowledge on various techniques of finding derivatives.
3. To familiar students 'how integration is associated with differentiation.'
4. To learn several techniques of finding integrals.
5. To prepare students to apply derivatives and integrals in engineering problems.

Course Outcomes (CO):

1. Demonstrate basic concepts of calculus.

2. Derive derivatives and integrals of polynomials, trigonometric functions, logarithmic function, exponential function, hyperbolic function etc.
3. Use derivatives and integrals to expand a function into a series and to find the relationship of Gamma and Beta function.
4. Evaluate motion of moving objects, area of a region and volume of space.
5. Apply derivatives and integrals when modelling engineering problems.

Course Content:

Differential Calculus: Introduction to Differential Calculus: Concepts of Set, Interval and absolute value, Cartesian product set, Relation, Functions, Domain, Co-domain and Range of a function, Types of functions, Graph of functions, Limit, continuity and differentiability, successive differentiation of various types of functions, Leibnitz's theorem, Rolle's theorem, Mean value theorem, expansion in finite and infinite forms, Lagrange's form of remainder, Cauchy's form of remainder, indeterminate form, Partial differentiation, Euler's theorem, tangent and normal, subtangent and subnormal in Cartesian and polar coordinates, maxima and minima of functions of single variables, curvature, asymptotes.

Integral Calculus: Introduction to integral calculus, Indefinite integral, Methods of integration: integration by substitution, integration by parts, standard integrals, integration by successive reduction, Integration by partial fraction, integration of hyperbolic functions, Definite integrals: Evaluation of definite integrals, properties of definite integral and its use, summing series, Walli's product formula, improper integrals, Beta function and Gamma function, double integral and multiple integral, Application of integration: area under a plane curve, area of the region enclosed by two curves, volume of solid of revolution, arc length, area of surfaces.

Reading Material:

1. Integral Calculus & Differential equation, (5th ed), Matin & Chakraborty, Standard Publication (2016).
2. Differential Calculus (5th ed), Matin Chakraborty, Dhaka Standard Publication (2015).
3. Calculus - Haward Anton (9th ed), Stephen Davis, Wiley (2012).
4. A Textbook on Integral Calculus (4th ed), Mohammad, Bhattacharjee & Latif, Dhaka (2010).

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3	✓											
CO4		✓										
CO5		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final

		Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: MAT 4203 Course Title: Ordinary and Partial Differential Equations
Credits: 3.00

Rationale of the Course:

To provide knowledge of formation, classification, order and application of Ordinary Differential Equation and Partially Differential Equation.

Course Objectives:

1. To introduce students with Ordinary Differential Equations and Partial Differential Equations.
2. To Increase understanding on two-dimensional as well as three dimensional measurements.
3. To teach solving techniques of ordinary differential equation.
4. To teach solving techniques of partial differential equations.
5. To prepare students so that they can create and analyse mathematical models for engineering problems such as harmonic oscillator, heat transfer and circuits using differential equations.

Course Outcomes (CO):

1. Demonstrate fundamental concepts of Ordinary Differential Equation and Partial Differential Equation.
2. Solve first and higher order ordinary differential equations by various methods.
3. Solve partial differential equations by various techniques.
4. Select and apply appropriate geometric formula to measure angles, perimeters, surface area and volume.
5. Formulate scientific problems into Differential Equation problems then analyse engineering problems using of solving methods of differential equations.

Course Content:

Ordinary Differential Equations: Introduction to Differential Equations, Formulation of Differential Equations; Degree and order of differential equations, Solution of first order differential equations by various methods, solution of first order higher degree ordinary differential equations, Solution of general linear equations of second and higher orders with constant co-efficient, Solution of homogeneous linear differential equations, solution of differential equations of higher order when dependent and independent variables are absent, solution of differential equations by the method of factorization of operators. Application of ODE.

Partial Differential Equations: Formation of differential equation eliminating constants and functions, four rules for solving simultaneous equations, Lagrange's method of solving PDE of order one. Integral surfaces passing through a given curve, non-linear PDE of order one (complete singular and general integrals), Standard forms, Charpit's method, second order PDE, its nomenclature and classification to canonical, parabolic, elliptic, hyperbolic, solution by separation of variables, linear PDE with constant coefficients. Application of PDE.

Reading Material:

1. Ordinary and Partial Differential Equations, Raisinghania.
2. A Textbook on Co-ordinate Geometry with Vector Analysis, Rahman & Bhattacharjee.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3	✓											
CO4		✓										
CO5		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: MAT 4305 Course Title: Linear Algebra, Coordinate Geometry and Complex Variables Credits: 3.00

Prerequisite: *Ordinary and Partial Differential Equations (MAT 4203)*

Rationale of the Course:

To provide knowledge of matrices and linear algebra which ultimately increase knowledge to formulate and solve engineering problems. It also provides knowledge of problem-solving regarding geometry.

Course Objectives:

1. Familiar students with basic ideas on Linear Algebra and coordinate geometry.
2. Display linear transformation and inner product spaces.
3. Clarify eigenvalues and eigenvectors with example.
4. Show application of linear algebra in engineering problems.
5. Demonstrate understanding of the basic concepts underlying complex number system.

Course Outcomes (CO):

1. Demonstrate basic ideas of matrix, linear algebra, coordinate geometry and complex numbers.
2. Demonstrate and apply linear transformation

3. Evaluate eigenvalues and eigenvectors and recognize characteristic matrix, characteristic polynomial, characteristic equation and Diagonalization.
Apply knowledge of linear algebra to formulate and solve engineering problems

Course Content:

Linear Algebra: Matrices and Determinants: Definition of matrices, order of a matrix, Row matrix, Column matrix, square matrix, Operations of matrices, determinant of square matrix, Minors and cofactors, Transpose of a matrix, Symmetric and skew-symmetric matrix, Cofactor and adjoint of a matrix, inverse of a matrix, Echelon Matrix, Canonical matrix. System of linear equations: Linear equations, Solution of a system of linear equations (Cramer's rule, Gauss elimination method, Gauss-Jordan elimination method, inverse matrix method), Application of system of linear equations. Vectors in Euclidean n-space, Basic properties of the vectors, Basic properties of dot product, Orthogonal vectors, Norm or length, Angle between two vectors, Mathematical problems. Vector spaces: Introduction, Definition and examples of vector spaces and subspaces, Linear combination, linear dependence and independence, Basis and Dimension, Rank of matrix. Linear transformation: Definition of linear transformation, properties of linear transformation, Kernel and image of linear transformation, Singular and non-singular linear transformation, Matrix representation of a linear transformation, Rank and Nullity of a linear transformation. Inner product spaces: Euclidean and Unitary spaces, orthogonality, Gram-Schmidt process and QR-Decomposition. Eigen values and Eigen vectors: Matrix polynomials, Polynomial of a linear operator, Eigenvalues and eigenvectors of a square matrix, Eigenvalues and eigenvectors of a linear operator, Eigenvalues and eigenvectors of linear transformation, Characteristic matrix, Characteristic polynomial and Characteristic equation, Diagonalization, Cayley-Hamilton theorem, Application of linear algebra in engineering problems.

Coordinate Geometry: Two-dimensional coordinate geometry: Changes of axes: Transformation of co-ordinates, simplification of equation of curves, Conic section (pair of straight line, system of circle, parabola, Ellipse, Hyperbola). Three-dimensional coordinate geometry: System of coordinate, distance between two points, section formula, projections, direction cosines, equations of planes and lines.

Complex Variables: complex number system, general function of a complex variable, limits and continuity of a function of a complex variable and related theorems, complex differentiation, the Cauchy-Riemann equation, mapping by elementary functions, line integral of a complex function, Cauchy's Integral Theorem and formula, Liouville's theorem, Taylor's theorem, Laurent's theorem, Singular points, Residue, Cauchy's Residue Theorem, evaluation of Residues, Contour integration, conformal mapping.

Reading Material:

1. Elementary Linear Algebra, Anton, H. & Rorres, C.
2. A textbook on Co-ordinate Geometry with Vector Analysis, Rahman & Bhattacharjee
3. Elementary Linear Algebra, Prof. Abdur Rahman

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: MAT 4407 Course Title: Fourier and Laplace Transform Credits: 3.00

Pre-requisite: *Ordinary and Partial Differential Equations* (MAT-4203)

Course Objectives:

a. To discuss the complex number system, different types of complex functions, analytic properties of complex numbers and theorems in complex analysis to carryout various mathematical operations in complex plane, roots of a complex equation.

b. To discuss limits, continuity, differentiability, contour integrals, analytic functions and harmonic functions.
c. To learn about Cauchy–Riemann equations in the Cartesian and polar coordinates, Cauchy's integral formula, Cauchy–Goursat theorem, convergence of sequence and series, Taylor series, Laurent's series.
d. To apply Integral transforms with a special focus on Laplace integral transform and Fourier transform.

Course Outcomes:

1. Describe geometrical interpretation of complex inequalities and find the 'n' th root of complex number.
2. Apply the methods of complex analysis to evaluate definite integrals and infinite series.
3. Explain the concept of integral transforms, e.g., Laplace, Fourier transforms and the related inverse transforms by using the following Partial fractions method, Convolution theorems and apply these transformations for engineering problems.

Course Content:

Laplace Transforms: Laplace transforms of some elementary functions, sufficient conditions for existence of Laplace Transforms, inverse Laplace transforms, Laplace transforms of derivatives, the unit step function, periodic function, some special theorems on Laplace transforms, partial fractions, solutions of differential equations.

Fourier Transforms: Real and complex form of Fourier series, finite transforms, Fourier Integral, Fourier transforms and their uses in solving boundary value problems of wave equations.

Reading Material:

1. Theory and problems of Complex Variables, Murry R. Spiegel
2. Mathematical Methods (volume Two), Professor MD. Abdur Rahman
3. Function of a Complex variable, M. L. Khanna

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: MAT 4509 Course Title: Probability & Statistics Credits: 3.00

Pre-requisite: MAT 4101 (Differential and Integral Calculus)

Rationale of the Course:

Engineering students always require knowledge of probability and statistical to investigate the feasibility of engineering equipment. To provide the knowledge of probability and statistics this course is designed for engineering students.

Course Objectives:

1. Teach common statistical techniques which are useful for engineering students.
2. Disseminate knowledge to display, summarize and analyse numerical data.
3. Explain fundamental ideas of probability and probability distributions.
4. Clarify how to test a hypothesis for large and small sample.
5. Discuss how statistical techniques and probability are used in statistical inference.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Identify statistical techniques for summarizing and analysing data.
2. Calculate statistics and parameters such as mean, median from data sets.
3. Apply statistics and parameters for social and business decision.
4. Analyse engineering problems in statistical context.

Course Contents:

Basic Statistics: Definition of statistics, Population and Sample, Parameter and Statistic, Data collection, Data Presentation, Measures of Central Tendency: Arithmetic Mean, median, mode, Geometric mean and harmonic mean. Measures of dispersion: Range, Mean deviation, Quartile deviation, Variance and Standard deviation, Moments, Skewness and Kurtosis, Correlation and Regression.

Probability: Set and Probability theory, Sampling Theory, Random Variables, Probability Distribution: Discrete Probability Distributions (Binomial, Poisson and negative binomial distribution), Continuous Probability distribution (Normal distribution), Sampling and Sampling Distribution, Estimation of parameters, Test of Hypothesis.

Text and Ref Books:

1. An Introduction to Statistics and probability M. Nurul Islam
2. Probability and Statistics for Eng. and Scientists Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye
3. Schaum's Outline Series: Theory and Problems of Statistics Spiegel, M. R. & Stephens,

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3		✓										
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: MAT 4509 Course Title: Probability & Statistics Credits: 3.00

Pre-requisite: MAT 4101 (Differential and Integral Calculus)

Rationale of the Course:

Engineering students always require knowledge of probability and statistical to investigate the feasibility of engineering equipment. To provide the knowledge of probability and statistics this course is designed for engineering students.

Course Objectives:

1. Teach common statistical techniques which are useful for engineering students.
2. Disseminate knowledge to display, summarize and analyse numerical data.
3. Explain fundamental ideas of probability and probability distributions.
4. Clarify how to test a hypothesis for large and small sample.
5. Discuss how statistical techniques and probability are used in statistical inference.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Identify statistical techniques for summarizing and analysing data.
2. Calculate statistics and parameters such as mean, median from data sets.
3. Apply statistics and parameters for social and business decision.
4. Analyse engineering problems in statistical context.

Course Contents:

Basic Statistics: Definition of statistics, Population and Sample, Parameter and Statistic, Data collection, Data Presentation, Measures of Central Tendency: Arithmetic Mean, median, mode, Geometric mean and harmonic mean. Measures of dispersion: Range, Mean deviation, Quartile deviation, Variance and Standard deviation, Moments, Skewness and Kurtosis, Correlation and Regression.

Probability: Set and Probability theory, Sampling Theory, Random Variables, Probability Distribution: Discrete Probability Distributions (Binomial, Poisson and negative binomial distribution), Continuous Probability distribution (Normal distribution), Sampling and Sampling Distribution, Estimation of parameters, Test of Hypothesis.

Text and Ref Books:

1. An Introduction to Statistics and probability M. Nurul Islam
2. Probability and Statistics for Eng. and Scientists Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye
3. Schaum's Outline Series: Theory and Problems of Statistics Spiegel, M. R. & Stephens,

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3			✓									
CO4				✓								

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: PHY 4101 **Course Title: Physics** **Credits: 3.00**

Rationale of the Course:

To perform experiments to verify practically the theories and concepts learned and understand the

difference between theory and application.

Course Objective:

1. To help learning how to be patient and careful while taking observation.
2. To provide knowledge on how to develop the ability of the students to conduct, observe, analyse and report an experiment.
3. To help to deal with physical models and formulas mathematically.

Course Content:

Waves and Oscillation: Differential equation of Simple harmonic oscillator, total energy and average energy, Combination of Simple harmonic oscillations, Lissajous figures; spring mass system, Calculation of time period of torsional pendulum; damped oscillations, determination of damping co-efficient forced oscillation, resonance, two body oscillations, reduced mass , differential equation of a progressive wave, power and intensity of wave motion, stationary wave, phase velocity and group velocity, Architectural acoustics, reverberation and Sabine's formula

Geometrical optics: *Combination of lenses:* equivalent lens and equivalent focal length, cardinal points of a lens, power of a lens. Defects of images: spherical aberration. Astigmatism, coma, distortion, curvature and chromatic aberration. Optical instruments, Compound microscope, polarizing microscope, resolving power microscope, camera and photographic techniques.

Wave mechanics: Principle of statistical physics: Probabilities, classical statistics, quantum statistics: Bose-Einstein statistics, Fermi-Dirac statistics and their applications. Fundamental postulates of wave mechanics, time dependent Schrodinger's equation, steady state Schrodinger's equation for one electron atom and its solution.

Thermal Physics: Heat and work- the first law of thermodynamics and its applications; Kinetic Theory of gases- Kinetic interpretation of temperature, specific heats of ideal gases, equipartition of energy, mean free path, Maxwell's distribution of molecular speeds, reversible and irreversible processes, Carnot's cycle, second law thermodynamics, Carnot's theorem, entropy, Thermodynamic functions, Maxwell relations, Clausius and Clapeyron equation.

Teaching-learning and Assessment Strategy:

Lectures, class performances, assignments, class tests, final exam.

Reading Material:

1. A Textbook of Optics, Brijlal and Subramannyam
2. A Textbook of Sound, Brijlal and Subramannyam
3. Waves and oscillation, Brijlal and Subramannyam
4. Physics part-I, Resnick and Haliday
5. Physics part-II, Resnick and Haliday
6. Fundamentals of Physics, Haliday, Resnick and Walker
7. Concept of Modern Physics, Arther Beiser; McGraw Hill
8. Perspective of Modern Physics, Arther Beiser; McGraw Hill

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											

CO2		✓										
CO3			✓									
CO4									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: PHY 4102 Course Title: Physics Sessional Credits: 1.50

Pre-requisite: None

Rationale of the Course:

To perform experiments to verify practically the theories and concepts learned and understand the difference between theory and application.

Course Objective:

1. To help learning how to be patient and careful while taking observation.
2. To provide knowledge on how to develop the ability of the students to conduct, observe, analyse and report an experiment.
3. To help to deal with physical models and formulas mathematically.

Course Outcomes (CO):

1. Demonstrate an ability to make physical measurements and understand the limits of precision in measurements.
2. Demonstrate the ability to use experimental statistics to determine the precision of a series of measurements.
3. Compare to use various methods of circuit analysis both for DC and AC networks and their solution methods.
4. Appraise the characteristics of different electronic devices.

Course Content:

Determination of specific resistance of a wire using a meter bridge, determination of the resistance per unit length of meter bridge wire, determination of mechanical equivalent (J) of heat by the electrical method, determination of electrochemical equivalent (ECE) of copper by the cooper Voltammeter, determination of resistance of a galvanometer by half deflection method, determination of high resistance by the method of deflection, determination of focal length of a concave lens by auxiliary lens method, determination of radius of a curvature of a Plano convex lens by Newton's ring method, determination of the refractive index of the material of a prism by spectrometer, determination of the specific rotation of sugar solution by Polarimeter, determination of frequency of a tuning fork by the Melda's experiment, determination of the spring constant and the effective mass of a loaded spring and hence to calculate the rigidity modulus of the spring, determination of the value of g acceleration due to gravity by means of a compound pendulum, determination of Young's modulus of a bar by bending method, determination of the modulus

of rigidity of a wire by statistical method.

Reading Material:

1. Fundamentals of Physics (10th ed), Halliday, Resnick and Walker, Wiley (2013).
2. A Textbook of Optics, Brijlal and N. Subrahmanyam, S. Chand (2006).
3. Physics for Engineers, Part-1 & Part-2 (4th ed), Dr. Giasuddin Ahmad, Hafiz Book Centre (2000).
4. Waves and oscillations (2nd ed), Brijlal and Subramannyam, Vikas Publishing House Pvt Ltd (1994).
5. Physics part-I & II (2nd ed), Resnick and Halliday, John Wiley & Sons, Inc. (1968).
6. Fundamental of Optics (3rd ed), Jenkine and White, McGraw Hill (1957).

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3			✓									
CO4									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: CHM 4201 **Course Title: Chemistry** **Credits: 3.00**

Pre-requisite: None

Rationale of the Course:

To provide a broad foundation in chemistry that stresses scientific reasoning and analytical problem solving with a molecular perspective.

Course Content:

Atomic structure, quantum numbers, electronic configuration, periodic table; Properties and uses of noble gases; Different types of chemical bonds and their properties; chemical bonding and concepts of conductor, insulator, semi-conductor; Molecular structure of compounds: Selective organic reactions; introduction to computational chemistry, Different types of solutions and their compositions; Phase rule. Phase diagram of mono component system; Properties of dilute solution; Thermo chemistry, chemical kinetics, chemical equilibrium; Electrical properties of Solution. pH and buffer solutions, Eletro-chemical cell reactions.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Identify the basic knowledge of different areas of chemistry.
2. Interpret the application of computer science in chemistry.
3. Apply the knowledge of chemistry for studying wide range of subjects such as materials science.

Course Objective:

1. To learn laboratory skills to design, conduct and interpret chemical research.
2. To expose the students to a breadth of experimental techniques using modern instrumentation.

Text and Ref Books:

1. Principles of Physical Chemistry – Haque & Nawab
2. Physical Chemistry - P. W. Atkins; Oxford University Press.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: CHM 4202 Course Title: Chemistry Sessional Credits: 0.75

Rationale of the Course:

To learn the basic concepts of inorganic, organic and physical chemistry.

Course Contents:

Volumetric analysis: Acid-base titration, Oxidation-reduction titration: Determination of Cu, Fe and Ca content volumetrically.

Course Outcomes (CO):

Upon completion of all sessional, the students will be able to:

1. Define the different parameters regarding acid and base neutralization, titration and quantitative analysis of metals etc. and others key words like primary standard substances, secondary standard substances, molarity, normality, indicator, equivalent weights and so on.
2. Explain the different phenomena regarding iodometric and iodometric method, complexometric titration etc.
3. Estimate zinc, ferrous content in water sample by using various titrimetric methods.
4. Summarize a report of any project work and apply in real life.

Text and Ref Books:

1. Principles of Physical Chemistry – Haque&Nawab; Students' Publications.
2. Fundamentals of Physical Chemistry- Samuel H. Maron& Jerome B. Lando; MacMillan
3. Publishing Co., Inc., Newyork.
4. Physical Chemistry P. W. Atkins; Oxford University Press.
5. Essentials of Physical Chemistry- B.S. Bahl& G.D. Tuli; S. Chand and Company Ltd.
6. General Chemistry- Ebbing; Houghton Mifflin Company.
7. Organic Chemistry – M. Ahmed &JabbarMian; Mrs. Sufia Ahmed and Mrs. Jahan-Ara Begum.
8. Organic Chemistry- IL Finar; ELBS Longman Group Ltd.
9. Organic Chemistry- Morison & Boyd; Prentice Hall of India.
10. Introduction to Modern Inorganic Chemistry – S.Z. Haider; Friend's International.
11. Modern Inorganic Chemistry – R. D. Madan; S. Chand and Company Ltd.
12. Advanced Inorganic Chemistry – F. Albert Cotton &Geoffey Wilkinson; John Wiley & Sons.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3									✓			

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

d. Humanities and Social Sciences (HASS) Courses

Course Code: HUM 4101 Course Title: Communicative English Credits: 3.00

Pre-requisite: None

Rationale of the Course:

To provide fundamental knowledge on how to improve spoken English skills and to enable me to write more effectively in English.

Course Objective:

1. Help students to improve skills as a writer, critical reader and thinker.
2. Enables students to learn the structure and style of effective sentences, paragraphs, and essays.
3. Enables students to focus on diction and spelling, punctuation and mechanics, and functional grammar in direct relation to their own writing.

Course Outcomes (CO): Upon completion of the course, the students will be able to:

1. Recognize and get a clear idea about how to speak, understand and speak English quickly and smartly.
2. Organise different writings within the shortest possible time to present their ideas and opinions.
3. Find out the main points of any long article within a very limited time.
4. Interpret the techniques of any effective writing.
5. Overcome language barrier.

Course Content:

Language: Introduction to Language, Methodologies/Approaches of English Teaching-Learning Process, Introduction to Phonetics, IPA Transcription and variation of accents. Grammar: Types of Sentences, Introduction to Tense, Use of Main Verbs, Auxiliaries, Modals & Semi modals, Causatives, Conditionals, Active / Passive Sentences, Forming Questions, Problems with Adverbs and Adverb related structures, Adjectives, Conjunction, Prepositions. Reading: Skimming, and Scanning, Reading Comprehension, Selected Short Stories. Writing: Writing Strategies. General Correspondence & Communication/Tech Writing: Introduction to General Correspondence, Formal Communication, Report Writing; How to write a curriculum vitae.

Reading Material:

1. English Grammar in Use (4th ed), Raymond & Murphy, Cambridge University Press (2012).
2. Introduction to Linguistics, Prof Dr. Maniruzzaman.
3. A Guide to Correct Speech (5th ed), S M Amanullah, Albatross Publication (2007).
4. Oxford Advanced Learners' Dictionary (7th ed), Oxford University Press (2007).
5. Prose of Our Time, Ahsanul Hoque (2nd ed), Serajul Islam Chowdhury and M Shamsuddoha.
6. From Paragraph to Essay, Maurice Imhoof and Herman Hudson, Longman, London (1976).
7. Headway Series Advanced Level (2 parts with CDs) (4^{th}ed), Oxford University Press Ltd.
8. Business correspondence and report writing, R. C. Sharma & Krishna Mohan.
9. Advanced learners Degree in general English, Chowdhury and Hossain.
10. The most common mistakes in English usage, Thomas Elliott Berry.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1												✓
CO2										✓		
CO3												✓
CO4										✓		
CO5										✓		

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: HUM 4102 Course Title: Communicative English Sessional Credits: 1.5

Pre-requisite: None

Rationale of the Course:

To help to improve spoken English skills and to enable to communicate more effectively in English.

Course Objective:

1. Provide confidence among students to read, write and speak English Language efficiently.
2. Help pupils learn how to present in groups and to think coherently.
3. Provide support for developing the English language skills (listening, speaking, reading and writing) that are necessary to be successful on the IELTS test.

Course Outcomes (CO):
Upon completion of the course, the students will be able to:

1. Determine and get a clear idea about how to speak, understand and speak English quickly and smartly.
2. Organise them within the shortest possible time to present their ideas and opinions.
3. Find out the main points of any long article within a very limited time.
4. Recognize the techniques of any effective writing.
5. Overcome language barrier.

Course Content:

Introduction to Language: Introducing basic skills of language; Phonetics, English Vowel and consonant sounds, Difference between different accents, Self-Introduction, how a speaker should introduce himself; Situational talks / dialogues; Speaking: IELTS speaking - Part 1, 2 & 3; Group discussion, taking participation in any discussion and drawing conclusion and giving recommendation; Brain storming: Principles of brain storming. How to think logically; Facing any problem, trying to find out possible solutions, drawing conclusion and giving recommendation; Individual / Group presentation, good presentation skills; Skimming, Scanning & Analytical Ability, techniques of skimming, scanning and generating ideas through purpose reading. Reading: Introducing IELTS academic reading comprehension, Listening and understanding, note taking and answering questions; Listening: Introducing IELTS / TOEFL listening section Academic Writing: Introducing IELTS academic writing, Public Speaking: Basic elements of good presentation or public speaking, some tips for good speech.

Tutorial Discussion – On a given topic to test the proper use of phonetics, pronunciation grammar, logic and confidence; Public Speaking – Demonstration by teacher for a short specific period, speaking by students (each student minimum twice) on different but easy given topic, well in advance as per a schedule maximum for 3 to 4 minutes for each student; Extempore – Minimum two presentations by each student for a duration of maximum 3 to 4 minutes; Debriefing on public speaking and extempore presentation ; Presentation – On a given professional topic or on a given research paper using power point followed by question and answer session. Group presentation or different given topics by the students using power point.

Reading Material:

1. Introduction to Linguistics, Prof Dr. Maniruzzaman.
2. A Guide to Correct Speech (5th ed), S M Amanullah, Albatross Publication (2007).
3. Oxford Advanced Learners' Dictionary (7th ed), Oxford University Press (2007).
4. English Grammar in Use (4th ed), Raymond & Murphy, Cambridge University Press (2012).
5. From Paragraph to Essay, Maurice Imhoof and Herman Hudson, Longman, London (1976).
6. Headway Series, Advanced Level (2 parts with CDs) (4thed), Oxford University Press Ltd.
7. IELTS and TOEFL Practice Book, Cambridge University Press.
8. English for Technical Students, David Bonamy, Longman Publisher.
9. Business Correspondence and Report Writing, R. C. Sharma & Krishna Mohan.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1												✓
CO2										✓		
CO3												✓
CO4										✓		
CO5										✓		

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Project
CO5	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: HUM 4203 Course Title: Bangladesh Studies and Society Credits: 3.00

Pre-requisite: None
Rationale of the Course:

This course has been designed to help the students in obtaining a comprehensive idea about the history, culture and heritage of Bangladesh. It will introduce students to the economy, society, politics, diplomacy and foreign policy of Bangladesh. Students will learn about the challenges and potentials of Bangladesh in shaping its peaceful and sustainable future. It will also assist the students in assessing the roles and contribution of Bangladesh in the regional and international bodies which are dedicated to establishing world peace.

Systematic study of social behaviour and human groups. It focuses on the influence of social relationships upon people's attitudes and behaviour and on how societies are established and changed. This course provides students with both methodologies and knowledge of the study of critical social issues ranging in scope from family to global.

Course Objective:

1. To enrich knowledge with brief history, culture and heritage of Bangladesh.
2. To provide in-depth knowledge on the major political events that shaped Bangladesh as an independent sovereign state.
3. To improve understanding on political, economic and social development of Bangladesh.
4. Introduce students with the rich history, culture and heritage of Bangladesh.
5. Providing them in-depth knowledge on the major political events that shaped Bangladesh as an independent sovereign state.

Course Outcomes (CO):

1. To think critically and comprehensively about foreign policy of Bangladesh, its relationship with other countries and its important roles in the international organizations like UN, Commonwealth and SAARC etc.
2. Correctly identify causes of critical social issues through a systematic study of social behaviour and social change.
3. Demonstrate comprehension of roles and functions of various social institutions and relationships among them.
4. Demonstrate understanding of several sociological theories and apply them to explain social phenomena or situations.
5. Demonstrate interest in taking part in social activities.

Course Content:

Ancient period and Muslim period of Bengal, British period, Pakistan period: an overview (1952-1971), The problem of national integration under Ayyub regime, Elite in crisis during Pakistan rule, nation-building in the new state, The ideals and philosophy of constitution making of Bangladesh, Study on the coup and assassination of Bangabandhu Sheikh Mujibur Rahman, Philosophy and fundamental changes of

Zia regime, Constitutional amendments of Bangladesh, Corruption and good governance in Bangladesh, Issues of governance of Bangladesh, Bangladesh's economy, Ideas on political and ethnic conflict in Bangladesh, Geographical setting of Bangladesh, environmental challenges of Bangladesh, Bangladesh foreign policy: realities and challenges, Foreign policy-decision-making process in Bangladesh, Bangladesh-soviet union relations.

Nature, scope and perspectives of sociology; stages of social research and research methods; culture and civilization; socialization and personality development; globalization; media and individual; social organization and social problem; social stratification; industrial revolution, capitalism and socialism; work and economic life; environment and human activities; climate change and global risk; population and human society; urbanization and city development; social change and technology.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1						✓						
CO2		✓										
CO3										✓		
CO4						✓						
CO5						✓						

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: HUM 4413 Course Title: Engineering Economics Credits: 3.00

Pre-requisite: None

Rationale of the Course: To learn the basic principle of Microeconomics and Macroeconomics and their application Engineering.

Course Objective:

1. Help students demonstrate their knowledge of the fundamental and technical concepts of economics.
2. Familiarise students with comparative statics and optimization problems in microeconomics and macroeconomics using matrix algebra and calculus

Course Outcomes (CO):

1. Explain the function of market.
2. Apply the concept of equilibrium to both microeconomics and macroeconomics.
3. Identify key macroeconomic indicators and measures of economics change, growth, and development.
4. Identify and explain major types of market failures.

Course Content:

Microeconomics: Definition of economics; Fundamentals of economics; Market and government in a modern economy; Basic elements of supply and demand; Choice and utility; indifference curve technique; Analysis of cost; Short run long run theory of production; Analysis of Market;
Optimization; Theory of distribution.

Macroeconomics: key concept of macroeconomics; saving, consumption, investment; National income analysis; Inflation, Unemployment; Fiscal and monetary policy. Development: Theories of developments; Economic problem of developing countries; Planning in Bangladesh.

Textbooks:

1. Economics by P. A. Samuelson and W. D. Nordhaus (7t h Edition)
2. An Introduction to Positive Economics By R G Lipsey (7t h Edition)
3. Introductory Economics by G. F. Stanlake and S. J. Grant (6t h Edition)

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1											✓	
CO2											✓	
CO3		✓										
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

e. Departmental Elective Courses

Course Code: AVE 4721 Course Title: Avionics Software Design Credits: 3.00

Pre-requisite: None

Rationale of the Course:
To learn the best practice and principle of avionics software design and development for real time and safety critical systems in the aviation and aerospace domain.

Course Objective:

1. Enable the students to explain the basic concepts of avionics software systems.
2. Facilitate the students to know algorithm for designing software
3. Help students to design different software for particular applications.

Course Outcomes (CO):

Upon completion of the course, the students will be able to

1. Apply the software development process in the avionics domain.
2. Evaluate the user requirements to design the system model.
3. Develop system prototypes by using software prototyping tool and evaluate
4. Produce a complex and safe avionics software using modern software techniques

Course Contents:
Introduction to software engineering in avionics perspective, Software development life cycle, Agile, Scrum, Overview of DO-178C including Supplements, System Aspects Relating to Software Development Software Life Cycle (SDLC)

Software Planning Process, Software Development Processes (with Exercise), Software Verification Process, Software Configuration Management Process, software project management

Text and Reference book:

1. SOFTWARE ENGINEERING: A PRACTITIONER'S APPROACH, / Roger S. Pressman, Y Mcgraw-Hil
2. DO-178C, Software Considerations in Airborne Systems and Equipment Certification.
3. NASA Software Safety Guidebook,2004

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1		✓										
CO2			✓									
CO3					✓							
CO4					✓							

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4723 Course Title: Autonomous Systems, Guidance & Control Credits: 3.00

Pre-requisite: Control Systems (AVE 4705)

Rationale of the Course:
This program is designed to train a world-class cohort of students in the theory and practice of a new generation of autonomous system applications including aerospace, missiles etc.

Course Objective:

1. Enable students to explain the basic concepts of network control systems.
2. Provide knowledge about the algorithm based on carrier phase
3. Empower students to design different controllers for particular applications.

Course Outcomes (CO):

Upon completion of the course, the students will be able to

1. Develop path planning systems for autonomous vehicle guidance;
2. Design and implement obstacle avoidance methods within an autonomous guidance regime;
3. Implement trajectory optimisation and smoothing for both discrete and continuous paths.

Course Contents:
Application Background and Significance: Wireless mobile Ad Hoc network, Tactical data link, Dynamic route planning, Dynamic target assignment, Formation configuration control, Network control system.
Basic of Co-operative Guidance and Control: Basic concept and definition, Typical flight area of winged missiles autonomous formation, Architecture of CGCS, Information acquisition system, Decision and management system, Flight control system of MAF, Support network system.

Information Acquisition System of MAF: Composition of IAS, Time-space registration for IAS, Relative navigation of MAF, Relative velocity algorithm based on carrier phase

Decision and Management System of MAF: Composition of DMS, Basic principle of MAF, Combat Effectiveness Indication of MADF, Task planning and dynamic weapon target assignment, Handover guidance and co-operative guidance.
Flight Control System of MAF: Compositions of FCSM, Formation generation and guidance, Loose formation control, Dense formation control.
Member Flight Control System: Configuration design of MFCS, Overlord Controller design of the missile, Network flight controller design of the missile, Support network system of MAF.

Text and Reference book:

1. Cooperative Guidance & Control of Missiles Autonomous Formation- Sentang Wu, Springer
2. Design of Guidance and Control System for Tactical Missiles-Q.I. Zaikang and Lin Defu

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1		✓										
CO2			✓									
CO3		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4725 Course Title: Robotics and Mechatronics Credits: 3.00

Pre-requisite:

Rationale of the Course:
The aim of this lecture is to expose students to the fundamentals of Mechatronics and Robotics system. Over the course of these lectures, topics will include how to interface a computer with the real world, different types of sensors and their use, different types of actuators and their use, and forward and inverse kinematics of simple two link robotic manipulators.

Course Objective:

1. Enable the student to understand the modern Robotics and Mechatronics components;
2. Present the underlying principles and alternatives for mechatronics systems design;
3. Provide the student with the opportunity for hands-on experience with the related components of the technology for diverse domains of application.
4. Develop the student's ability to evaluate appropriate technology and create and devise realistic industrial systems.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Identify, select, and integrate mechatronic components to meet product requirements
2. Develop kinematic, dynamic and control models for robots
3. Use commercial software tools for modelling and simulation of mechatronic systems
4. Design, analyse, and optimise mechatronic products

5. Write technical reports and present engineering design solutions efficiently

Course Contents:

Robot components (sensors, actuators, and end effectors, and their selection criteria)
Basic categories of Robots (serial and parallel manipulators, mobile robots); mobility/constraint analysis; workspace analysis; rigid body kinematics (homogeneous transformation, angle and axis of rotation, Euler angles, cylindrical and spherical coordinates);
Manipulator kinematics and motion trajectories (displacement and velocity analyses, differential relations, Jacobian matrix); non-redundant and redundant sensing/actuation of manipulators; manipulator statics (force and stiffness); singularities; and manipulator dynamics.
Mechatronics Introduction: Fundamentals of Mechatronics, Examples of different Mechatronic systems, Scope and applications of Mechatronics.
Sensors and Transducers: Introduction to sensors and transducers, classification of sensors, sensor terminologies, sensor characteristics, different types of mechanical and electrical sensors; Proximity sensors.
Signal Conversion and Processing: Basic principles of potentiometer, Wheatstone bridge, op-amps, signal conditioning and data acquisition.
System Modeling and Control: Introduction to signals and systems, Modeling of Mechanical, Electrical, Fluid and Thermal systems, Linearization of nonlinear systems, Rotational-translational systems, Electro-Mechanical systems and Hydraulic-Mechanical System. Basic components of Control system, Types of control system, System representation, System responses, Time constant, Measurement of system performance, Illustrative examples.
Actuation Systems: Introducing the Linear and rotary actuators, AC and DC motors, Solenoids, Stepper motor, Fluid power actuators and Smart actuators. Recent trends in Mechatronic systems.
Introduction to Instrumentation: Characteristics of instruments, Errors, Propagation of errors, Errors types, Reduction of errors.

Text and Ref Books:

1. Paul, R.P., Robot Manipulators - Mathematics, Programming and Control. MIT Press.
2. Craig, J.J., Introduction to Robotics: Mechanism and Control. Addison-Wesley Publishing Co.
3. Fundamentals of Mechatronics," Musa Jouaneh, Cengage Learning, 2011.
4. Mechatronics," Sabri Cetinkunt, Wiley, 2006.
5. Mechatronics: a Foundation Course", Clarence de Silva, CRC Press, 2010.
6. Mechatronics Systems Fundamentals", Rolf Isermann, Springer,2005.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1		✓										
CO2			✓									
CO3					✓							
CO4			✓									
CO5										✓		

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4729 Course Title: Remote sensing and Digital Image Processing
Credits: 3.00

Pre-requisite: None

Rationale of the Course:
To learn the technologies of Remote sensing and Digital Image Processing used in many real life applications.

Course Objective:

1. Present the basic principles of remote sensing.
2. Help the students to understand the different remote sensing platforms.
3. Enable the students interpret, process and integrate remote sensing digital Images.

Course Outcomes (CO):

Upon completion of the course, the students will be able to

1. To know the basic principles of remote sensing.
2. Understand the remote sensing platform.
3. Able to interpret images acquired by remote sensing platforms.
4. Apply digital image processing technique to find insights in image data.

Course Contents:

Introduction to Remote Sensing: Radiometry, Electromagnetic Spectrum, Radiant and Spectral quantities,

Remote sensing systems, ground-truthing procedures, air photo interpretation, satellite image processing and classification, radar imagery, applications of remotely sensed imagery, remote sensing, and geographic information systems.

Text and Reference book:

1. Remote sensing and image interpretation by Lillesand, Thomas M.; Kiefer, Ralph W.; Chipman, Jonathan W.
2. Remote Sensing: Principles, Interpretation, and Applications, by Sabins & Ellis. 4th edition, 2020.

3. Microwave radar and radiometric remote sensing by Fawwaz T. Ulaby; David G. Long; William J. Blackwell; Charles Elachi; Adrian K Fung; Chris Ruf; Kamal Sarabandi; Howard A. Zebker; Jakob Van Zyl 2014.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3		✓										
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4731 Course Title: Avionics Human Factors Engineering Credits: 3.00

Pre-requisite: None

Rationale of the Course:
To learn the details of human performance and human errors.

Course Objective:

1. Present various incidents and accidents happened previously for human performance and limitations.
2. To Explain the causes that may lead to hazard.
3. To Evaluate this knowledge of human performance and its limitations in their real-life work field to avoid any kind of undesirable incident.
4. To Know about recognizing and avoiding hazards, dealing with emergencies.

Course Outcomes (CO):

Upon completion of the course, the students will be able to

1. Know about various incidents and accidents that happened previously for human performance and limitations.
2. Explain the causes that may lead to hazard.

3. Evaluate this knowledge of human performance and its limitations in their real life work field to avoid any kind of undesirable incident.
4. Know about recognizing and avoiding hazards, dealing with emergencies.

Course Contents:

Fundamental Human Factors Concept: Understand the term human factor, the need to take human factors into account, incidents attributable to human factors/human error, human factors applications in aviation operations.

Human performance and limitation: Vision, Hearing, Information and protection, memory, claustrophobia and physical access. Social Psychology & Responsibilities: Individual and group, motivation and de-motivation, peer pressure, culture issues, team working, management, supervision and leadership.

Factors affecting performance: Fitness/health, Stress: domestic and works related, time pressure and deadline, workload, overload, sleep and fatigue, shift work, alcohol, medication, drug use, use of psychoactive, substances, restriction on exercising privileges of license/ authorization under influence psychoactive substance.

Physical Environment, Management and Organization: Noise and fumes, illumination, climate and temperature, motion and vibration, working environment, management 's contribution to safety, allocation of resources, safe and unsafe organization. Takes: physical work, repetitive tasks, visual inspection, and complex systems.

Communication: Within and between teams, work logging and recording, keeping up to date currency, dissemination of information, terms and organizational issues in aircraft maintenance.

Human Error: Error models including the SHEL and Reason models, and theories, Murphy's law, human error in aircraft maintenance inspection including selected case studies, implications of error, error prevention considerations and strategies, avoiding and managing errors. Hazards in workplace: Recognizing and avoiding hazards, dealing with emergencies.

Text and Reference book:

1. Human Factors for Aviation Maintenance - An EASA Part 66/147 approved manual on Human Factors; Aircraft Technical Book Co.
2. Human Performance and Limitations - Trevor Thom
3. Human Performance and Limitations in Aviation – R. D. Campbell, Michael Bagshaw

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3						✓						
CO4	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4733 Course Title: RF Integrated Circuits Credits: 3.00

Prerequisites: Electronic Circuits II (AVE 4401)

Rationale of the Course:

The course aims to expose students to the RF Integrated Circuit design techniques from the system and functional module perspective and appreciate the many advances in the IC technology that has facilitated its advancement. But an essential aspect of the course is for the student to understand the trade-offs in design specifications like noise, power, bandwidth etc.

Course Objective:

1. Present the concepts of enabling RF Capability on a chip
2. Enable students to understand the design of modern RF and wireless communication integrated circuits.
3. Enable students to understand the relationship and limitations of circuit topology and device characteristics to achieve competitive specifications.

Course outcomes:

After the successful completion of the course, the student will be able to:

1. Understand a range of technologies used for RF circuit implementation on chip.
2. Understand the principle of operations of Transceivers Systems, noise, linearity, etc.
3. Analyse and design low noise amplifiers and mixers
4. Analyse and design RF power amplifiers and voltage-controlled oscillator
5. Analyse and design PLL

Course Contents

Introduction to RF communication systems: non-linearity, cascaded stages; Noise models, device noise, Noise figure calculations, Friis eqn.; Dynamic range; RF passive IC components.

Revision of transmission line: Principles; Smith chart; S-parameter; matching networks; LNA: intrinsic 2 port MOSFET noise parameters, power match vs noise match, power constrained optimization, design examples, power gain, stability.

Mixers fundamentals: Nonlinear system based multiplier, potentiometric mixers, diode mixers, noise

analysis of mixers; Oscillators: voltage controlled, phase noise; RF power amplifiers, PLL.

Textbooks:

1. RF Microelectronics, 2nd Edition, by Behzad Razavi, Prentice Hall, ISBN: 978-0137134731
2. Lee: Design of CMOS RF integrated circuits, Cambridge,2004, 2nd edition.

Reference Books:

1. D. Leenaerts, et.al.: Circuits design for RF transceivers, Kluwer 2001
2. Pozar: Microwave and RF design of Wireless Systems, Wiley 2001
3. Everhard: Fundamentals of RF Circuit Design with low noise oscillators, Wiley, 2001
4. Ludwig & Bretchko: RF Electronics – Theory and Applications, Prentice Hall 2000

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3			✓									
CO4			✓									
CO5			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4821 Course Title: Introduction to Satellite Communication Engineering Credits: 3.00

Pre-requisite: None

Rationale of the Course:
To learn the technologies of satellite communication engineering used in many real-life applications.

Course Objective:
Upon completion of the course, the students will be able to:

1. Teach the basics of satellite communication systems.

2. Enable the students to understand the satellite link budget
3. Present modulation and multiple access techniques used in satellite communication.

Course Outcomes (CO):
Upon completion of the course, the students will be able to

1. Gain Knowledge about the components of a satellite communication system.
2. Calculate satellite link budget
3. Evaluate modulation and multiple access techniques for satellite communication.
4. Analyse satellite IP networking
5. Understand Non-GEO satellite system

Course Contents:

Introduction to Satellite Systems- space and ground segment components, earth stations, bus and payloads, antennas and coverage, transparent and non-transparent transponders. FSS, MSS, BSS applications areas, GEO, MEO, LEO etc. Satellite Orbit, satellite look angles and ranges. Doppler effect.

Satellite Link Budget: Basic transmission theory, gain, radiation patter, eirp, Noise sources, noise temp, noise figure, sky noise G/T ratio and calculation C/N for up-path and down-path. Intermodulation, back-off, interference and C/I calculation. Calculation of margins. Link budget with overall C/N and availability.

Modulation Techniques: Introduction to various modulation used in satellite systems, such as QPSK, OQPSK, MPSK, APSK.

Multiplexing and Multiple Access Techniques - FDMA, TDMA, MF-TDMA, CDMA

Satellite Networking: Review of Internet Protocol (IP) over satellite. IP packet encapsulation. Satellite IP networking, TCP slow start and congestion avoidance schemes, TCP enhancement for satellite networks – PEP.

Non-GEO satellite systems: Satellite constellation review and properties, GPS.

Reference and Textbook:

1. Satellite Communications Systems: Systems, Techniques and Technology by Maral, Gérard; Sun, Zhili; Bousquet, Michel
2. Satellite Communications by Timothy Pratt, Jeremy Allnutt, 3rd edition, 2020
3. Satellite networking: principles and protocols, by Zhili Sun, 2014.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3				✓								
CO4		✓										

CO5	✓											

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4822 Course Title: Introduction to Satellite Communication Engineering Sessional Credits: 0.75

Pre-requisite: None

Rationale of the Course:
To perform experiments with the technologies of satellite communication systems used in many real-life applications.

Course Objective:

1. Teach the basics components of satellite communication systems.
2. Enable the students to understand the satellite link budget
3. Present modulation and multiple access techniques used in satellite communication.

Course Outcomes (CO):
Upon completion of the course, the students will be able to

1. Know the components of a satellite communication system.
2. Experiment and calculate satellite link budget
3. Evaluate modulation techniques for satellite communication.
4. Troubleshooting the Satellite Repeater, Earth Station Transmitter and Receiver

Course Contents:

Experiment-1: Satellite Communication Systems

Experiment-2: Satellite Earth Stations Payloads and Telemetry

Experiment-3: Satellite Orbits and Coverage

Experiment-4: Antenna Alignment, Power Gain and Antenna Parameters for Geostationary Satellites

Experiment-5: Noise and the Link Budget for Satellite Communication

Experiment-6: Analog and Digital Baseband Processing and Modulation

Experiment-7: Troubleshooting the Satellite Repeater, Earth Station Transmitter and Receiver.

Reference and Textbook:

1. Satellite Communications Systems: Systems, Techniques and Technology by Maral, Gérard; Sun, Zhili; Bousquet, Michel
2. Satellite Communications by Timothy Pratt, Jeremy Allnutt, 3rd edition, 2020
3. Satellite networking: principles and protocols, by Zhili Sun, 2014.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3			✓									
CO4				✓								

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Project

Course Code: AVE 4823 Course Title: Space Avionics Credits: 3.00

Pre-requisite: None

Rationale of the Course:
This module aims to introduce the student to the operating principles, design and operation of spacecraft avionics systems able to support a wide variety of spacecraft missions – primarily in Earth orbit.

Objective:

1. Enable students to classify space mission on avionics requirements
2. Provide knowledge to the students about implementation of simulated avionics components on design tool

Course Outcomes (CO):
Upon completion of the course, the students will be able to

1. Analyse spacecraft avionics requirements
2. Design flight avionics systems

3. Evaluate flight avionics requirements
4. Recognize threads and requirements for on board electronics components

Course Contents:

Introduction

Classification of spacecraft functions depending of mission profile and identification of requirements and functions of on-board data handling systems

Architecture

Typical spacecraft structure, system and major subsystem, redundancy management, data flow, telematics, service module, payloads

Space environment threads to electronics systems and mitigation tactics

On board electronics susceptibility to space radiation environment, radiation hardness, radiation mitigation techniques, HW and SW error detection and correction

Components and subsystems

On board microprocessors and microcontrollers, on board communication buses and interfaces, mass memories, attitude and orbit control subsystems, payloads data processing, telemetry and telecommand

Standards and system modelisation

Modelisation of flight avionics systems, spacecraft onboard interface services SOIS, Standard Space links protocols, standard data units, spacecraft synchronization time, buses and networks

Reference and Textbook:

1. Spacecraft Systems Engineering-Fortescue, Peter W.; Stark, John; Swinerd, Graham
2. Space Mission Analysis and Design- Wertz, James R.; Larson, Wiley J.
3. Principles of Space Instrument Design– Cruise, A. M.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1		✓										
CO2			✓									
CO3				✓								
CO4						✓						

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4824 Course Title: Space Avionics Sessional Credits: 0.75

Pre-requisite: Space Avionics

Rationale of the Course:
This seasonal course is designed to introduce the student to the operating principles, design and operation of spacecraft avionics systems able to support a wide variety of spacecraft missions practically.

Course Objectives:
1. Enable students to evaluate flight avionics requirements
2. Help students to classify space mission on avionics requirements

Course Outcomes:
1. Analyse spacecraft avionics requirements
2. Design flight avionics systems
3. Recognize threads and requirements for on board electronics components

Course Contents:
Experiment no. 1: Design a low-cost platform that interfaces various sensors and buses.
Experiment no. 2: Test and verify common space avionics designs features.
Experiment no. 3: Design of a spacecraft on-board computer to interface to particular subsystems in a robust manner.
Experiment no. 4: Design function of TT&C system: launch and early operation phase (LEOP), end-of-life operations.
Experiment no. 5: Ground-station design, Mission Operations Control Centre (MOCC), ground-support equipment, use during AIT and LEOP, orbit determination.

Text and Reference Book:
1. Spacecraft Systems Engineering-Fortescue, Peter W.; Stark, John; Swinerd, Graham
2. Space Mission Analysis and Design- Wertz, James R.; Larson, Wiley J.
3. Principles of Space Instrument Design– Cruise, A. M.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1		✓										
CO2			✓									
CO3						✓						

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: AVE 4825 **Course Title: VLSI Design** **Credits: 3.00**

Pre-requisite: Electronic Circuits II (AVE 4401), Digital Systems and Logic Design (AVE 4407)

Rationale of the Course:
To achieve a proper knowledge on VLSI IC design and to build a good CMOS chip.

Course Objectives:

1. Enable students to learn the basic VLSI design approach and basic CMOS circuits, technology trends and design styles.
2. Enable students to learn the concepts of designing VLSI subsystems.
3. Present the estimate of the response and delay as well as power dissipation for CMOS circuits.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Identify the various IC fabrication methods.
2. Express the Layout of a simple MOS circuit using Lambda based design rules.
3. Apply the Lambda based design rules for subsystem design.
4. Differentiate various CMOS architectures.
5. Design an application using the latest technology tools.

Course Contents:

VLSI technology: Top-down design approach, technology trends and design styles.

Review of MOS transistor theory: Threshold voltage, body effect, I-V equations and characteristics, latch-up problems, NMOS inverter, CMOS inverter, pass-transistor and transmission gates.

CMOS circuit characteristics and performance estimation: Resistance, capacitance, rise and fall times, delay, gate transistor sizing and power consumption.

CMOS circuit and logic design: Layout design rules and physical design of simple logic gates.

CMOS subsystem design: Adders, multiplier and memory system, arithmetic logic unit. Programmable logic arrays.I/O systems.VLSI testing.

Text books:

1. Basic VLSI Design by Douglas A. Pucknell; Prentice Hall of India private Ltd.
2. CMOS VLSI Design - A Circuits and System Perspective by N. H. E. Weste and D. Harris.

Reference books :

1. Introduction to VLSI – D. Bricius; McGraw-Hill international.
2. CMOS Circuit Design, Layout and Simulation by R. Jacob Baker, Harry H. Li, David E Boyce, Wiley–IEEE Press 3rd Edition, 2010.
3. CMOS digital Integrated circuits: Analysis and design: Sung-Mo Kang and Yusuf Leblebici.
4. Analysis and design of digital integrated circuits:D.A.Hodges, Jackson and Saleh.
5. Microelectronics-Circuit Analysis and Design by Donald A. Neamen

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3			✓									
CO4				✓								
CO5			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4826 Course Title: VLSI Design Sessional Credits: 0.75

Pre-requisite:

Rationale of the Course:
To learn and familiarize with the basics of VLSI circuit design as well as the analysis and simulation of VLSI circuits.

Course Objectives:

1. Enable students to understand the use of Microwind, DSCH2, Quartus II and FPGA board.
2. Present schematic and layout using Microwind and DSCH2.
3. Present Verilog HDL code and test digital circuits on FPGA.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Analyse the CMOS layout levels, how the design layers are used in the process sequence, and resulting device structures (i.e. cross-sectional views)
2. Implement digital logic designs of various types (i.e. combinational logic, multiplexers).
3. Develop professional IC layout using Microwind.
4. Design prototypes of different practical systems by working in collaboration.

Course Contents:

Experiment no. 1: Spice simulation of logic gates

Experiment no. 2: Layout design of a CMOS inverter.
Experiment no.3: Layout design of Boolean algebra using Euler's Formula.
Experiment no. 4: Layout design using pass transistor logic.
Experiment no. 5: Design of a 4-bit full adder by using one-bit full adder in DSCH2.
Experiment no. 6: Design of a 3-bit comparator by using one bit comparator in DSCH2.
Experiment no. 7: Schematic Design of a 4-bit A.L.U. (Arithmetic Logic Unit)
Experiment no. 8: Introduction to Verilog HDL, Quartus II and FPGA board.
Experiment no. 9: Design of a Full-adder, 8:1 Multiplexer and 1:8 Demultiplexer using Verilog.
Experiment no. 10: Design of a Finite State machine using Verilog in Quartus II.

Text books:

1. Basic VLSI Design by Douglas A. Pucknell; Prentice Hall of India private Ltd.
2. CMOS VLSI Design - A Circuits and System Perspective by N. H. E. Weste and D. Harris.

Reference books:

1. Introduction to VLSI – D. Bricius; McGraw-Hill international.
2. CMOS Circuit Design, Layout and Simulation by R. Jacob Baker, Harry H. Li, David E Boyce, Wiley–IEEE Press 3rd Edition, 2010.
3. CMOS digital Integrated circuits: Analysis and design: Sung-Mo Kang and Yusuf Leblebici.
4. Analysis and design of digital integrated circuits: D.A.Hodges, Jackson and Saleh.
5. Microelectronics-Circuit Analysis and Design by Donald A. Neamen

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1		✓										
CO2		✓										
CO3						✓						
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Project

Course Code: AVE 4827 Course Title: Optical Fibre Communication System Credits: 3.00

Pre-Requisite: Communication Systems

Rationale of the Course: This course is designed to understand the concept of optical fibre communication and its advantages, disadvantages, applications, construction, index profiles & operation.

Course Objective:

1. Enable the students to learn about the historical development, advantages, disadvantages, applications, construction, index profiles & operation of optical fibre communication.
2. Present Modulation techniques of different modulators such as LASER and LED.
3. Provide knowledge about different types of photodetector's operation, classification, pros & cons.

Course Outcome:

1. Understand the basic and fundamental concept of optical fibre communication system, its advantages, disadvantages, applications, construction, index profiles & operation.
2. Explain the characteristics of optical sources, detectors and optical amplifiers and their economical use.
3. Solve numerical problems to find the different parameters of an optical fibre system.

Course Content:

Light propagation through optical fibre: Ray optics theory and mode theory.
Optical fibre: Types and characteristics, transmission characteristics, fibre joints and fibre couplers. **Light sources:** Light emitting diodes and laser diodes.
Detectors: PIN photodetector and avalanche photodetectors.
Receiver analysis: Direct detection and coherent detection, noise and limitations.
Transmission limitations: Chromatic dispersion, nonlinear refraction, four waves mixing and laser phase noises.
Optical amplifier: Laser and fibre amplifiers, applications and limitations.
Multi-channel optical system: Frequency division multiplexing, wavelength division multiplexing and optical CDMA. Radio on fibre technology, Fibre optic access networks.

Ref and Textbooks:

1. Fiber-Optic Communications - Joseph C. Palais.
2. Optical Fibre Communications: Principles & Practice - John M. Senior; Prentice Hall of India.
3. Fibre Optic Communications - D C Agrawal; Wheeler Publishing.
4. Fibre Optic Communication System - Gerd Keiser; McGraw-Hill International.
5. Modern optical Engineering the Design of Optical Sys. – J. Smith; SPIE Press McGraw-Hill.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam

CO3	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4828 Course Title: Optical Fibre Communication System Sessional Credits: 0.75

Pre-requisite: Communication Systems (AVE 4601)

Rationale of the Course:
This seasonal course is designed to familiarise with the optical fibre system and performance criteria for data communication.

Course Objectives:
1. Enable the students to calculate different types of loss of an optical fibre cable.
2. Enable the students to calculate the required bandwidth of an optical fibre system.

Course Outcomes:
1. Analyse the relation among attenuation, numerical aperture and core area.
2. Calculate the bandwidth for a length of fibre and verify bending losses using relative power measurements.

Course Contents:

Experiment no. 1: Setting up Fibre Optic Analog Link.
Experiment no. 2: Setting up Fibre Optic Digital Link.
Experiment no. 3: Study of Propagation Loss in Optical Fibre.
Experiment no. 4: Determine the scattering and absorption loss of optical fibre.
Experiment no. 5: Determine the bending loss and dispersion of an optical fibre.

Ref and Textbooks:

1. Fibre-Optic Communications - Joseph C. Palais.
2. Optical Fibre Communications: Principles & Practice - John M. Senior; Prentice Hall of India.
3. Fibre Optic Communications - D C Agrawal; Wheeler Publishing.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam

Course Code: AVE 4829 **Course Title: Avionics Cyber Security** **Credits: 3.00**

Pre-requisite:

Rationale of the Course:

This course provides the scientific foundations and engineering principles needed to advance the field of aviation cybersecurity, with focus on fundamentally new, principled approaches to deter, protect, detect, and respond to actions that compromise or threaten to compromise the confidentiality, integrity, or availability of computer- and network-based aviation systems.

Course Objective:

1. Identify the principles and consequences of cyber security in the aviation environment.
2. Describe how cyber security impacts different actors in aviation.
3. Select an appropriate standard, or set of standards, to adopt for specific aviation purposes.
4. Describe different cybersecurity issues.

Course Outcome:

Understand electromagnetic radiation mechanism and its physics and be able to compute radiation form several common antenna structures.

1. Be able to counter various aviation security issues.
2. Gain access to the tools and understanding to use available standards to manage cyber risk in an aviation context in a standards-led way.
3. Design drone surveillance system.
4. Implement security layers in wireless and satellite communication systems.

Course Content:

1. Counter unmanned aerial vehicle for aviation security
2. Formal verification of safety-critical software using SPARK
3. Drone surveillance system—RF/WiFi-based drone detection,localization, and tracking: a survey
4. Cybersecurity for the L-band Digital Aeronautical Communications System (LDACS)
5. Using terrestrial ranging to monitor GNSS for targeted RF interference
6. Security risk assessments and countermeasures for future aeronautical communication network architecture
7. Physical layer security in wireless HetNets avionics and satellite communication systems
8. Cybersecurity and privacy issues
9. Workforce training and development in cybersecurity
10. Aviation cybersecurity: a cyber-physical systems perspective

Textbooks and Ref:

1, Aviation Cybersecurity: Foundations, principles, and applications (Radar, Sonar and Navigation) - Houbing Song

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1						✓						
CO2					✓							

CO3			✓									
CO4							✓					

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: AVE 4830 Course Title: Avionics Cyber Security Sessional Credits: 0.75

Pre-requisite:

Rationale of the Course:

This course provides practical foundations and engineering principles needed to advance the field of aviation cybersecurity, with focus on fundamentally new, principled approaches to deter, protect, detect, and respond to actions that compromise or threaten to compromise the confidentiality, integrity, or availability of computer- and network-based aviation systems.

Course Objective:

1. Train students to identify the principles and consequences of cyber security in the aviation environment.
2. Present cyber security defence mechanism in aviation.
3. Enable students to select an appropriate standard, or set of standards, to adopt for specific aviation purposes.
4. Train about Mitigating different cybersecurity issues.

Course Outcome:

Understand electromagnetic radiation mechanism and its physics and be able to compute radiation form several common antenna structures.

1. Be able to counter various aviation security issues.
2. Gain access to the tools and understanding to use available standards to manage cyber risk in an aviation context in a standards-led way.
3. Design drone surveillance systems.
4. Implement security layers in wireless and satellite communication systems.

Course Content:

Experiment no. 1: Counter unmanned aerial vehicle for aviation security
Experiment no. 2: Formal verification of safety-critical software using SPARK
Experiment no. 3: Drone surveillance system—RF/WiFi-based drone detection, localization, and tracking
Experiment no. 4: Cybersecurity for the L-band Digital Aeronautical Communications System (LDACS)
Experiment no. 5: Using terrestrial ranging to monitor GNSS for targeted RF interference

Experiment no. 6: Physical layer security in wireless HetNets avionics and satellite communication systems
Experiment no. 7: Aviation cybersecurity: a cyber-physical systems perspective

Textbooks and Ref:

1, Aviation Cybersecurity: Foundations, principles, and applications (Radar, Sonar and Navigation) - Houbing Song

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1							✓					
CO2					✓							
CO3			✓									
CO4						✓						

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: AVE 4831 Course Title: Wireless Communication Credits: 3.00

Pre-Requisite: Communication Systems (AVE 4601)

Rationale of the Course: To introduce the student to the principles and building blocks of wireless communications.

Course Objective:

1. Present the limitations of fixed networks; the need and the trend toward mobility; the concepts portability and mobility
2. Describe and analyse the network infrastructure requirements to support mobile devices and users.
3. Illustrate the concepts, techniques, protocols and architecture employed in wireless local area networks, cellular networks, and perform basic requirements analysis.

Course Outcome:

1. Apply cellular concepts to evaluate the signal reception performance in a cellular network and traffic analysis to design cellular networks with given quality of service constraints.
2. Determine the type and appropriate model of wireless fading channel based on the system parameters and the property of the wireless medium.
3. Analyse and design receiver and transmitter diversity techniques.

Course Content:

Introduction

Evolution of wireless (mobile) communications, worldwide market, examples, Comparison of available wireless systems, trends, Trends in cellular radio (2G, 2.5G, 3G, beyond 3G) and personal wireless communication systems

Cellular mobile communication concept

Frequency re-use and channel assignment strategies, Handoff strategies, types, priorities, practical considerations, Interference and system capacity, co-channel and adjacent channel interference, power control measures, Grade of service, definition, standards, Coverage and capacity enhancement in cellular network, cell splitting, sectoring, repeaters, microcells.

Radio wave propagation in mobile network environment

ReviewFree space propagation model, radiated power and electric field, ReviewPropagation mechanisms (large-scale path loss) - Reflection, ground reflection, diffraction and scattering, Practical link budget design using path loss models. Outdoor propagation models (Longley-Rice, Okumura, Hata, Walfisch and Bertoni, microcell), Indoor propagation models (partition losses, long-distance path loss, multiple breakpoint, attenuation factor), Small scale fading and multipath (factors, Doppler shift), Impulse response model of multipath channel, multipath measurements, parameters of mobile multipath channel (time dispersion, coherence bandwidth, Doppler spread and coherence time), Types of small-scale fading (flat, frequency selective, fast, slow), Rayleigh and Ricean fading distribution.

Modulation-Demodulation methods in mobile communications

Review of amplitude (DSB, SSB, VSB) and angle (frequency, phase) modulations and demodulation techniques, Review of line coding, digital linear (BPSK, DPSK, QPSKs) and constant envelop (BFSK, MSK, GMSK) modulation and demodulation techniques, M-ary (MPSK, MFSK, QAM and OFDM) modulation and demodulation techniques, Spread spectrum modulation techniques, PN sequences, direct sequence and frequency hopped spread spectrums, Performance comparison of modulations techniques in various fading channels.

Equalization and diversity techniques

Basics of equalisation. Equalisation in communications receivers, linear equalisers, Non-linear equalisation, decision feedback and maximum likelihood sequence estimation equalizations,Adaptive equalisation algorithms, zero forcing, least mean square, recursive least squares algorithms, fractionally spaced equalisers, Diversity methods, advantages of diversity, basic definitions, Space diversity, reception methods (selection, feedback, maximum ratio and equal gain diversity),Polarization, frequency and time diversity, RAKE receivers and interleaving

Speech and channel coding fundamentals

Characteristics of speech signals, frequency domain coding of speech (sub-band and adaptive transform coding) Vocoders (channel, formant, cepstrum and voice-excited), Linear predictive coders (multipulse, code and residual excited LPCs), Codec for GSM mobile standard, Review of block codes, Hamming, Hadamard, Golay, Cyclic, Bosh-Chaudhary- Hocquenghgem (BCH), Reed-Solomon (RS) codes,Convolutional codes, encoders, coding gain, decoding algorithms (Viterbi and others), Trellis Code Modulation (TCM), Turbo codes.

Multiple Access in Wireless communications

Frequency Division Multiple Access (FDMA), principles and applications, Time Division Multiple Access (TDMA), principles and applications, Spread Spectrum Multiple Access, Frequency Hopped Multiple Access, Code Division Multiple Access, hybrid spread spectrum multiple access techniques, Space Division Multiple Access, Standards for Wireless Local Area Networks

Wireless systems and standards

Evolution of wireless telephone systems: AMPS, PHS, DECT, CT2, IS-94, PACS, IS-95, IS-136, IS-54 etc.Global system for Mobile (GSM): Services and features, system architecture, radio sub-system, channel types (traffic and control), frame structure, signal processing, example of a GSM call CDMA standards: Frequency and channel specifications, Forward and Reverse CDMA channels, WiFi, WiMAX, UMB, UMTS, CDMA-EVDO, LTE, and recent trends, Regulatory issues (spectrum allocation, spectrum pricing, licensing, tariff regulation and interconnection issues).

Ref and Textbooks:

1. Wireless Digital Communications - K. Feher.
2. Wireless Communications - T. Rappaport
3. Mobile Communications - J. Schiller
4. Analog and Digital communication systems - B.P.Lathi
5. Communication Systems II – D. Sharma

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1			✓									
CO2				✓								
CO3			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: AVE 4832 Course Title: Wireless Communication Sessional Credits: 0.75

Pre-requisite: Communication Systems (AVE 4601)

Rationale of the Course:
This course is designed to teach about practical experiments on principles and building blocks of wireless communications.
Objective:

1. Train the students to implement various modulation techniques.
2. Enable the students to understand the security issues in the wireless and implement the security algorithms.
3. Present various protocols of physical, MAC and routing layers.

Course Outcomes (CO):

1. Ability to comprehend and appreciate the significance and role of this course in the present contemporary world
2. Modify the characteristics of wireless channels and develop a new channel model.
3. Experimentally understand the QoS parameters of the network for different applications.

Course Contents:

Experiment No 1 Characterization of Wireless Channels (Simulation/Experiment)

Experiment No 2 Equalization Techniques for Wireless Channels
Experiment No 3 Simulation / Implementation of Multicarrier Modulation
Experiment No 4 Simulation / Implementation of Space Time Block Codes
Experiment No 5 Performance Studies of Adaptive Modulation and Coding
Experiment No 6 Performance Studies of Random MAC Protocols
Experiment No 7 Performance Studies of LLC Protocols
Experiment No 8 Wireless Routing Protocols
Experiment No 9 Wired and Wireless Packet Analysis using Open-Source Tools.
Experiment No 10 Network Security Protocols.

Ref and Textbooks:

1. Wireless Digital Communications - K. Feher.
2. Wireless Communications - T. Rappaport
3. Mobile Communications - J. Schiller
4. Analog and Digital communication systems - B.P.Lathi
4. Communication Systems II – D. Sharma

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

f. Non-Departmental Elective Courses

Course Code: ASE 4551 **Course Title: Aircraft Systems** **Credits: 3.00**

Pre-requisite: None.

Rationale of the Course:

To introduce the students with various systems associated with aircraft operations and safety issues.

Objective:

1. Enable the students to create an idea about the operation of different systems of aircraft.
2. Enable the students to visualise the importance of different systems for aircraft safe operations.
3. Provide knowledge on various features of the systems and subsystems associated with aircraft control and navigation

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Demonstrate an understanding of hydraulic system,
2. Understand the principles of safe operation to aircraft cabin atmospheric control systems, ice protection systems, position and warning systems,
3. Understand modern aircraft electrical system power generation and distribution principles
4. Knowledge about Flight Data Recorders, Cockpit Voice Recorders.
5. Determine the fuel system of an aircraft

Course Contents:

Hydraulic Systems: Fundamentals of hydraulic systems: components, hydraulic systems controllers, fluid reservoirs, oil coolers, valves and actuators
Landing Gear System: Anti-skid system, shock absorbers, maximum braking performance.
Airplane Control System: Push pull rod system, Power assisted and fully powered flight controls, digital fly by wire system**s**
Engine Systems: Starting and ignition systems, Fuel systems of piston and jet engine, multi-engine fuel systems, Fuel system operating modes
Air Conditioning and Pressurization System: Basic air cycle systems, Oxygen systems, Deicing and anti- icing system
Electrical Systems: AC and DC power generations and supply in aircraft, aircraft batteries, external power supplies, Auxiliary Power Unit (APU), Components of power distribution, safety requirements, aircraft electrical wiring and lighting system
Avionics Systems: Flight Data Recording System, Cockpit Voice Recording System, Cockpit Display System, Glass Cockpit, HUD, HDD, HMD, Warning Systems, Fire detection and suppression Emergency power sources, Emergency landing, Full Authority Digital Engine Control (FADEC) System.

Text and Ref Books:

1. Aircraft Power Plants- Mekinley, J.L. and R.D. Bent; McGraw Hill 1993.
2. Aircraft Fuel Systems -Roy Langton, Chuck Clark, Martin Hewitt, Lonnie Richards; WILEY Publications.
3. Aircraft Maintenance & Repair- Mckinley, J.L. and Bent R.D; McGraw Hill.
4. Aircraft Instruments & Principles- Pallet, E.H.J; Pitman & Co 1993.
5. Advanced Aircraft Systems – David A. Lombardo; Mc-Grew Hill.
6. Aircraft Landing Gear Design: Principles and Practices (AIAA Education Series) - Norman S. Currey; American Institute of Aeronautics and Astronautics.
7. Aircraft Hydraulic Systems: An Introduction to the Analysis of Systems and Components – W.L Green, Wiley Publication.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2						✓						
CO3					✓							
CO4		✓										
CO5		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO5	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: ASE 4552 Course Title: Aircraft Systems Sessional Credits: 0.75

Pre-requisite: None.

Rationale of the Course:

To design, analysis and selection of commonly used mechanical components subject to static and dynamic loads.

Objective:

1. To calculate various loads as applied to shaft, and specify appropriate design stresses for Shaft.
2. To specify suitable keys and couplings for shaft and other type of machine elements
3. To analyze and design spur gear, helical gear and bevel gear
4. To analyze and design of rolling element bearings
5. To analyze and design of lubrication and sliding bearings

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Understand the theories relating to power screws, shaft, keys, springs, bearings, gears, brakes and clutches.
2. Apply the knowledge to design such machines like power screws, shaft, keys, springs, bearings, gears, brakes and clutches.
3. Evaluate the design requirement of various engineering machines.
4. Analyse the design parameters of various engineering machines.

Course Contents:

Introduction to machine design. Design of basic machine elements like power screws, shaft and hole systems, keys and couplings, rivets, springs, bearings, gears, brakes and clutches. Design with composite materials.

Text and Ref Books:

1. A Textbook of Machine Design - R. S. Khurmi, J. K. Gupta

2. Fundamentals of Machine Component Design - Robert C Juvinall.
3. Design of Machine Elements (4th Ed) - Virgil Moring Faires
4. Mechanical Engineering Design (7th Edition) - Joseph E Shigley, Charles R Mischke & Richard G Budynas.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3			✓									
CO4			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam
CO4	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam

Course Code: ASE 4557 Course Title: Aerospace Vehicle Dynamics and Control Credits: 3.00

Pre-requisite: None

Rationale of the Course:

The main course's objective is to teach students about the stability, the trim and the control characteristics of the aircraft.

Objective:

1. Impart knowledge to define the longitudinal static stability and longitudinal control of the aircraft
2. Enable the students to solve the longitudinal equations of motion which result in an exact description of the stability and response characteristics of the aircraft.
3. Present the difference between, and factors affecting, static and dynamic stability

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Describe aircraft dynamic modes.
2. Analyse contribution of various aircraft components to longitudinal, directional and lateral stability.
3. Evaluate aircraft stability and appreciate relevant flight and handling qualities
4. Develop the equations of motion of an aircraft.

Course Contents:

Importance and significance of flight stability and control: Static Longitudinal, Directional and Lateral stability with respect to the aircraft axis systems; Effect of various wings design and secondary control surfaces; Origin of symmetric forces and moments; Static and manoeuvring longitudinal stability, equilibrium and control of rigid aircraft; Effects of various major components on Static Stability, Critical flight conditions and controls requirement.

Dynamic Stability: The Axes Systems (Inertial, Body and Stability axes) and their Transformations; Treatment of Aircraft Equations of motion / linearization; Aerodynamic load effects of wings, stabilizers, fuselages and power plants; Trailing edge aerodynamic controls; Trimmed equilibrium condition; Static margin; Effect on static stability of free and reversible controls.

Introduction to automatic flight control: Setup of the flight control system, System performance specification: - Requirements on flying and handling qualities and Parameters.

Stability augmentation systems: - Dampers-Acquiring static stability, Feedback-Acquiring static stability. Basic autopilot systems- Basic Longitudinal and Lateral autopilot systems; Navigational autopilot systems- Longitudinal and Lateral autopilot systems.

Text and Ref Books:

1. Aircraft Performance Stability and Control, Vol-I, James D; Lang United States Air Force Academy.
2. Automatic Control of Aircraft and Missiles, Col. John H, Blakelock
3. Airplane Performance, Stability and Control, Perkins and Hage
4. Dynamics of Flight, Bernard Etkin

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3				✓								
CO4				✓								

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

Course Code: ASE 4797 **Course Title: Orbital Mechanics** **Credits: 3.00**

Prerequisite: Mathematics, Physics

Rationale of the Course:
To have introductory understanding of orbits, orbital mechanics and interplanetary missions

Objectives:

1. Teach the fundamental laws that govern the orbital dynamics, equations for satellite dynamics, basics of different orbits
2. Help the students to learn the different types of satellite orbit transfer from one orbit to an another, basics of interplanetary trajectory

Course Outcomes (CO): Upon the completion of the course, student will be able to-

1. Understand basics of orbital mechanics, orbits, trajectories.
2. Understand the orbital properties in detail.
3. Understand orbital transfer and interplanetary missions

Course Contents:

Orbital Mechanics Fundamentals; Laws of orbital dynamics, Solar system, Earth geometry and structure of upper atmosphere, multi-body problem, two body problem, Equation of motion, Orbit equation, in circular, elliptical orbits, parabolic, hyperbolic orbits, Interplanetary trajectories.

Orbits in Three Dimensions: Coordinate systems, Time systems, Keplerian Orbital elements, Relations between position and time, Kepler‟s equation, Effects of the earth‟s oblateness, Types of satellite orbits, their characteristics and applications, Orbit perturbation due to third body, Orbit decay and lifetime.

Orbital Transfer and Powered Flight: Rocket equation, application to orbit transfer, velocity requirement, different propulsion systems, Propellant, Single and two Impulse transfer, Hohmann transfer, One tangent manoeuvre, Plane change manoeuvre, Bielliptic manoeuvre, Phasing manoeuvre and interplanetary missions.

Text and Ref Books:

1. Orbital Mechanics for Engineering Students, Howard D. Curtis, Elsevier Butterworth-Heinemann, 2005
2. Rocket Propulsion and Space Dynamic, Cornelisse, J.W, Schoyer H F R, and Wakker K F, Pitman Publishing Co., 1979
3. Fundamentals of Astrodynamics, Bate R. R., Mueller D. D., and White J. E., Dover Publications, New York, 1972
4. Atmospheric and Space Flight Dynamics, Ashish Tewari, Birkhauser Boston, 2007
5. Rocket and Spacecraft Propulsion, Martin J L Turner, Springer Praxis Publishing Co, Chichester, UK, 2001
6. Fundamentals Astrodynamics and applications, David A Vallado, Space Technology Series, McGraw Hill, 1997
7. Orbital Mechanics, Vladimir A Chobotov, AIAA Education Series, 2002

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2	✓											
CO3		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: CSE 4781 **Course Title: Machine Learning** **Credits: 3.00**

Pre-requisite: None

Rationale of the Course:

Machine learning provides appropriate learning algorithm to best suit the current need and enhance the learning parameters for maximum performance.

Objective:

1. Help students learn paradigms in different environmental setting and apply the appropriate learning algorithm to best suit the current need.
2. Familiarize with the techniques associated with enhancing the learning parameters to achieve maximum performance.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Discuss the different learning paradigms in different environmental setting
2. Select and apply the appropriate learning algorithm to best suit the current need.
3. Enhance the learning parameters to achieve maximum performance.

Course Content:

Introduction to Machine Learning; Regression analysis: Logistic Regression, Linear Regression; Supervised and Unsupervised learning; Bayesian Learning; Decision Tree Learning; Rule based learning; Instance based learning; Neural Nets; Support Vector Machine; Genetic Algorithms; Reinforcement learning; Ensemble learning; Hidden Markov Models; Maximum Likelihood Estimates, Parameter Estimation; Computational learning theory.

Teaching-learning and Assessment Strategy:

Lectures, class performances, assignments, class tests, final exam.

Text and Ref Books:

1. Pattern Recognition and Machine Learning - Christopher M. Bishop; Springer
2. Machine Learning - Tom Mitchell, McGraw Hill
3. Pattern Recognition –Sergios Theodoridis and Konstantinos Koutroumbas; Elsevier Inc.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2					✓							
CO3			✓									

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: CSE 4783 Course Title: Artificial Intelligence Credits: 3.00

Pre-requisite: None

Rationale of the Course:
Artificial intelligence is the beginning of revolution for rational behaviour of intelligent agents along with knowledge perception, representation, planning, reasoning, learning and understanding ideas to solve real life complex situations

Course Objective:

1. Present the difference between the notions of rational behaviour and intelligent agents.
2. Develop a general appreciation of the goals, subareas, achievements and difficulties of AI.
3. Impart knowledge of methods of blind as well as informed search in case of knowledge representation, planning, learning, robotics and other AI areas and ability to practically apply the corresponding techniques.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Remembering and understanding the notions of rational behaviour, goals, subareas, achievements and difficulties of AI agents.
2. Able to apply problem solving methods (informed, uninformed, local search, adversarial search and CSP) of single or multi agents to solve real life problems
3. Able to apply major concepts and approaches of knowledge representation, planning and learning for improving machine intelligence.
4. Able to develop communication skills by presenting topics on Artificial Intelligence.

Course Contents:

Introduction: Overview of AI and intelligent agents. Problem Solving: Review of Uninformed Search Strategies and game playing; Informed search Strategies: A*, Heuristic functions, Memory Bounded Search (IDA*, SMA*), Iterative improvement Search, adversarial search, local search Constraint satisfaction problems.
Knowledge representation: Review of Propositional logic, first order Logic,
Planning: Introduction to Planning, Partial Order Planning.
Reasoning: Bayesian Rule and its use in probabilistic reasoning.
Learning: Belief Networks and Decision Networks; Learning Decision Trees; Learning General Logical Descriptions-Hypothesis. Introduction to Natural Language Processing.

Text and Ref Books:

1. Artificial Intelligence: A Modern Approach (4th Edition) – Stuart Jonathan Russell, Peter Norvig; Prentice Hall (2020)
2. Artificial Intelligence: A New synthesis – Nils J. Nilsson; Routledge

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2		✓										
CO3				✓								
CO4										✓		

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Class Test /Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: CSE 4785 Course Title: Introduction to Quantum Computing Credits: 3.00

Pre-requisite: None

Rationale of the Course:
The goal of the developing field of quantum computation is to create efficient atomic-scale machines that take use of the parallelism of the quantum mechanical laws. While this may sound far-fetched, quantum computers are already a reality and have the power to completely change computation in the near future.

Course Objectives:

1. Help student demonstrate basic understanding of Quantum Information Processing
2. Provide knowledge on the hardware and software architecture design of a Quantum Computer
3. Share insights on Quantum Space Communication.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain basic principles behind Quantum Information Processing
2. Design a Quantum Circuit
3. Simulate different Quantum Algorithms
4. Describe different protocols for Quantum Communication

Course Contents:

Introduction: Vector space, orthonormal basis, dirac braket, basis, basis vectors, inner products, postulates

of quantum mechanics, quantum measurements, Changing Basis and Heisenberg Uncertainty Principle, Bra Ket Operations and Vector Normalization, Pauli Spin Matrices, Eigenvalues and Eigenvectors, Operators, Matrices, Adjoint and Hermitian Matrices, Bra Ket Equations for Operators, Matrix Diagonalization and Finding Eigenvalues and Eigenvectors, Unitary Matrix and Transformation Matrix Construction.

Quantum gates and circuits, Programming using IBM Quantum Experience and Circuit Composer, Programming using Qisit, Quantum Algorithms: Deutsch Jozsa Algorithm, Bernstein Vazirani Algorithm, Grover's Search, Shor's Algorithm, NISQ-era quantum algorithms.

Quantum Space Communication: Entangled States, teleportation protocol, Quantum Cryptography, Satellite Quantum Communication

Text and Ref Books:

1. Introduction to Quantum Computing: From a Layperson to a Programmer in 30 Steps, Hiu Yung Wong (Author), Springer; 1st ed. 2022 edition
2. Quantum Computation and Quantum Information: 10th Anniversary Edition, Michael A. Nielsen (Author), Isaac L. Chuang (Author), Cambridge University Press
3. Quantum Mechanics 2nd Edition by Amit Goswami (Author), Waveland Pr Inc; 2nd edition

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2			✓									
CO3				✓								
CO4							✓					

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO2	Standard Lecture (Teacher)	Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: EEE 4881 Course Title: Electrical Power Systems Credits: 3.00

Pre-requisite: Electric Machines and Drives (AVE 4503)

Rationale of the Course:

This course is designed to teach fundamental concepts of power systems, fault analysis, different types of power plants, different kinds of switchgear elements with their use in our modern power system.

Course Objectives:

1. Familiarise students with the terminology associated with the knowledge on the power system.
2. Provide basic knowledge of switchgear and protection.
3. Develop the ability to understand the different criteria to detect power system faults.
4. Help them to acquire the knowledge of different types of protection schemes.

Course Outcome:

1. Develop basic understanding of power system
2. Explain different types of power plants and the usage of switchgear and protection devices and schemes.
3. Analyse different problems of circuit breakers, relays and protection schemes.
4. Calculate and illustrate symmetrical and unsymmetrical faults and power related problems.

Course Content:

Fundamental Concepts and Fault Analysis: Fundamental concepts and definitions and basic knowledge on power system, basics of different power plants, effect of variable load on power plants, load curve, detail of power factor issues, economics of power generation, Tariff, Per Unit System, Single Line Diagram, Faults in Power System, calculation of symmetrical faults, unsymmetrical faults, single line to ground fault, double line to ground fault, line to line fault.

Transmission lines cables: Overhead and underground transmission line
Stability: Swing equation, power angle equation, equal area criterion, step by step solution of swing equation. AC transmission system (FACTS). High voltage DC transmission system.
Distribution System: DC distribution and AC distribution.

Switchgear and Power System Protection: Purpose of switchgear and power system protection. Unit protection schemes: Generator, transformer, motor, bus bar, transmission and distribution lines. Miniature circuit breakers and fuses. Circuit breakers: Principle of arc extinction, selection criteria and ratings of circuit breakers, types - air, oil, SF6 and vacuum.

Ref and Textbooks:

1. Principles of power system- V. K. Mehta, Rohit Mehta
2. Switchgear Protection and Power Systems- Sunil S Rao.
3. Elements of Power System Analysis – William D Stevenson
4. Power System Analysis – Hadi Saadat
5. Power System Protection and Switchgear- B. A. Oza, N-K. C. Nair, R. P. Mehta, V. H. Makwana.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	✓											
CO2						✓						
CO3				✓								
CO4		✓										

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam

CO2	Standard Lecture (Teacher)	Class Test/Assignment, Mid Term Exam, Final Exam
CO3	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam
CO4	Standard Lecture (Teacher)	Class Test/Assignment, Final Exam

Course Code: EEE 4882 Course Title: Electrical Power Systems Sessional Credits: 0.75

Pre-requisite:

Rationale of the Course:
This course is designed to learn about the elements of the power system and their impacts on the power transmission and distribution system practically.

Course Objectives:
1. Enable, train and evaluate the ability of the students to perform the analysis of power system related instruments.
2. Provide the knowledge to solve real power system problems effectively.
3. Help them to analyse the operating characteristics of different kinds of circuit breakers and fuse wire.
4. Provide an overview of how the equipment related to a power system can be protected using modern protective apparatus.

Course Outcomes (CO):
1. Familiarize with different types of power system protection schemes and their basic characteristics through laboratory experimental work.
2. Determine rating of the circuit breakers by analysing symmetrical short circuits.
3. Visualise the practical operation of all kinds of fuses, relays, circuit breakers with observing their actual or perfect field of application.

Course Contents:
Experiment No 1 Overview of the pros & cons and safety of the operator in the power system lab.
Experiment No 2 Study and observation of a daily load of a power system.
Experiment No 3 Experiment on power factor improvement using synchronous motor
Experiment No 4 Performance of transmission line under different loading conditions.
Experiment No 5 Study of symmetrical short circuit.
Experiment No 6 Determination of circuit breaker rating.
Experiment No 7 Determination of time current characteristics (TCC) curve of a rewire able fuse & MCB.
Experiment No 8 Performance study of a different protection system used for the protection of transformers.
Experiment No 9 Study of different types of motor protection system using PLC.

Text and Ref Books:

1. Principles of power system- V. K. Mehta, Rohit Mehta
2. Switchgear Protection and Power Systems- Sunil S Rao.
3. Elements of Power System Analysis – William D Stevenson
4. Power System Analysis – Hadi Saadat
5. Power System Protection and Switchgear- B. A. Oza, N-K. C. Nair, R. P. Mehta, V. H. Makwana.

Mapping of Course Outcomes to Program Outcomes:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1					✓							
CO2		✓										
CO3						✓						

Mapping Course Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam, Final Exam
CO2	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Mid Term Exam
CO3	Standard Lab (Teacher)	Quiz, Viva, Lab Report, Final Exam

PART **D**

23. Grading/Evaluation

(I) Grading Scale

The grading scale for the program of B.Sc. in Avionics Engineering is 4.0.

(II) Grades

The total performance of a student in each course is based on a scheme of continuous assessment, for theory courses this continuous assessment is made through a set of quizzes, class tests, class evaluation, class participation, homework assignment/case studies and a semester final examination. The assessments for sessional courses are made by evaluating performance of the student at work during the class, viva-voce during laboratory hours, quizzes and a detailed report. Besides, at the end there will be a final lab test. Each course has a certain number of credits, which describes its corresponding weightages. A student's performance is measured by the number of credits completed satisfactorily and by the weighted average of the grade points earned. A minimum grade point average (GPA) is essential for satisfactory progress. A minimum number of earned credits must also be acquired in order to qualify for the degree. Letter grades and corresponding grade points will be given as follows:

Score	Grade	Grade Point
80% and above	A+	4.00
75% to below 80%	A	3.75
70% to below 75%	A-	3.50
65% to below 70%	B+	3.25
60% to below 65%	B	3.00
55% to below 60%	B-	2.75
50% to below 55%	C+	2.50
45% to below 50%	C	2.25
40% to below 45%	D	2.00
below 40%	F*	0.00
Incomplete	I	-
Withdrawal	W	-
Project/ Thesis continuation	X	-

* Subjects in which the student gets F grade shall not be regarded as earned credit hours for the calculation of Grade Point Average (GPA).

(III) Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA)

Grade Point Average (GPA) is the weighted average of the grade points obtained of all the courses passed/completed by a student. For example, if a student passes/completes *n* courses in a semester having credits of C_1, C_2, ..., C_n and his grade points in these courses are G_1, G_2, ... G_n respectively, then

$$GPA = \frac{\text{Grade points earned in the semester}}{\text{Credits completed in the semester}}$$

$$\Rightarrow GPA = \frac{\text{Summation of (credit hours in a course} \times \text{grade points earned in that course)}}{\text{Total number of credit hours completed}}$$

$$\Rightarrow GPA = \frac{\sum_i^n C_i G_i}{\sum_{i=1}^n C_i}$$

The Cumulative Grade Point Average (CGPA) is the weighted average of the GPA obtained in all the terms passed/completed by a student. For example, if a student passes/ completes n semesters having total credits of TC_1, TC_2, ... , TC_n and his GPA in these semesters are GPA_1, GPA_2,... , GPA_n, respectively then

$$CGPA = \frac{\sum_{i=1}^{n} TC_i GPA_i}{\sum_{i=1}^{n} TC_i}$$

Suppose a student has completed eight courses in a semester and obtained the following grades:

Course	**Credits, C_i**	**Grade**	**Grade Points, G_i**	**C_I*G_i**
AVE 4302	1.50	A-	3.50	5.250
ASE 4101	3.00	A+	4.00	12.000
CHM 4201	3.00	A	3.75	11.250
MAT 4203	3.00	B	3.00	9.000
HUM 4101	3.00	B-	2.75	8.250
HUM 4313	3.00	B	3.00	9.000
PHY 4101	3.00	A+	4.00	12.000
CSE 4202	1.50	A	3.75	5.625
Total	**21.00**			**72.375**

GPA = 72.375/21.00 = 3.45

Suppose a student has completed four semesters and obtained the following GPA.

Semester	**Credit Hours Earned, TC_I**	**GPA Earned, GPA_i**	**GPA_i*TC_i**
1	21.00	3.73	78.330
2	20.50	3.93	80.565
3	19.75	3.96	78.210
4	20.25	4.00	81.000
Total	**81.50**		**318.105**

CGPA = 318.105/81.50 = 3.90

(IV) Course Withdrawal

If a student is unable to complete the Term (Semester) Final Examination due to serious illness or severe accident, he/she may apply to the Head of the degree awarding department for a total withdrawal from the semester **before the commencement of the term final examination**. However, application may be considered during the semester final examination only on special cases. The application must be supported by a medical certificate from a Medical Officer of a medical facility recognised by BSMRAAU. **The concerned student may opt for retaining the sessional courses of the semester** The Academic Council will take the final decision about such cases. However, the total duration for graduation will not exceed 6 academic years.

(V) Incomplete (I) Courses

If a student fails to attend 60% of the classes of any registered course in a semester whatever be the reasons, the registration shall be cancelled for that course and the course will be treated as Incomplete (I) course. In a special case such students may be allowed to appear in the examination with the permission of the Vice Chancellor and it must be approved by the academic council.

(VI) Retake

If a student who has registered for a course for any semester is unable to attend the semester final examination due to (flood/illness/strike), they may retake the exam with full credit by applying within one week of the exam's conclusion and providing documentation of their justification. The Academic Council will then approve their request. However, this chance will only come up once throughout the semester, and regardless of whether the semester covers one or more subjects, it must be attended in conjunction with the subsequent semester that is already in progress. There won't be a chance to take it alone.

(VII) Grade Improvement

If a student's cumulative grade point average (CGPA) is less than 3.00, they may take an exam again to raise their grade in the course. Undergraduate students can take up to 2 subjects, and postgraduate students can take up to 1. The student will be given a maximum grade point of 3.00 in this scenario.

(VIII) Dropout

If any graduating student clears his/her failed course(s) in an Odd Semester and his/her graduation requirements are fulfilled, his/her graduation will be effective from the result publication date of that Odd Semester and that student will be allowed to apply for a provisional certificate.

24. Rules and Regulations for B.Sc. in Avionics Engineering Program

(I) Overview

BSMRAAU has introduced Course Curriculum for undergraduate studies from the academic session 2019-2020. Therefore, the rules and regulations mentioned in this paper will be applicable to students for administering undergraduate curriculum through the Course Curriculum. This will aim at creating a continuous, even, and consistent workload throughout the semester for the students.

(II) The Course Curriculum

The salient features of the Course Curriculum are as follows:

a. The number of theory courses will be generally limited to 5 for each semester. However, with the recommendation of the Course Coordinator and the Head of the Department, the Honourable Vice Chancellor of BSMRAAU may allow relaxation in this regard. This relaxation is to be duly reported to the Academic Council of BSMRAAU.

b. Students will not face any level repeat for failing.

c. Students might get the opportunities to improve their grades.

d. Availability of numerous electives to enable the students to select courses according to their individual needs and preferences.

e. Continuous evaluation of students' performance and achievements.

f. Promotion of effective Lecturer-Content and Faculty-Student interactions.

Beside the professional courses pertaining to each discipline, the undergraduate curriculum gives a strong emphasis on acquiring thorough knowledge in the basic sciences of mathematics, physics, and chemistry. Due importance is also given on the study of several subjects in language, humanities, and social studies. The first two years of bachelor's degree programs generally consist of courses on basic and engineering sciences, and language, humanities, and social studies, while the third and subsequent years focus on specific disciplines with the respective core and elective courses.

(III) Course Semesters

BSMRAAU follows the Semester System for the purpose of conducting lectures and assessments. An academic year consists of two semesters each of which having fourteen weeks of instruction. The undergraduate program will have a total of 8 semesters in 4 years. Semesters are named as Spring (Odd Semester) and Fall (Even Semester) with a provision of a concise Summer Semester (Short Term) where a limited number of courses can be offered as required.

a. Duration of Semesters

The duration of each semester can be extended up to a maximum of 21 working weeks. Following table has the breakdown of a typical term.

Events	Durations
Classes before Midterm	7 weeks
Mid Term Vacation	1 week
Classes after Mid Term	7 weeks
Makeup Classes and Preparatory leave	2 to 3 weeks
Term Final Assessment and Evaluation	2 to 3 weeks
Term End Vacation	4 to 7 weeks

(IV) Course Pattern and Credit Structure

The undergraduate program is covered by a set of theoretical courses along with a set of sessional/laboratory courses to support them.

a. Course Designation System

Each course is designated by a maximum of three letter code identifying the department offering the course followed by a four-digit number. All courses offered by the Department of Avionics Engineering will be prefixed by the department code 'AVE'. The course code contains four numeric digits WXYZ. The interpretation of WXYZ is as follows:

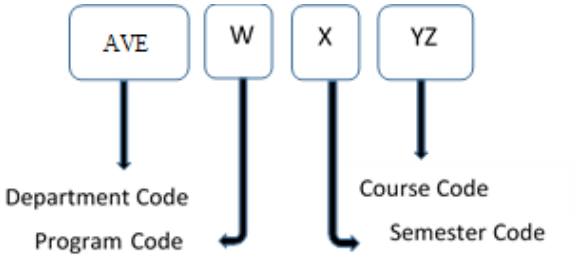
a. The first digit corresponds to the duration of the program. For a 4-year program such as B.Sc. Engineering, the first digit is 4, e.g., course code is AVE 4XYZ. Courses offered by other departments will be prefixed by the department codes; ASE for Aerospace Engineering, EEE for Electrical & Electronic Engineering, CSE for Computer Science and Engineering and MCE for Mechanical Engineering.

b. The second digit X represents semester (1 to 8). For example, a course in the third semester of a four-year program will be ASE 43YZ.

c. Third and fourth digits YZ represent type of course as follows:

Course Code Designation System		
3rd and 4th Digit	**Meaning**	**3rd Digit**
01 to 19	Compulsory Core Courses	0 or 1
20 to 79	Departmental Elective Courses	2, 3, 4, 5, 6, 7
80 to 99	Non-Departmental Elective courses	8 or 9

Odd fourth numeric digit Z represents theory course and even fourth numeric digit Z represents sessional course. The numbering policy can be summarised in the following figure:



For general courses like Mathematics, Physics, Chemistry, Humanities, the prefixes will be as follows; MAT for Mathematics, PHY for Physics, CHM for Chemistry and HUM for Language, Humanities and Social Studies.

b. Assignment of Credits

The assignment of credits to a theoretical course follows a different rule from that of a sessional course.

a. Theoretical Courses. A 'One contact-hour' per week per semester is equivalent to one credit hour.

b. Sessional Courses. A 'Two contact-hours' per week per semester is equivalent to one credit hour.

Credits are also assigned to project and thesis work taken by the students. The amount of credits assigned to such work varies from discipline to discipline.

c. Types of Courses

The types of courses included in the undergraduate curriculum are divided into the following groups/categories:

a. Core Courses. In each discipline, several courses are identified as core courses, which form the nucleus of the respective bachelor's degree program. A student must complete all the designated core courses belonging to his/her discipline.

b. Co- and/or Pre-requisite Courses. Some of the core courses are identified as Co-requisite and/or Pre-requisite to a specific course.

c. Elective Courses: Apart from the core courses, the students can choose from a set of Elective Courses. A required number of Elective Courses from a specified group must be undertaken.

(V) Course Offering and Instruction

The courses to be offered in a semester are announced and published in the Course Catalogue along with the tentative semester Schedule before the end of the previous semester. The courses to be offered in any semester will be decided by the respective department.

Each course is conducted by a course teacher who is responsible for maintaining the expected standard of the course and for the assessment of students' performance. Depending on the class size (i.e. on the number of students) enrolled for the course, the concerned teacher might have course coordinators and Teaching

Assistants (TA) to aid in teaching and assessments.

(VI) Faculty-Student Interaction

The course curriculum encourages students to come in close contact with the teachers. To promote a high level of Faculty-Student Interaction, each student is assigned to an adviser and the student is free to discuss all academic matters with his/her adviser throughout the duration of his/her program. Students are also encouraged to meet with other teachers for assistance and guidance in academic matters at any time through prior arrangement. The Heads of the Departments, the Director of Administration, the Director of Students Welfare (DSW) and the Dean may address the students at intermittent intervals However, students are not allowed to interact with their teachers during and after the moderation process of the questions.

(VII)Student Adviser

One adviser is normally assigned for a group of students by the concerned department. The adviser advises each student about the courses to be taken in each semester by discussing the academic program of that particular semester with the student.

However, it is also the student's responsibility to keep regular contact with his/her adviser who will review and eventually approve the student's specific plan of study (Academic Progression Form) and monitor subsequent progress of the student.

For a student from the second and subsequent terms(semesters), the number and nature of courses for which he/she can register is decided based on his/her academic performance/standing during the previous semester. The adviser may allow the student to drop one or more courses based on his/her previous academic performance/standing. Students' academic performance/standing can be expressed in three ways, namely, Good Standing, Academic Probation and Academic Warning.

(VIII) Course Registration

Any student making use of classroom, laboratory facilities and/or faculty-time is required to register through proper channels. Upon admission to BSMRAAU, each student is assigned to an adviser. Advisers provide necessary guidance to the students in choosing and registering courses.

a. Registration Procedure

At the commencement of each semester, each student must consult with his/her adviser for guidance and assistance before he/she gets registered for the courses. An academic Progression Form could be handy for a student to register for the selection of his/her proposed or projected courses. The date, time and venue of registration are announced in advance by the Registrar's Office. Counselling and advising are done at the same time. It is essential that all students be present for registration within the specified time frame.

b. Pre-conditions for Registration

a. For the first-year students, department-wise enrolment/admission is mandatory prior to registration. At the beginning of the first semester, an orientation program will be conducted where students are handed in with the registration package on submission of the enrollment slip.

b. Any student, other than the new batch, with outstanding dues to the BSMRAAU or a hall of residence is not allowed to register. Each student must clear their dues and obtain a clearance certificate, upon production of which, he/she will be given necessary Course Registration Forms to complete the course registration process.

c. A student can register in a course subject to the class capacity constraints and satisfactory outcome of the pre-requisite courses. However, if a student fails in a pre-requisite course in any semester,

the concerned department may allow him/her to register for a course which requires a pre-requisite course condition provided that his/her class attendance, involvement and performance during the continuous assessment process for that particular pre-requisite course is found to be satisfactory.

c. Registration Deadline

Each student must register for the courses to be taken before the commencement of each semester. Late registration is permitted only during the first week of classes. Late registration after this date will not be accepted unless the student submits a written application to the Registrar through the concerned Head of the department explaining the reasons for delay. Acceptable reasons could be on medical grounds with supporting documents from the Medical Officer of a Medical Facility recognised by BSMRAAU or any other arguably acceptable academic commitments that prohibit enrolment prior to the last date of registration.

d. Penalty for Late Registration

Students who fail to register during the designated dates for registration are charged a **late registration fee of Tk. 200.00 (Taka two hundred only) per course.** Penalty for late registration will not be waived.

(IX) Limits on the Credit Hours to be taken

A student should be enrolled for **at least 15 credit** hours and can take a maximum of 24 credit hours. Relaxation on minimum credit hours may be allowed. A student must enrol for the sessional courses prescribed in a semester within the allowable credit hour limits.

In special cases where it is not possible to allot the minimum required 15 credit hours to a student, the concerned department may permit with the approval of the Vice Chancellor, a lesser number of credit hours to suit individual requirements based on a student's academic standing, such as, Academic Probation and/or Academic Warning. Such cases are also applicable to students of Level 4 requiring less than 15 credit hours for graduation.

(X) Course Add/Drop

A student has limited options to add or drop courses from the registration list. **Addition of courses** is allowed **only within the first two weeks of a regular** semester and **only during the first week of a short** semester. **Dropping a course** is permitted **within the first four weeks of a regular** semester and **two weeks of a short** semester.

Any student willing to add or drop courses must fill in a Course Adjustment Form. This must also be done in consultation with and under the guidance of the student's respective adviser. The original copy of the Course Adjustment Form must be submitted to the Registrar's Office, where the required numbers of photocopies are prepared for distribution to the concerned adviser, Head of the Departments, Dean of the Faculty, Controller of Examinations and the students.

(XI) Distribution of Marks

a. Theory

Forty percent (40%) of marks of a theoretical course shall be allotted for continuous assessment, i.e., quizzes, home assignments/case studies, class tests (CTs), observations/class participation and midterm examination. These marks must be submitted to the Office of the Controller of Examinations before the commencement of the final exam. The rest of the marks will be allotted to the semester Final Examination. The duration of the final examination will be three (03) hours. The scheme of continuous assessment that a teacher would follow for a course will be announced on the first day of the classes. Distribution of marks for a given course per credit is as follows:

Class Performance	5%
Class Assignments/Case Studies/Class Test	20%
Mid Term Exam	15%
Final Exam	60%
Total	**100%**

b. Sessional

Sessional courses are designed and conducted by the concerned departments. Examination on sessional/practical subjects will be conducted by the respective department before the commencement of semester final examination. The date of practical examination will be fixed by the respective department. Students will be evaluated in the sessional courses based on the followings (all or as decided by the Examination Sub-Committee):

Class Performance	15%
Quiz	10%
Report Writing	15%
Mid Term Evaluation	20%
Final Evaluation	30%
Viva Voce	10%
Total	**100%**

c. Sessional Course in English

Students in English sessional course shall be evaluated based on the following:

Class Participation/Observation	5%
Written Assignment	20%
Oral Performance	25%
Listening Skill	10%
Group Presentation	30%
Viva Voce	10%
Total	**100%**

(XII) Collegiate and Non-Collegiate

Students having class attendance of **75% or above (≥75%)** in an individual subject will be **treated as collegiate** and **less than 75% and up to 60% (≥60%, <75%) will be treated as non-collegiate** in that subject. The non-collegiate student(s) may be allowed to appear in the examination subject to payment of **non-collegiate fee/fine** of an amount fixed by BSMRAAU. Students having class **attendance below 60%** (**<60%**) will be treated as **dis-collegiate** and **will not be allowed to appear in the examination and treated as a failure**. But in a special case such students may be allowed to appear in the examination with the permission of the Vice Chancellor and it must be approved by the Academic Council.

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(XIII) Impacts of Grade Earned

The courses in which a student has earned a 'D', or a higher grade will be counted as credits earned by him/her. Any course in which a student has obtained an 'F' grade will not be counted towards his/her earned credits or GPA calculation. **However, the 'F' grade will remain permanently on the Grade Sheet and the Transcript.**

A student who obtains an 'F' grade in a core course will have to repeat that course. However, if a student gets an 'F' in an optional course, he/she may choose to repeat that course or take a substitute course if

available. When a student will repeat a course in which he/she has previously obtained an 'F', he/she will not be eligible to get a grade better than 'B+' in that repeated course.

If a student obtains a grade lower than 'B+' in a course, he/she will be allowed to repeat the course only once for the purpose of grade improvement. However, he/she will not be eligible to get a grade better than 'B+' for an improvement course.

A student will be permitted to repeat for grade improvement purposes a maximum of 2 courses in BSc. Engineering program.

If a student obtains a 'B+' or a better grade in any course, he/she will not be allowed to repeat the course for the purpose of grade improvement.

(XIV) Classification of Students

At BSMRAAU, regular students are classified according to the number of credit hours completed/earned towards a degree. The following classification applies to all the students:

Level	Credit Hours Earned
Level 1	0.0 to 36.0
Level 2	More than 36.0 to 72.0
Level 3	More than 72.0 to 108.0
Level 4	More than 108.0

However, before the commencement of each semester all students other than new batch are classified into three categories:

a. **Category 1 (Good Standing).** This category consists of students who have passed all the courses described for the semester. A student belonging to this category will be eligible to register for all courses prescribed for the upcoming semester.

b. **Category 2 (Academic Probation).** This category consists of students who have earned a minimum of 15 credits but do not belong to category 1. A student belonging to this category is **advised to take at least one course less** since he might have to register for one or more backlog courses as prescribed by his/her adviser.

c. **Category 3 (Academic Warning).** This category consists of students who have failed to earn the minimum required 15 credits in the previous semester. A student belonging to this category is **advised to take at least two courses less** than a category 1 student subject to the constraint of registering at least 15 credits. However, he/she will also be required to register for backlog courses as prescribed by the adviser.

a. Graduating Student

Graduating students are those who will have ≤ 24 credit hours left for completing the degree requirement.

(XV) Performance Evaluation

The performance of a student will be evaluated in terms of two indices, viz. Semester Grade Point Average and Cumulative Grade Point Average which is the grade average for all the semesters completed.

Students will be making normal progress toward a degree if their Cumulative Grade Point Average

(CGPA) for all work attempted is 2.20 or higher. Students who regularly maintain a semester GPA of 2.20 or better are making good progress toward the degrees and are in good standing with BSMRAAU. Students who fail to maintain this minimum rate of progress will not be in good standing. This can happen when any one of the following conditions exists.

a. The semester GPA falls below 2.20.
b. The Cumulative Grade Point Average (CGPA) falls below 2.20.
c. The earned number of credits falls below 15 multiplied by the number of semesters attended.

All such students can make up for their deficiencies in GPA and credit requirements by completing courses in the subsequent semester(s) and backlog courses, if there are any, for better grades. When the minimum GPA and credit requirements are achieved, the student is again returned to good standing.

(XVI) Application for Graduation and Award of Degree

A student who has fulfilled all the academic requirements for bachelor's degree will have to **apply to the Controller of Examinations** through his/her Adviser for graduation. Provisional Degree will be awarded by BSMRAAU on completion of credits and GPA requirements.

(XVII) Attendance, Conduct and Discipline

BSMRAAU has strict rules regarding the issues of attendance in class and discipline.

a. Attendance

All students are expected to attend classes regularly. The university believes that attendance is necessary for effective learning. The first responsibility of a student is to attend classes regularly, and one is required to attend the classes as per BSMRAAU rules.

b. Conduct and Discipline

During their stay at BSMRAAU all students are required to abide by the existing rules, regulations and code of conduct. Students are strictly forbidden to get involved in forming student organisations or political parties, clubs, society etc., other than those set up by BSMRAAU authority in order to enhance student's physical, intellectual, moral and ethical development. Zero tolerance is strictly observed in and around the campus with regards of sexual abuse and harassment in any forms and drug abuse and addiction.

(XVIII) Absence during a Semester

A student should not be absent from quizzes, tests, etc. during the term(semester). Such absence will naturally lead to reduction in points/marks, which count towards his/her final grade. Absence in the Semester Final Examination will result in an F grade in the corresponding course. A student who has been absent for short periods, up to a maximum of three weeks due to illness, should approach the course teacher(s) or the course coordinator(s) for make-up quizzes or assignments immediately upon returning to classes. Such requests must be supported by a medical certificate from competent authority (e.g. BSMRAAU Medical Officer).

(XIX) Recognition of Performance

As recognition of performance and ensure continued studies BSMRAAU awards medals, scholarships and stipends will be given as per existing rules and practices.

(XX) Types of Different Examination

Following different types of final Examinations will be conducted in BSMRAAU to evaluate the students

of Undergraduate Programs:

a. Regular Examination: It denotes that a student has not failed any courses and that the semester or academic year was successfully completed within the allotted time (registration).

b. Irregular Examination: It shall be regarded as such if a student obtains F Grade in any subject or refers, repeats, or has a backlog in any semester. No grade higher than B will be given in this situation, and the course must be finished within the allotted time.

c. Referred Examination: If a student fails no more than two subjects in a calendar year, such as two subjects in the odd semester and none in the even semester, or one subject in any semester and one failed subject in the previous semester, he may take the referred exam in addition to the final exam for the following semester. No grade higher than B will be awarded in this instance, though.

d. Repeat Examination/Re-Exam: With the Academic Council's permission, a student who does not enrol in the course during any semester other than the first semester may retake the exam during the following semester. No grade higher than B will be awarded in this situation, unless necessary for health. It must, however, be validated by the university doctor and permitted by the vice chancellor if there are medical reasons. A student may only repeat an academic year once during their time in the program.

e. Supplementary Examination:

1. Applicable to students who are graduating: If a student satisfies the requirements listed below, they may register for a supplementary examination:

a. Students who have already registered and failed one or two theory courses but still need them to be graduated.
b. Students who finished all their courses, such as sessional industrial training, practical survey, etc., before the semester of graduation.
c. Supplementary exams may not be utilized to raise a course's grade, and no student may get a grade higher than the possible B+ on this exam.
d. A written application should be submitted to the Head of Department through appropriate Adviser in the prescribed format.
e. Students who failed on the supplementary exam may still take the semester final exam.

2. For regular students only: If a student satisfies the requirements listed below, they may register for a supplementary examination:

a. Students at this university who have earned at least five semesters of credits in undergraduate engineering program are only permitted to enrol for supplementary examinations. They can take two theory courses out of all theory courses they failed in previous semesters for the supplemental examination at that semester. Other students are limited to enrolling in one theory course only.
b. The concerned program will choose the subjects for the supplementary test.
c. A written application should be submitted to the Head of Department through appropriate Adviser in the prescribed format.
d. A student may enrol in the course during the normal semester after registering for a course supplementary examination. The enrolment for the course will be automatically cancelled from the regular semester if the course is passed in the supplementary exam.
e. Students who did not pass the supplementary exam may still take the semester final exam.

3. For all students:

a. Supplementary Exams may not be utilized to raise a student's grade in any course, and the highest possible grade on one of these exams is B+.
b. The supplementary exam will take place after two weeks of the even semesters
c. The supplementary exam must be finished in one week.

d. A maximum of one supplementary examination may be held each year.

f. Short Term Examination: Subject to the approval of the Academic Council during the next two weeks after completing a course of 3-6 weeks, a student is prevented from passing/graduating to the next higher step if they fail/do not register in two or more courses/subjects (other than Sessional) without Backlog must complete the final exam. This test must be given in successive even semesters. It should be emphasized that the prior continuous evaluation from the short-term examination—for which 40% of the possible points were awarded—will be cancelled. No grade higher than B will be awarded on a new continuous evaluation (40% marks). There won't be any short-term examination if the topic is a sessional one.

g. Backlog: A student will advance to the next higher step if they fail no more than one course or subject after retake/supplementary exam. Failures will be treated as Backlog subjects, requiring action during the subsequent re-examination (except for short term). If not, the student will be dismissed from the university.

h. Improvement Examination: If a student's cumulative grade point average (CGPA) is less than 3.00, they may take an exam again to raise their grade in the course. Undergraduate students can take up to 2 subjects, and postgraduate students can take up to 1. The student will be given a maximum grade point of 3.00 in this scenario.

(XXI) Conduction of re-examination:

a. A candidate may be permitted to retake a course if they score "F" on one of their exams if they receive at least "D" on all their other exams that semester. On the other hand, if a candidate receives "D" or "F" in one or more exams, they will only be granted the chance to retake the exam for that topic.
b. If a student fails more than two courses or subjects in two normal semesters, he or she must retake the exam in the following regular semester or may retake it later.
c. Students in the second semester of every level, regardless of the first semester's results.
d. A student may voluntarily withdraw from the semester and reapply for the academic year the following year (Academic with the Council's approval); in this case, a full grade will be awarded if the student misses more than 60% of classes due to illness or is unable to participate in a semester examination.
e. All exams, except for regular exams, must be finished within a week.

Note: For conduct of examinations, the examination policy of the university will be the main guiding policy.